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Digital Curation in the Digital Humanities

Preserving and Promoting Archival and Special Collections

Arjun Sabharwal



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About the author

Arjun Sabharwal joined the University of Toledo Library faculty in January 2009 as assistant professor and digital initiatives librarian. He holds a Master of Library and Information Science and a Graduate Certificate in Archival Administration in addition to the previously earned graduate degrees. He oversees the digital preservation of archival collections, manages the Toledo's Attic virtual museum Web site, designs virtual exhibitions, leads the planning and implementation of UTOPIA (The University of Toledo OPen Institutional Archive) and the University of Toledo Digital Repository at the university, and manages digitization projects. Current professional interests include archiving, digital humanities, digital history, and developing thematic research collections. He has authored several research articles and reviews, and presented at conferences on work related to archives and digital libraries. Since 2010, he has engaged in digital scholarship via his international blog on ResearchGate titled *Digital Humanities and Archives*.

Preface and Acknowledgments

In the digital age with mobile technology, ubiquitous computing, digital humanities, and trustworthy repositories entering our discourse on archives and preservation, few are aware that archives are among the legacies of Sumerian civilization (born over five millennia ago), which left behind such literary masterworks as *The Epic of Gilgamesh,* and institutional records of administrative and cultural significance (see Robson, 2003). Perhaps it takes someone interested in Sumerian culture and cuneiform writing to share such an obscure factoid! While this book is not about the history of archives, it suffices to say that social, intellectual, political, and cultural forces that have shaped the world since the Sumerian civilization have also influenced the development of archives as we see them today. Records tell more than meets the eye!

The paradigm shifts spanning especially the past two centuries—from the French Revolution to the present (Cook, 2013)—have ushered in new and more exciting (as well as challenging) times for archives, bringing greater visibility and interaction with the profession. The third and fourth paradigms in the past century alone placed archivists in public role as mediators, interpreters, and even as activists who shape public memory and community identity through dedication to history, cultural heritage, and public knowledge. The focus on preserving and promoting archives and special collections has benefited from new technologies as well as new cultural, social, and political directions. The emergence of the public and digital humanities along with timely concerns about preserving the digital record naturally places archivists and digital curators in the middle of the emerging digital landscape. This background is the preamble for my book, which builds on personal and professional interests as well as interdisciplinary approaches to curating cultural heritage collections.

Personally, I have always held a profound interest in the Humanities—from Antiquity and Classics to more contemporary forms in literature, arts, folklore, and ethnic traditions. Professionally, my interests in the digital humanities motivated me to bring digital technology further into my work at the Ward M. Canaday Center for Special Collections and for Toledo's Attic, a pioneering historical hypertext project, which started in 1995 and produced a Web site in 1997. Working as a digital initiatives librarian with archival training and interests puts me into a two-way epistemic gateway where I work with knowledge in nondigital and digital collections. This position enables me to map across two somewhat overlapping worlds and help researchers identify resources on the other side from where they may be. From the "digital" side of things, I must remind digital humanists that there is a lot more out there (or actually *in there* since archival collections are mainly in boxes) than meets the computer—and thereafter the eye. At the same time, on the "nondigital" side, I must ensure that I

reach out to the digital humanists and historians who are doing wonderful things with technology and scholarship, and see how I can help their work. My book respects the legacy and accomplishments of the archival profession and its continuing importance into the future. Nevertheless, it recognizes the broader implications of digital curation for digital humanities and other disciplines. This is a mutually engaging space—a digital curation workspace—where collaboration and discourse continue to present new knowledge regardless of the format of the primary source. Therefore, these two worlds cannot be mutually exclusive!

This book considers a broader theoretical and practical framework for digital curation, which expands from its association with the core aspects of archiving and digital preservation to supporting innovative and creative projects framed around specific historiographical and humanistic question and hermeneutic activities. The scope of this material is not prescriptive, as I ardently recognize and advocate the uniqueness and relevance of every institution's archive, preservation priorities, and curation programs regardless of size and support. The intended audience for this book, therefore, includes archivists, historians, digital humanists, educators (including those teaching online), digital curators, special collection librarians, historically "wired" information architects interested in cultural heritage, project managers, administrators at heritage institutions, citizen scholars, and hopefully some others not mentioned here. The book focuses on digital curation within the broader framework of the digital humanities, digital history, and archival preservation of digital heritage collections. It discusses the integral role of metadata, information architecture, social curation, and collaboration in the preservation and promotion of archives and special collections.

Curators may consider a technorealist position taken by Cohen and Rosenzweig (2006) in their approach to digital history but with preservation in mind. Tibbo and Lee (2007) address the need for open standards in digital curation, which are crucial to the long-term viability of preservation efforts pursued at institutions. Serious and long-term commitments to open-source operating systems, software, and servers may foster productive relationships between archivists, librarians, technologists, and faculty. At the same time, such commitments should not mean or suggest the replacement of valued nondigital collections; there is a continuing demand for such collections for research purposes. Curators must develop a cautious and realistic outlook and hindsight: while DSpace, ArchiveSpace, Fedora Commons, Omeka, and some others not listed here are committed to long-term digital preservation, their installation, and maintenance will require specialized IT skills and dedicated support that simply are not available (or made available) in some institutions opting for commercial platforms. More affordable solutions (usually through consortia) may pose limits on collection size and options to customize despite the good will from developers who have generously helped me in the past. Another source of concern is the inability or inflexibility of some commercial services, hosted products, and emerging technologies to support the production, preservation, and accessibility of obsolete digital file formats, storage media, software, and operating systems. Although commercial and hosted services may be practical for small- to mid-sized institutions, commercial interests ultimately define and narrow the windows of usability, rendering older content, software, and storage obsolete and useless. If digital curation aims to preserve

both digitized and born-digital content, strategies for preservation and accessibility will require greater harmonization across existing barriers. Digital curation standards, therefore, should extend to all platforms as to reduce the effects of obsolescence.

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Several individuals provided varying levels of support for the projects referenced in this book, but first I would like to thank my wife and love of 25 years, Kim Chi Tran, for supporting my work on the book and everything else. I also thank my colleagues at the Ward M. Canaday Center for Special Collections—Barbara Floyd, Tamara Jones, Sara Mouch and Patrick Cook—for input and participation in developing the digital collections and virtual exhibitions. I also thank my good colleagues in the University Libraries for their input and collaboration on many projects—namely Christine Rigda, Gerald Natal, Wade Lee, Sheryl Stevens, and Laura Kinner. Special thanks also go out to the OhioLINK Digital Resource Commons Team for its most generous assistance with the University of Toledo Digital Resource Commons collections between 2007 and 2013. Tedd Long's role was instrumental in getting the Toledo's Attic project moved to the Joomla platform with which he has selflessly assisted in many ways. I would also like to thank the Toledo's Attic Steering Committee and its co-chairs Barbara Floyd and Dr. Earnest Weaver for their support. Last but not least, I must acknowledge the direct and profound influence of the Scholarly Dashboard workshop series led by Dr. Andrew Schocket, Associate Professor of History and American Culture Studies and Director of American Culture Studies at Bowling Green State University, organized by OhioLINK, and funded by the National Endowment for the Humanities. In addition to introducing me to the digital humanities, the discussions at these workshops also helped me recognize a greater role for archives and libraries in the digital humanities through innovative projects.

Introduction

Conceptualizing the framework for digital curation

Archives and special collections departments have a long history of preserving and providing long-term access to organizational records, rare books, and other unique primary sources including manuscripts, photographs, recordings, and artifacts in various formats. The careful curatorial attention to such records has also ensured that such records remain available to researchers and the public as sources of knowledge, memory, and identity. In the past semicentury, various analog and digital technologies have emerged, introducing at least as many new media formats, standards, devices, and production methods. Archivists closely following this trend have also taken notice of the exponentially growing volume of born-digital contents included with accessioned collections, which required new methods of preservation in the digital environment. The international conversation on digital preservation and curation at conferences, workshops, and meetings in the early 2000s was in direct response to these trends and developments. These early discussions involved archivists, librarians, technologists, and researchers aiming to ensure long-term access to valuable historical records (evidence), data, heritage collections, and scholarship in digital forms. The audience for digitally produced and preserved content includes digital humanists, scholars from other disciplines, open access communities, institutions, and educators interested in integrating digital resources into curricula and coursework, and businesses (e.g., Lyrasis, DuraSpace, OCLC, Bepress, and others) providing repository services and storage. Archives remain vital to this emerging intellectual landscape focused on using, reusing, and producing data, information, and knowledge in digital forms.

Digital curation presents an important framework for the continued preservation of digitized and born-digital collections, given the ephemeral and device-dependent nature of digital content. With the emergence of analog and digital media formats in close succession (compared with earlier paper- and film-based formats) came new standards, technologies, methods, documentation, and workflows to ensure safe storage and access to content and associated metadata. Researchers in the digital humanities have extensively applied computing to research; for them, continued access to primary data and cultural heritage means both the continuation of humanities scholarship and new methodologies not possible without digital technology. Digital curation and digital humanities, therefore, comprise a joint framework for preserving, promoting, and accessing digital collections.

This framework has multiple components bound together by theory, practice, and social relationships: in addition to digital curation and digital humanities, information architecture, social media, and knowledge management significantly contribute to the cohesiveness of this framework. In a sense, there is a curation ecosystem—that is (among other things), “web content which is cataloged, cross-referenced, rated, filtered, and otherwise

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managed by human beings" (Jones, 2011)—in place, which reinforces various conceptual and physical relationships among the components of digital curation environment. This ecosystem holds together a knowledge architecture—a concept that brings together people, contents, and technologies (Applehans, Globe, & Laugero, 1999) into meaningful strategic relationships throughout the digital content lifecycle. These knowledge architectures naturally vary by institutions and departments with differing finances, staffing priorities, and politics to shape the collaboration landscape. The book does not explore institutional politics, however.

The Digital Curation Center at the University of Edinburgh has produced a digital curation lifecycle model—a practical framework for conceptualizing, planning, and implementing digital curation initiatives at institutions of all sizes. The model brings the various components of the aforementioned knowledge architectures into working relationships throughout the curation lifecycle. While the model presents details about the cycle itself, there is much to say about the conceptualization part, which only gets a spot in the model but in fact, may inform everything else in the rest of the cycle. An organization's disciplinary focus, curricular structure (if it is an academic institution), community (with interests in local cultural heritage), stakeholder interests, and many other factors must come into consideration during conceptualization. Questions about metadata schema, repository platform, content structure, and controlled vocabularies must receive attention at the conceptualization (including scoping) meetings well before implementation. At this stage also, the various specialists must meet to inform the process, but making such connections may be a matter of organizational politics rather than considering the real needs for a long-term project.

This book explores at great length the conceptualization of digital curation projects with interdisciplinary approaches that combine the digital humanities and history, information architecture, social networking, and other themes for such a framework. The individual chapters focus on the specifics of each area, but the relationships holding the knowledge architecture and the digital curation lifecycle model together remain an overarching theme throughout the book; thus, each chapter connects to others on a conceptual, theoretical, or practical level. The uniqueness of these frameworks is important, however; this book treats every institution, archival collection, digital repository, staff, and curation program as inherently unique regardless of size, funding, staffing, or available technologies. No two cultural heritage collections merit qualitative comparison!

Figure I.1 presents the framework of various relationships that shape the conceptual foundation for this book. The modified digital curation lifecycle model occupies a significant place in this framework with the content at the very center of the process surrounded by associated and interrelated processes, key participants including archivists and researchers, and digital resources and social media. The model, in fact, outlines the various knowledge architectures (people, content, and technology) present at various levels, as their role is to shape the curation process. Archivists and digital curators play a central role in this model, as they apply technologies such as digitization tools, digital repositories, social media, and visualization tools to create, preserve, and curate the content in collaboration with digital humanities researchers. Historians, other digital humanists, and scientists, in turn, shape the disciplinary context within which this model evolves.

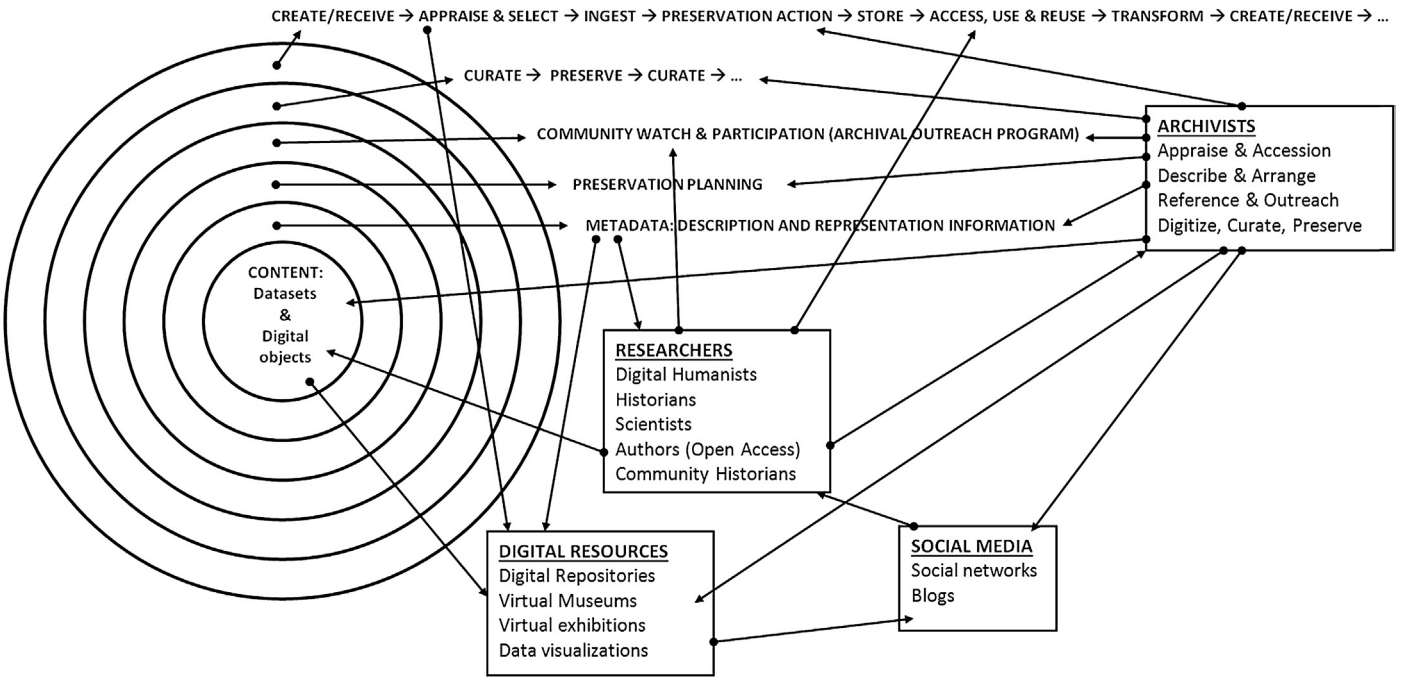


Figure I.1 Digital curation framework.

Archivists and librarians may also be specialists and even experts in these fields and may collaborate on a higher level of curation.

Figure I.1 also presents archivists in an important role as shown by the corresponding box on the top right with arrows point to several regions of the curation lifecycle model (representing routine operations), and other arrows pointing to researchers (representing collaborative relationships outside the department), and technologies utilized in the curation process. Although social media is not a part of the preservation/curation routine, it is an important part of outreach efforts as well as what researchers need as potential data source. Chapters 1 and 2 focus attention on the role of archivists in digital curation and the digital humanities while Chapter 7 addresses the impact of social media on digital curation. Finally, the box showing digital resources list some of the products (or deliverables) in an expanded digital curation practice. Digital repositories, virtual museums, virtual exhibitions, and data visualizations provide support for essential archival and library functions such as reference, instruction, and outreach. Virtual exhibitions and virtual museums (which may bring several institutions into collaborative relationships) are extensions of archival public services and outreach strategies aiming to preserve and promote collections especially after physical exhibitions end. Historians and other scholars in the Humanites can use or contribute to these resources, resulting in data visualizations available for analysis, interpretation, and scholarly communication purposes. Therefore, the digital curation framework presents a rich context for archival and scholarly practices associated with digital curation.

The theoretical relationships in this framework not only reinforce some of the practice-based relationships discussed above, but also serve as frameworks themselves for future relationships or conceptualizations for future projects. Chapter 5 relates the outermost region of the lifecycle model to archival theory and practice related to appraisal (Cox, 2000). Other areas of archival theory and practice can also comprise this framework; in fact, Mortensen (1999) explores the place of archival theory in archiving practice, shunning the "positivist idea of science" (p. 1), which ignores the historical, social, cultural, and political significance of archival work. Cook's article on the changing paradigms in archival work (2013) points out such dimensions, thus placing archival theory (not science) in an intellectual region bordering the humanities and social sciences. Various works (Cook, 2001a; Cook, 2001b; Greene, 2002; Griffin, 2010) address the effects of postmodernism on archival theory and practice. The effects of cultural diversity have not only changed the collections in archives but have transformed the curatorial balance between the archives and the public (Cook, 2013).

Another critical aspect of archival theory is related to provenance, which lends credibility to the archival record, and is crucial to the contextualization of manuscript collections for historiographical purposes: provenancial properties present evidence to historians placing records into historical, social, cultural, political, and organizational contexts (Bearman & Lytle, 2000; Sternfeld, 2011). For instance, redacted information may not be visible in digital documents but researchers accessing the physical records associated with a particular organization may find related records to support theories behind top-level decisions in an organization. Most digitization projects have passed through an initial selection process, and for a number of reasons may not always include every piece of paper in the archival box. As for a theory of

digital preservation, therefore, Moore (2008) argues that such a theory is necessary for purposes of documenting the process in the preservation (and curation) environment and decisions by curators. Watry (2007) writes,

> *A theory of preservation extends the concept of digital preservation from one focused on sending the records (metadata) into the future to one that can also send along the documentation of the environment that is being used to manage and read the records. The true test of a preservation environment is whether it describes the entire preservation information context sufficiently well that the records can be migrated into an independent preservation environment without loss of authenticity or integrity. This requires migrating not only the records, but also the characterizations of the preservation environment context.*
>
> *p. 42*

The resulting documentation is present in descriptive, preservation, and technical metadata fields created prior to, during, and after collection development and migration as well. These theoretical foundations provide the framework for archival activities indicted by the arrows between the Archivists' box and the activities various regions in Figure I.1.

Chapter 4 focuses on information architecture—a field focused on designing information spaces (Rosenfeld & Morville, 2002)—which brings the various digital domains (digital repositories, virtual exhibitions, data visualizations, finding aids, library catalogs, and other resources) in digital curation work into virtual relationships via hyperlinks. This brings information architecture into a theoretical relationship with hypertext theory (Aarseth, 1994, 1997; Landow, 1992, 1994, 2006; Nelson, 1992) as well as social network theory (Kadushin, 2004, 2012) both of which have evolved around node–link relationships.

Scholars in the digital humanities and (equally importantly) history also construct theoretical frameworks for analysis and interpretation based on data and information in digital repositories, research databases, and data visualization tools. One of the most contentious issues is whether the work that digital humanists perform is scholarly and theoretical in scope (Davidson, 2012; Drucker, 2012; Gibbs, 2011; Gold, 2012b; Hall, 2011; Schmidt, 2011). Stephan Ramsey (as cited in Gold, 2012b) advocates for the recognition of repository building (which involves search, information retrieval, and discovery as scholarly work in the digital initiatives). Others (Ramsey & Rockwell, 2012) point out that digital artifacts such as platforms, software, and digital collections are prototypes that use specific metadata fields to validate theories and explain observations. Digital artifacts, they argue, are hermeneutical instruments for purposes of interpreting events, phenomena, or answer humanistic questions and are suitable for hermeneutic activity since they present contextual data (that is, metadata) for intersubjective reading, analysis, discussion, and interpretation. Thus, they become "theory frameworks" for interpreting.

In fact, visualization tools are also hermeneutical instruments that present analytical and interpretive information using the metadata record. For instance, location data and dates allow the spatial and chronological mapping of data sets and inform new humanistic question. The scholarly and theoretical scope of collection building activity, therefore, evolves in the theoretical frameworks of disciplines (such as

Digital Humanities and History) requiring such data-driven computing activities. In the history area, digital historical representations such as digital collections, virtual museums, data visualizations, and other digital resources (Sternfeld, 2010, 2011) support digital historiography—an interdisciplinary field bringing archival metadata into the theoretical framework of historical analysis, interpretation, and writing. While narrative historical writing may not always require structured digital data other than those found in digitized records and publications, historical computing approaches will directly benefit from the reaffirming relationship between archives and the discipline of history and other humanities. Hence the arrow between the Researcher box and content and metadata regions of the curation lifecycle model (Figure I.1)!

A different perspective on or approach to digital curation may have benefitted from integrating general systems theory in the framework for this book; for instance, works on interdisciplinary methods (Klein, 1990, 1996) cite system theory as framework for interdisciplinary research. At the most basic level, the approach taken in this book may benefit from Ludwig von Bertalanffy's (1969) general system theory, which scientifically investigates "wholes" or "wholeness" through isomorphisms (formal or structural correspondences between elements of two or more sets) that comprise systems. The three aspects of general systems theory—system science, system technology, and system philosophy—comprise a separate framework for studying the relationship between digital curation and digital humanities. The first aspect is system science, which investigates the theory of systems in the associated disciplines and principles applicable to all systems. Understanding the relationship between archival theory and digital humanities will require a systematic study involving several adjoining theoretical frameworks. System technology presents a more specific agenda due to the specific role technology plays in digital curation and Digital Humanities (DH). In the present context, the scope of system technologies is not limited to hardware, programs, automation, and software; it embraces the study of social networks, Web 2.0 tools, hypertext structures, metadata, ontologies, and various preservation and visualization technologies that support analytic and hermeneutic activities in the DH community.

The system approach is evident throughout this book, as the chapters focus on the role of archives and digital curation in Digital Humanities and History. Developing a system of information architecture (Morville, 2011) also demonstrates the need for a system approach, which also frames the hypertextual networks and social relationships for other various methods of digital humanities curation. The social curation ecosystem explored in Chapter 6, for instance, communicates the interrelationships of various interconnected social networks that comprise a whole, but is more typically a fragmented environment. Finally, system philosophy—or the "reorientation of thought and world view … as a [then] new scientific paradigm" (Bertalanffy, 1969, p. xxi) presents a new philosophical approach through system ontology, or the definition and description of system building blocks and their relationships on various levels. System epistemology refers to the interaction between the knower and known, which specifically holds as digital curation aims to preserve now only the bit-level data but also the information and knowledge needed to support digital humanities projects and research. Digital curation and DH comprise a system in which their

ontological and methodological relationships bring various theories and practices into close relationships.

Finally, the framework greatly depends on social relationships. Chapter 6 investigates collaboration and the importance of knowledge architectures in digital curation. Chapter 7, in contrast, focuses on the impact of social media and social networks on digital curation. Kadushin (2004, 2012) approaches social networks from a sociological perspective to build a theoretical framework based on various types of node–link relationships. Similar node–links relationships have become the building blocks of George P. Landow's hypertext theory (2006). Both social and hypertext relationships form a dynamic link in the social curation ecosystem discussed in Chapter 7.

The focus on Web-based social networks should not communicate the message that electronic communication may replace, displace, or otherwise substitute verbal and personal communication and close collaboration at the workplace or in the scholarly communication system. In fact, personal digital archiving—the latest trend in personal history curation—in social networks present significant challenges to standard archiving practices needed for long-term preservation and curation. These challenges have three dimensions: first, the file formats are low-resolution lossy files with a short life span. This is because content produced with some current technologies and applications may not easily be converted into such broadly recognized archival lossless formats as TIFF, PDF/A, JPEG2000, and other preferred file types simply because of their file size, resolution, and the bandwidth required to transmit such huge files. Another issue is provenance and the history of ownership, which may raise questions about privacy and copyright in the context of digital communication networks. While the Digital Millennium Copyright Act and Technology, Education and Copyright Harmonization (TEACH) Act have directly addressed the copyright status and the use of digital content on the Web, concerns about privacy, libel, and other issues will surface as soon as research about ownership of digital files produces tangible and verifiable results. While his book does not investigate the legal issues surrounding digital content, this is an important matter that archives and libraries are handling or will eventually do. The third and recently debated issue is the use of social tagging in contrast to professionally developed classification systems and related ontologies.

The line of opposition between those preferring ontologies and taxonomies and those advocating the use of social tags is clear since tags are more popular in comparison to formal ontologies considered rigid. Gruber (2007b) notes that the "attack on 'ontology' is really an attack on top down categorization *as a way of finding and organizing information,* and the praise for folksonomies is really the observation that we now have an entirely new source of data for finding and organizing information: *user participation*." ("Comparing apples," para. 4). Peterson (2006) warns about the use of folksonomies, which are open-ended, unsystematic, and often deemed by some information professionals to be unsophisticated. Some authors (Lee & Neal, 2010; Weinberger, 2005) point out the strongly subjective nature of social tagging, which impede information retrieval and affect the interpretation of social tags; for instance, a wonderfully restored historic building may be described minimalistically, using irrelevant tag keywords pertaining to personal reaction or lack of knowledge. How can a preservationist locate potentially useful information using practically useless

information representation? In defense of social ontologies, Avery (2010) emphasizes the relationship of social tagging to collaborative indexing and group intelligence, which underscores the value behind collaborative (and distributed) curation practices common to social networking. These competing description strategies may prevail in curation approaches where subject metadata fields contain values following professional standards. Ding et al. (2010) present a hybrid method called Upper Tag Ontologies to bridge the cultural and ontological gap between the two competing communities. Thus, social networks have a significant effect on digital curation.

The sequence of the chapter topics is important, as it outlines the conceptual framework for this book. Chapter 1 focuses on defining digital curation and relating various levels of curation to the work of digital humanities scholars. Chapter 2 shifts focus toward the digital humanities in general while focusing on the role of archives and libraries in the digital humanities landscape. The discussion on the debates and the scholarly nature of digital humanities projects is important, as it shapes the discourse on the discipline as well as the highly visible contributions of archivists and librarians playing various leading and support roles. Chapter 3 focuses on digital history, historiography, and the continuing (even transforming) role of archives. As archivists have emerged throughout the history of the profession, shedding the image of invisible record keepers and taking on new roles in the community, the paradigm shift has presented new opportunities for them.

Chapter 4 brings information architecture into the book's framework because not only have archivists contributed to or, in some cases, created (often single handedly) various works of historical hypertext which required good design, but some knowledge of information architecture (a huge field in itself with other practices) can help archivists and historians understand and document the details of digital historical representations (Web sites, digital collections, visualizations, etc.) within the code, and that is a part of the curation process. Chapters 1 – 4, therefore, are vital to the conceptualization part of the lifecycle model while Chapter 5 presents the curation lifecycle model as the practical framework that sets the process into motion after conceptualization. The chapter addresses routine aspects of a team-driven approach. Chapter 6 places this information in the broader organizational context where collaboration develops around the various knowledge architectures in place at the level of archives, academic library, institution, and the surrounding community. This chapter relates directly to the community involvement as aspect of the lifecycle model: stakeholders (including community members) monitor and participate at various stages of the lifecycle model. Finally, Chapter 7 brings in the public into social curation. The paradigm shifts (Cook, 2013) leading to a highly visible role of archivists and digital curators places them into a close working relationship with the public. Some heritage institutions, such as the Brooklyn Museum engages the public in a crowd-curation process. Archivists, along with historians and history teachers, are highly visible during National History Day event organizing and judging. The use of social media enables heritage institutions and their curators to interact with the global public; in turn, the social media sites enable the public to follow the work of the curators, interact with the content, and network with one another. This process not only leads to increased hits on Web sites or physical traffic to the museum, archive, or library; it engages academics, students,

and the public in new ways of constructing meanings and enriches discourse in relation to exhibits and unique collections. The remaining challenge is to understand how to archive social media and other Web contents (including personal digital archives) in ways that preserve the quality and hypertextual relationship of their content so that researchers remain aware of context. This book presents the tip of the iceberg surrounded by an ocean of knowledge published in other works. The bibliography presents recently important sources on this topic with thousands more deserving equal recognition. There is no conclusion: how could there be if the whole enterprise of digital curation seeks to make unique primary sources, data, and knowledge accessible in the future? With the evolution in technology (and amidst the concomitant concerns), the major underlying theme here is continuity.

Defining digital curation in the digital humanities context

1

Digital curation involves the preservation, promotion, and providing long-term access to born-digital and digitized collections of heritage material, data, and publications supporting research with surviving (albeit considered obsolete), current, and emerging digital technologies. As with the term "archive," "digital curation" needs a semantic clarification, as it has different meanings in the context of archival profession and digital humanities, given also the distinct perspectives of collaborating archivists and digital humanists. Digital archivists focus on preserving digital content in the context of archiving whereas some digital humanists, on creating thematic collections to create new interpretations, theoretical frameworks, and knowledge. There is also social curation, which involves community and public feedback using various social media platforms; its aim is to add meanings to the collections and enrich public discourse on collection or exhibition themes. The overlap in the definitions and applications of digital curation in these related contexts brings archivists, digital humanists, and the public together, and thus enhances collaboration at various levels of curation. This chapter calls this collaborative framework the digital curation workspace because it expands the meaning of "digital curation" to represent the works of collaborating archivists, librarians, digital humanists, technologists, information architects, and the public in different—perhaps intersubjective—contexts.

In his Introduction and Welcome talk at a Seminar in London titled "Digital Curation: Digital archives, libraries and e-science seminar" sponsored by the Digital Preservation Coalition and the British National Space Centre, Beagrie (2001) ascribed the emergence of "digital curation" to the continuing interdisciplinary dialogue between scientists and librarians. The association of digital curators with scientific work by Lord and Macdonald (2003) has placed digital curators in an active role of preserving and adding value to collections for the public good by promoting new science and maintaining a solid community of scientists. In the sciences, curation refers to the maintenance and publishing of databases containing knowledge and evidence, annotations, linkage, management, validation, and editorial input providing value to the digital library. In the digital humanities context, however, the definition of curation—rooted in fourteenth-century practices and associated primarily with museum artifacts—has undergone significant changes due to the influence of emerging technologies and the rise of interdisciplinary scholarship. "Digital curation has added a new dimensionality to the mix, which is technical knowledge, but even here technological knowledge is key but not a requirement" (Tebeau, 2011, "Digital Humanities Curation," para. 4).

The interdisciplinary scope of digital humanities not only spans the humanities and technology, however, but as Flanders and Muñoz (2011) point out, it also covers archival science, library and information science, computer science, systems, and records management. The digital humanities have also introduced new methodologies for the

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analysis, interpretation, and visualization of humanities data, which present a separate and new level of curation in addition to existing practices. The mutual relationship between digital humanities and digital curation is explained by the digital humanities' role to provide an interdisciplinary framework to support collaboration among scholars, archivists, librarians, and technologists on the one hand and to promote the role of digital curation for the long-term preservation of and access to resources needed in the digital humanities on the other.

This chapter focuses on digital curation as a practical framework for preserving and promoting cultural heritage collections, data, and other forms of digital content as well as discussing the various levels of curation aiming to preserve the quality and integrity of those collections and data. Although scholarly work in the digital humanities may extend beyond the physical archives, many projects will continue to use primary sources and data from archives and the digital repositories they maintain. The chapter first reviews foundational definitions of curation and digital curation, followed by discussions of the digital content lifecycle, levels of curation, and levels of representation essential to understanding the digital curation process. Then the chapter explicates digital humanities data curation and various aspects of treating and interpreting humanities data. Finally, the chapter concludes with a discussion on linked open data, including curating heritage collections, archives, and libraries. Although mashups present both primary and secondary data to create new services, the curation of such data also serves an important purpose: to preserve the relationship of data and collections from multiple sources and to build a broader ontological framework for preserving knowledge. The preservation of metadata in this context not only enhances the lifecycle of digital contents but also the continued accessibility of humanistic and scientific data across multiple generations of data models, file types, and other obsolete material. To this end, archives may become the epistemic bridge between the world of print on one side and digital content on the other, ensuring a continuum of knowledge transfer from print and analog to digital. However, this model does not advocate abandoning nondigital collections for the sake of emerging popular technologies.

Foundational definitions for curation

The lexical definition for *curation*, offered in *The Compact Edition of the Oxford English Dictionary* (1971), is "guardianship," which falls under the purview of the curator defined in *Oxford* as the "officer in charge of a museum, gallery of art, library; a keeper, custodian" (p. 625). While the online version of Oxford Dictionaries does not define *curation* per se, it is derived from *curate*, which means "select, organize, and look after the items in (a collection or exhibition)" and "select, organize, and present (suitable content, typically for online or computational use)" (Oxford, 2013, "curate"). The *Museum Curation Community* (2013) Web site defines "museum curation" as "The practice of managing historically valuable collections of artifacts," but adds that "museum curators should not be confused with museum archivists; a museum archivist usually only works with valuable documents" ("museum curation," para. 1). The distinction applies to archivists in general but not to repositories that accept three-dimensional objects as part of a larger donation of private collections, historical manuscripts, and organizational records.

The semantic clarification of curation, preservation, and archiving has been the work of Lord and Macdonald (2003) addressing the differences between these terms describing three curatorial activities:

> *Curation: The activity of, managing and promoting the use of data, from its point of creation, to ensure it is fit for contemporary purpose, and available for discovery and re-use. For dynamic datasets this may mean continuous enrichment or updating to keep it fit for purpose. Higher levels of curation (as in the Digital Humanities) will also involve maintaining links with annotation and with other published materials.*
>
> *Archiving: A curation activity which ensures that data is properly selected, stored, can be accessed and that its logical and physical integrity is maintained over time, including security and authenticity.*
>
> *Preservation (after Hedstrom): "An activity within archiving in which specific items of data are maintained over time so that they can still be accessed and understood through changes in technology."*
>
> *As cited in Lord and Macdonald (2003, p. 12)*

The *Glossary of Archival and Records Terminology* (Pearce-Moses, Ed., 2012) offers no definition for *curation* but defines *curator* as "An individual responsible for oversight of a collection or an exhibition" or "The administrative head of a museum or collection" ("Curator"). The definition for *manuscript curator* is "An individual responsible for appraising, acquiring, arranging, describing, preserving, and providing access to a collection of original documents; an archivist" ("Manuscript curator"). The curation of manuscripts, photographs, nontextual (e.g., audio–visual) medium, and three-dimensional artifacts may fall under the purview of specialists at larger repositories.

Digital curation

Curation has its origins in the fourteenth century in the context of healing and later as guardianship of personal affairs. In scientific usage during the 1960s and 1970s, "curation" referred to the systematic care of specimens. The linguistic turn of the recent decades introduced yet another shift from referring to an officer of a museum or a library to specialists interested in the continued access to the material for purposes of reuse, which eventually extended to the preservation of data for continued access and reuse and public involvement in collaborative approaches to curation. The linguistic turn contributed to the introduction of "digital curation" in 2001 as a cross-institutional and cross-disciplinary concept, but differences between definitions of "digital preservation," "digital archiving," and "digital curation" still required further clarification.

Tibbo (2012) acknowledges interchangeable uses of "digital preservation" and "digital curation" in earlier definitions; she addresses the conceptual and technical dimensions of digital curation, which distinguishes it from digital preservation. "The

term naturally blends both technical (digital) and conceptual (curation) facets [where] 'digital' is a modifier while 'curation' is the head word that carries most of the weight in the term" (p. 190). While digital curation remains curation, it does require the knowledge of applicable technologies that were not included in predigital curation practices, and involves a lifecycle.

An early definition of digital curation by Beagrie (2004) covers a broad base for data curation so as to address scientific use in addition to preserving historical, humanistic, and cultural heritage in archives. He writes,

> *[Digital curation refers to] the actions needed to maintain digital research data and other digital materials over their entire lifecycle and over time for current and future generations of users. Implicit in this definition are the processes of digital archiving and digital preservation, but it also includes all the processes needed for good data creation and management, and the capacity to add value to data to generate new sources of information and knowledge.*
>
> *p. 7*

Yakel (2007) regards digital curation as the "umbrella concept that includes digital preservation, data curation, electronic records management, and digital asset management" (335). Lee and Tibbo (2007) define digital curation as

> *Stewardship that provides for the reproducibility and re-use of authentic digital data and other digital assets. Development of trustworthy and durable digital repositories; principles of sound metadata creation and capture; use of open standards for file formats and data encoding; and the promotion of information management literacy are all essential to the longevity of digital resources and the success of curation efforts.*
>
> *"Opportunities and challenges," para. 3*

What are trustworthy or trusted repositories, however? Must digital repositories comply to a different set of standards and expectations than do traditional repositories with physical collections? Jantz and Giarlo (2006) offer some clarifications for "trusted repositories" applicable to both physical and digital repositories. The first requirement is that trusted repositories must come with preservation policies, content description standards (such as metadata), and proper physical infrastructure (such as climate control, security, and disaster planning) needed for long-term preservation. Second, the digital repository must use reliable systems of software and hardware that can follow certain rules and standards. Not only do preservation standards improve the quality of localized practices and resources but they also support global protocols for interoperability such as the Open Archives Initiative Protocol for Metadata Harvesting (OAI-PMH).

Digital preservation

Jones and Beagrie (2001) define digital preservation as "the series of managed activities necessary to ensure continued access to digital materials for as long as necessary" (p. 10). Digital preservation is a critical activity to counter the impending threats regarding the longevity of digital contents:

> *[Digital] information will not survive and remain accessible by accident: it requires ongoing active management from as early in the lifecycle as possible. The information and the ability to read it can be lost in a few years. Storage media such as punched paper tape, floppy disks, CD-ROM, DVD evolve and fall out of use. Digital storage media have relatively short archival life-spans compared to other media. As the volumes, heterogeneity, and complexity of digital information grows, this requirement for active management becomes more challenging and more critical to a wider range of organisations.*
>
> *p. 10*

Watry (2007) approaches preservation as a necessary step toward ensuring the survival of collections in the future. He postulates a "concept of digital preservation from one that is focused on sending the records (metadata) into the future to one that can also send into the future a description of the environment that is being used to manage and read the records" (p. 42). This approach involves a combination of preservation, technical, and administrative metadata to inform future curators about the context within which the records were created. Moore (2008) presents what preservation may directly offer to digital humanists as a channel of "communication with the future" which "corresponds to moving records onto new choices of technology and as the validation of communication from the past" (p. 64). In fact, both definitions accurately represent the objectives in curation.

With respect to the long-term preservation of digital content, Hedstrom (2001) addresses temporal interoperability as an important issue related to accessing heterogeneous content over time. She writes, "By temporal interoperability, I mean the ability of current systems or legacy systems to interoperate with future systems that may use new formats, data models, languages, communication protocols, and hardware" (para. 1). This is an escalating problem especially since digital curation aims to ensure long-term access to digital content, during which file types, software, hardware, operating systems, and metadata standards undergo successive revisions, changes, not to mention the effects of commercial actions such as mergers, failures, and other compatibility issues affecting accessibility to legacy files. There is a need for open standards and open source strategies to bridge various gaps across file types and the metadata that describes them. Hedstrom identifies several issues related to (and affecting) interoperability: technical, representational, and social, all of which may evolve over any given time frame, affecting practices, industries, and other legal, social, and professional frameworks, which in turn have varying implications for digital curation.

Lifecycle of digital contents

The lifecycle of digital contents has been included in definitions for digital curation by Beagrie and the Digital Curation Centre (DCC), and such a lifecycle presents the practical framework for several related activities occurring in a temporal sequence. The DCC has expanded Beagrie's earlier definition of digital curation, which began to refer to the "maintaining, preserving and adding value to digital research data throughout its lifecycle" ("What is digital curation?"). The DCC's definition has two parts: (1) "The active management of research data [that] threats to their long-term research value and

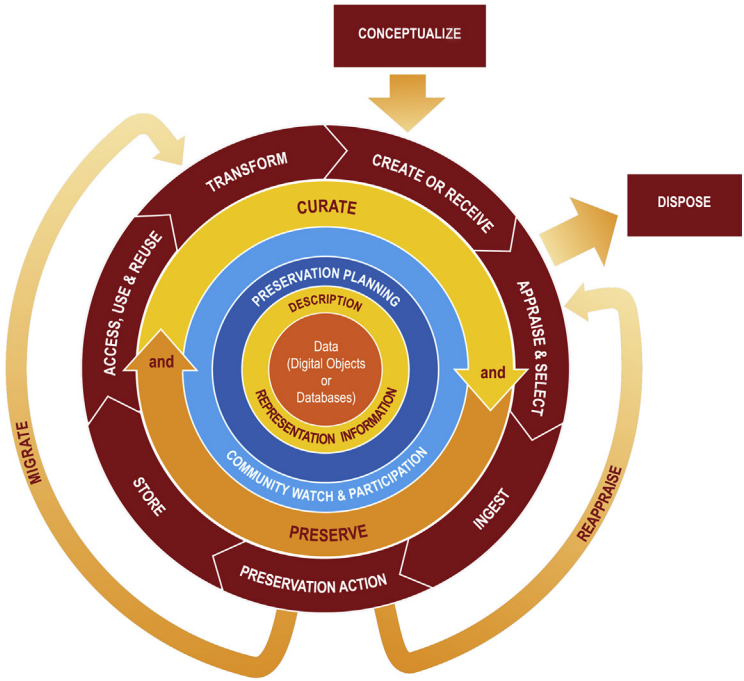


Figure 1.1 DCC curation lifecycle model.
Credit: Digital Curation Centre, University of Edinburgh.

mitigates the risk of digital obsolescence;" and (2) "reducing the duplication of effort in research data creation (in order to enhance) the long-term value of existing data by making it available for further high quality research" ("What is digital curation?"). If digital curation is to achieve the longevity—that is, over the lifecycle—of digital material, the need for a robust preservation plan involves the use of trustworthy repositories, open file format, and the promotion of digital contents. The digital curation lifecycle model in Figure 1.1 uses concentric circles to represent critical steps in the digital curation process. These are: (1) conceptualize (planning); (2) create (production of object and metadata); (3) access and use (accessibility and policy); (4) appraise and select (assess curation and preservation needs); (5) dispose (document the process of disposal); (6) ingest (archive digital objects to a trusted digital repository); (7) preservation action (retention); (8) reappraise (if validation fails); (9) store (keep a secure manner); (10) access and reuse (set access policy as needed); and (11) transform (create new digital objects).

Levels of curation

There are multiple levels in digital curation, given the structure of the content and the context of practice. In some cases, references to low- and high-level curation appear in association with preserving the physical medium and working with annotation

and meaning (Lord & Macdonald, 2003), respectively. IT-level operations such as migration, backup, upgrades (to the platform), and various bit-level preservation routines like checksum reporting and more comprehensive active preservation activities occupy different levels and areas of curation. Strategies for other levels involve the curation of metadata, bringing the content into relationship with information such as provenance, preservation actions, copyright, and other administrative requirements but not generating meaning or producing interpretations at this level. Finally, there are designated levels of curation for activities that produce annotations and add meanings to images, and this occurs in digital humanities scholarship and teaching practices. Examples of curation at these higher levels get special attention in the next chapter.

Describing different levels as low or high has been commonplace in such blogs as Content Rambler (2012), comparing curation to a social media ladder with creators, critics, collections, readers, and other categories occupying separate rungs of the ladder. Blogging in this context adds meaning to contents, data, and existing discourse, and, as such, occupies high levels in curation. The context and scope of curatorial activities, however, are the ultimate frameworks within which some activities receive priority over others. An electronic publishing environment will consider curation priorities differently from a museum, research library, historical society, digital humanities communities, or—for that matter—blogs of established scholars and dilettantes.

Curation practice varies at different points in a digital content's lifecycle, and shapes the archiving, preservation, and curation processes. Each stage in the digital content lifecycle presents different responsibilities and collaborative relationships for curators; for instance, conceptualization, appraisal, selection, access, and preservation (i.e., activities at the conceptual level) are very different from creating, ingesting, storing, disposing (or migrating), and transforming. These activity groups represent three distinct levels of curation: one level for working with the digital files and the technology to preserve them, another one to exercise intellectual control, and a third one for scholarly processing. Other factors such as hardware, software, network, file format, metadata schema, and interoperability standards will also have implications for curation practices and curatorial control at these levels. However, there is no monolithic framework for identifying these levels, and there have been several approaches to classify curatorial authority in relation to digital curation.

Lord and Macdonald (2003) describe curation on three levels in the context of e-science publishing with a focus on curating content and data. The authors describe the research process common in traditional academic information flows wherein research produces primary (raw) data needed for analysis to generate secondary (results) data published in monographs and journals. Scholarly work, in turn, becomes available for future researchers in archives, libraries, Internet, government, and other sources. The cycle repeats here, and represents Level 1 curation. Level 2 curation involves the archiving and preservation of primary and secondary research data, supporting data-driven research aiming at new discoveries and insights that can enhance the research process. Metadata (prepared by data archivists) plays a more important and obvious role in the process, as it informs new research. The annotations and links to other research enable researchers to cross-reference, validate, and recontextualize research results. Last, level 3 curation involves a greater curatorial role for the data archivist who ensures

the constant availability of primary and secondary data as well as metadata via data repositories, and through them to the public, government agencies such as the National Science Foundation (NSF), the Higher Learning Commission (HLC), and National Institutes of Health (NIH), as well as libraries, industries, and other data sources.

Digital curation in the digital humanities evolves along a similar model with the added benefit of social curation to expand and enrich public and scholarly discourse on humanistic research. The role of the archivist-curator using social media and Web 2.0 tools is to add value to the collections by further supporting the contextualization and recontextualization effort. Additional forms of curation are mashups and data visualization methods that may inspire yet newer ways to think about, analyze, interpret, and discuss data for continued humanistic scholarship. Perhaps, these developments represent yet another level for curatorial practice since knowledge and data flow in different directions often outside the purview of publishers and libraries. Open access publishing and data curation today, in the current decade of 2010, on the Web do not necessarily frame academic information flows in the way Lord and Macdonald have described a decade earlier, and the publishing paradigm continually changes.

In significantly greater detail, Lee (2012) describes a framework for digital curation on eight levels (from 0 to 7), each corresponding to a distinct level of representation. Lee (2012) writes, "Supporting the meaningful use of digital objects over time—a set of activities that has recently come to be called 'digital curation'—requires an understanding and appreciation of the various layers [or levels] of representation through which meaning can be conveyed in digital systems" (p. 507). The eight levels pertain to the digital curation of digital resources, ranging and progressing from microscopic and physical attributes to various aggregated forms of digital resources. Each level of representation addresses different structural characteristics of digital contents concerning different areas of digital curation. These levels of representation also mean different ways of accessing digital resources, depending on the file systems, hardware, software, operating system, or storage mechanism. Digital preservationists will view all levels as an interconnected system (from bit-level data to data sets and aggregations), whereas humanists will naturally focus on levels with data presented (and represented) in varying contexts of other data.

This framework is meaningful to historians who regard contextualization crucial to historical analysis; level 7 in this hierarchy presents content in context as it would appear in a finding aid of an archival collection. Levels 5 through 7 are particularly important (however, dependent on previous levels) to scholarship in the digital humanities, as all analyses and interpretive activities must utilize data at these levels of representation. In his presentation, Lee (2011) points to archivists' decision to preserve content and meaningful information at different levels of representation, but there is a reason for concern. According to Lee, the "lifting problem" (slide 9) occurs once a digital file (born-digital or digitized copy) is published. Lee refers to the effects of files separated from their original context, which can inadvertently place the files into a new context. Lee advises to set a balance between providing too little and too much contextual information, both of which have implications for digital humanities practice. One the one hand (as in the former case), too little information will cause the content to separate from accurate information supplied by archivists; on the other hand

(as in the latter case), too much information will saturate discourse to unnecessary extents with too much—perhaps ultimately meaningless—information. Throughout the development of the archival profession, archivists have entered a period where they have more influence on shaping public memory through various historical representations (via exhibits, finding aids, and publications); thus a similar balance will help humanities researchers continue working with the collections.

Elsewhere, Lee, Woods, Kirschenbaum, and Chassanoff (2013) present a framework for digital curation from bit-level preservation to preserving large digital collections in file systems, digital asset management systems, and digital repositories. "Digital materials can be considered and encountered at multiple levels of representation, ranging from aggregations of records down to bits as physically inscribed on a storage medium; each level of representation can provide distinct contributions to the informational and evidential value of the materials" (p. 3). Thus, describing the curation process through levels of representation enables archivists to approach digital curation far more systematically, providing a higher level of support for preserving cultural heritage collections, institutional records, and private material for a long time under current industry and professional standards. There is no separate monolithic model, but some models present great granularity and detail and may be applicable, therefore, to a greater range of curatorial activities. Lee's framework of multilevel representation reifies the principle that archives offer vital services and functions needed by digital humanists undertaking projects; this approach provides a more systematic support for long-term access to new knowledge in digital formats.

Digital humanities data curation

In contrast to the scope of digital curation in the archival context, digital humanities data curation (often by researchers and not archivists or librarians) is selective in order to support research around specific humanistic and historiographical questions. This approach usually results in separate data sets, Web sites, and digital collections around specific topics in thematic research collections. In *An Introduction to Humanities Data Curation*, Flanders and Muñoz (2011) write,

> *[Humanities] data is presented in specialized aggregations that themselves have significance for understanding, using, and curating the data. Some of these aggregations are digital extensions of long-standing traditional forms: for instance, finding aids, concordances, and scholarly editions, which have a long analog history. Others, like the thematic research collection or digital text corpus, are products of new digital research methods.*
>
> *"Unique Features," para. 14*

The emphasis on the curation of data, information, and knowledge (Gladney, 2012) has received significant attention in published works on the digital humanities, which provides new context for information in legacy digital collections related to cultural heritage. The rising importance of linked open data in archives and libraries also presents archivists,

librarians, and other researchers an opportunity to add such data to the growing mix; thus, digital curation must also span linked open data. Data curation, therefore, has become closely associated with digital humanities curation as well as the digital preservation and the curation of cultural heritage material. Cragin et al. defined data curation as "the active and ongoing management of data throughout its entire lifecycle of interest and usefulness to scholarship" (as cited in Flanders and Muñoz, 2011, para. 4), but Flanders and Muñoz point out that the semantic overlap of such terms as digital curation, digital archiving, and digital stewardship (or preservation) may lead to some confusion since curation refers not only to protection of materials in museums and rare book libraries, but also to "amelioration, contextualization, and effective exposure to an appropriate set of users" (para. 2). With respect to the difference between digital curation and data curation, the authors write,

> *[Digital curation is] the curation of digital objects and information…but "data curation" shifts the emphasis and brings an additional consideration into play. The curation of research data—raw and abstracted material created as part of research processes and which may be used again as the input to further research—carries with it the burden of capturing and preserving not only the data itself, but information about the methods by which it was produced.*
>
> *Flanders and Munoz (2011), para. 3*

Data curation involves the following activities and processes: description (documenting the context and relationship of various forms and of research data), annotation (enhanced information on the data with more granularity and context), collection/aggregation (connecting data and teams), storage (maintaining a platform for stable and accessible data), and migration (to ensure continued access via emulation or preservation). Data curation, therefore, involves routine digital archiving processes to ensure the stability and accessibility of research data over an extended period. These processes are related to what Bhargava (2011) calls "the 5 models of content curation" rooted in commercial marketing practices but are valuable to promoting cultural heritage contents in digital environments as well.

1. Aggregation ("the act of curating the most relevant information") requires the selection of the most relevant contents that can be included in digital collections, virtual exhibitions, and publications.
2. Distillation ("the act of curating information into a more simplistic format") is necessary to give the curated content clearer structure and transparency, although the results may not need to be simplistic if used in academic context to support analysis and hermeneutic agenda.
3. Elevation ("curation [to identify] a larger trend or insight") involves the analysis and interpretation of data and information gathered through aggregation and distillation.
4. Mashup ("unique curated juxtapositions [to merge and use existing content] to create a new point of view") involves data representations in various new contexts.
5. Chronology ("a form of curation that brings together historical information organized based on time") involves creating a timeline for historical analysis and representations of closely or loosely related events.

Archives and libraries alike employ these models of content curation in order to promote cultural heritage material; moreover, combinations of these models may afford heritage institutions to provide new kinds of resources. For instance, the use

of aggregation, chronology, and distillation may suit virtual museums and exhibitions centered around historical themes where the use of timelines, maps, and other visualization technologies such as topic maps can support researchers' work. The data digital humanists' work will be closely defined with the technologies (software, operating systems, platforms, and information politics) at their respective institutions, and it is up to discussion whether such data are equivalent to or conducive of hermeneutic interpretation. In his article, Owens (2011) reconciles this difference of interpreting "data" across the traditional and digital humanities practices. He writes,

> *We can choose to treat data as different kinds of things. First, as constructed things, data are a species of artifact. Second, as authored objects created for particular audiences, data can be interpreted as texts. Third, as computer-processable information, data can be computed in a whole host of ways to generate novel artifacts and texts which are then open to subsequent interpretation and analysis.*
>
> *"What is Data to a Humanist," para. 1*

While Owens recognizes that data *per se* is not necessarily evidence, each of these areas that data represent can lead to evidence, and nowhere that is clearer than in archives. As part of collection management (including processing and curation), important analytical, descriptive, and interpretive data come into existence, which leads researchers to physical evidence. Owens does point to the "artifactual quality" of data, and this is where historians and other humanists can regard data as artifact; for instance, the recent release of the 1940 Census data in the United States enables scholars, administrators, civilians, and others to launch inquiry into the nation's history from an academic, personal, political, and commercial perspective. As a historical document containing data in print format, the 1940 Census is also an artifact, and not just data resulting from abstraction.

Treating data as interpretable text is also close to the mission of archives active in curation and outreach. Exhibitions, lecture series, presentations, and other educational events at archives place data into a historical context. Researchers, public historians, and educators enrich public discourse on the subject matter. Virtual exhibitions, social media, and other digital channels extend the socialization of data from the repository to the Web. The visualization of data on maps and timelines further assists in the hermeneutic study of associated narratives resulting in new texts. The distributed, decentralized, and networked form of modern hypertext presents new ways of locating, interpreting, and sharing data, information, and knowledge across technological domains. This underscores the third quality of data as computer-processable information, which could benefit from a broader application context familiar to archives.

Archives have been developing digital finding aids using SGML/XML documents as well as adding XML metadata to images, and collaborating with Text Encoding Initiative (TEI) XML data for textual analysis in the digital humanities. The purpose of developing and curating data in XML is specifically to allow computers to recognize, retrieve, and visualize data in meaningful ways. Maps and timelines may allow historians to discover relationships among events, people, and organizations in temporal and spatial proximity—nexuses that were left unrecognized, unprocessed, and unaddressed by historians for a number of possible reasons. Archives also maintain provenance data in their accession

files, which may also be an important missing link for some historical interpretations. Humanities scholars must, therefore, come to terms with the future likelihood of seeing artifacts in digital form regardless of some missing data. Since some artifacts are restricted to public view researchers will travel to view such artifacts on site, but for most researchers the digital representation will be sufficient. However, the reliability of such data remains an issue: how do we know that the document appearing in the results of a Web search is a true copy of the original that it represents? Humanists justifiably continue to remain skeptical, as they have been in the past. Can a search engine or online encyclopedia put a stamp of authenticity (by independent and autonomous experts) on its content? The archival profession has been aware of theft and forgery, and may be able to verify the documents as original in an effort to certify digital copies of databases, digitized images, and texts (including books). Archives contribute to identifying what Owens (2011) calls the "evidentiary value of data." He writes,

> *As a species of human artifact, as a cultural object, as a kind of text, and as processable information, data is open to a range of hermeneutic tactics for interpretation. In much the same way that encoding a text is an interpretive act, so are creating, manipulating, transferring, exploring, and otherwise making use of data sets. Therefore, data is an artifact or a text that can hold the same potential evidentiary value as any other kind of artifact. That is, scholars can uncover information, facts, figures, perspectives, meanings, and traces of thoughts and ideas through the analysis, interpretation, exploration, and engagement with data [which is a]…a potential source of information that can hold evidentiary value.*
>
> *"Data Can Hold," para. 1*

The highly interdisciplinary and heterogeneous nature of digital humanities, however, presents unique challenges to curating research collections, data, and heritage material. Flanders and Muñoz (2011) describe the following types of research collections common in digital humanities scholarship, including challenges to their curation: scholarly editions, text corpora, marked-up digital texts, thematic research collections, research data, and archival finding aids. Scholarly editions are highly structured but variable due to disciplinary differences, and pose the question whether curation practice should treat each edition as a distinct, separate entity. Given the individuality of editor perspectives and the presenttation method, Flanders and Muñoz believe that curators should approach their work individually and on the merit of their content and structure. Scholarly editions differ from text corpora, which have been endemic to humanities research in print as well as early digital formats. The digitally reformatted texts present curators with the responsibility of making important decisions: with metadata describing entire collections of texts, curation should aim to support analysis across such interrelated sets. Unlike the previous two genres in predominantly PDF formats, these digital texts use XML markup following the Text Encoding Initiative schema. Curators must consider the fact that marked-up texts usually come with highly structured databases and variations in the application of the TEI schema.

Thematic research collections are highly structured aggregations of XML data, image files, style sheets, and configuration files containing and representing the editorial choices and process including strikeouts, insertions, deletions, and other modifications done prior to publication. Removing any element from the curatorial process will prevent the

reconstruction of material in the proper contexts. Data enhanced with analysis and annotations are critical resources for humanities as "primary" digital objects. This is because humanities scholarship requires the original works as well as interpretations thereof for analysis and new interpretations. The curation of such enhanced data must also account for variations in readings, interpretations, and editorial work beyond the author's (or authors') original intentions. Finally, archival finding aids and bibliographies are also highly structured information maps that enable researchers to locate resources. Given the emergence of Encoded Archival Description (EAD) finding aids, however, the variation of finding aids presents curatorial challenges requiring some standardization to support long-term access. Finding aids are increasingly using EAD and Encoded Archival Context - Corporate bodies, Persons and Families (EAC-CPF) schemata and the Describing Archives: A Content Standard (or DACS), but many institutions are still in the process of transforming legacy finding aids to new, more consistent standardized formats. The ArchiveGrid finding aid discovery system, for instance, requires the addition of metadata for harvesting purposes, which involve updating document headers in the case of PDF documents. The data from EAD finding aids exist in XML format, which can be easily accessible across multiple applications and platforms. Curating EAD finding aids may require the standardization in content description but the XML data structure, which represents information at the record group, series, collection, and (in some cases) item levels, will support analysis across a larger corpora of finding aids.

Flanders and Muñoz (2011) present and describe three methods for the treatment of data: interpretive layering, data capture and preparation, and capturing scholarly agency (such as responsibility, editorial voice, and debate). Interpretive layering involves locating and curating interpretive information in the markup, metadata, annotations, and in style sheets as core material needed for humanistic study. The documentation of data capture and preparation involves such curatorial decisions as choosing the source for digital representation, technical information (such as the calibration of photoreproduction tools), method of capturing and transcribing the data, details about the encoding scheme, data quality, editorial oversight, and curation process. Finally, capturing scholarly agency is vital to the hermeneutic aspect of humanities work, as the debates, editorial voice, and other metanarratives provide context for the original works. These three methods of treating the data are specific to various levels of curation. Whereas data capture and preparation correspond to where metadata and other forms of documentation are prepared for long-term preservation and curation, interpretive layering and capturing scholarly agency clearly belong to the highest level of curation where curators are most focused on hermeneutic work and scholarly discourse. The outcomes of curation include the production and dissemination of new knowledge resulting in new contexts and interpretation of existing data, scholarly work, and cultural heritage material.

Using linked open data in digital curation

As archives and libraries develop and expand digital heritage collections, data sets and other forms of digital assets generate valuable data that provide operational and/or scholarly context for some of that material. The long-term preservation of such data may not

only assist in discovering related content establishing relationships between collections and data that were archived, migrated, or even removed for economic, logistical, and political reasons well familiar to archivists. To this end, libraries have begun using mashups—Web applications that present new relationships between data from several sources (Lewis, 2009)—in order to add value to data in repositories. For instance, many research libraries in the United States have explored and utilized Viewshare, a service of the Library of Congress, in order to display their data in spatial and/or chronological arrangements, which requires a systematic inclusion of date and geolocation values throughout their metadata records. Many institutions have explored using Google Maps to plot locations of historical events and institutions, which illustrates the use of geospatial data for aggregation and related curation practices. ArchiveGrid is another example of a mashup using geospatial, temporal, and information from the document metadata to arrange and display information on maps and in textual form. Registries such as Registry of Open Access Repositories and Directory of Open Access Repositories play an important role in harvesting data from archives and libraries with digital collections; they act like intermediaries in the process but this role simplifies the work for libraries as long as they follow a single metadata schema such as Qualified Dublin Core.

Linked open data improve access to digital primary source collections (digitized and born-digital) and, thus, serve as a bridge between the humanities disciplines and underutilized digital collections. Lindquist, Dulock, Törnroos, Hyvönen, and Mäkelä (2013) report, "humanities faculty and students [most using search engines] still face challenges finding and contextualizing online primary sources" (p. 917) due to inadequate bibliographical data, a lack of organization by subject, time period, geographical data, and other missing contextual information. Search results in search engines are unsystematic due to other priorities in the search algorithms. Since one of the aims of data curation is to support research across multiple data sets, collections, and text corpora, it remains an essential function (and, therefore, expectation) of repositories to present resulting in humanities-relevant contexts such as time period, location, author, provenance, and subject. Additional contexts, such as stylistic periods in literature, art, music, and theater, may require highlighting and indexing additional fields.

Lindquist et al. (2013) emphasize that linked open data can be especially valuable for locating historical data such as names, places, events, and topics "buried in within the sources, particularly when drawing on ontologies and other specialized vocabularies that impart meaning to these concepts and the [complex, often nonlinear] relationships among them in a given historical domain" (p. 916). Additional benefits of linked data include the ability to locate historically varied, multilingual, and alternate forms of personal and geographical names as well as searching along broader, narrower, and related subject terms found in the *Library of Congress Subject Headings* or other classification systems.

In his case studies for using linked open data, Mitchell (2013) explores Europeana, the Digital Public Library of America, and BIBFRAME—a Library of Congress initiative, which made extensive use of linked open data and linked open vocabularies to aggregate and publish data within their repositories. The discussion of the individual building blocks is beyond the scope of this book, but the case studies demonstrate the ability of the building block framework to support extensive curation of humanities data, including cultural heritage information. Europeana is a digital library with

a multinational scope, bringing cultural heritage information submitted by curators from all participating countries in an aggregated form. The Digital Public Library of America is similar to Europeana in that it focuses its cultural heritage collection on historical materials in archives, libraries, and museums of US-based institutions. BIBFRAME aims to transform catalog records into a linked data model, thus opening researchers to a wealth of humanities information in digital, print, and other legacy formats. Many archival records may still be in nondigital format not accessible via digital repositories, but the data about them inform advanced researchers and direct them to digital finding aids. These services provide additional avenues for archives and libraries to curate cultural heritage materials, manuscript collections, and humanities data. The value of mashups becomes evident as researchers across the globe access and use these resources.

Conclusion

Digital curation presents a practical framework for preserving and promoting humanities and scientific data and digital objects in cultural heritage collections. The definitions and discussions on curation, digital curation, digital preservation, and digital humanities data curation present an interconnected network of practices and theories, which serves as a framework for digital humanities and scientific projects. Linked open data represent an extension of digital curation practices, and will enable archives and libraries to open their digital resources to researchers around the globe with the added vision that these methods will provide the critical contexts within which humanities data can and should be analyzed and interpreted. This chapter described the equal and comparable importance of activities at all levels of curation. Without a robust and trustworthy repository, there is no reliable scholarship in the digital humanities, and without a well-supported digital humanities community, there is little context for digital curation.

Archives and special collections in the digital humanities

2

Digital Humanities presents an emerging interdisciplinary framework for integrating digital technologies and engaging archives in humanities research and teaching. For several centuries now, archives have housed and preserved institutional records, special manuscript collections, and rare books of cultural, historical, and literary significance, which have been the foundation of historic partnership between humanities scholars and archivists. Recent advances in technology and various debates in the digital humanities have, however, placed increasing emphasis on computing, digital technologies, and social media, bringing into question the scholarly and theoretical scope of digital humanities practices as well as the relationship between archives and the humanities. Digital Humanities projects, such as thematic research collections, continue to demonstrate the value of archives through this continuing relationship because archivists have expanded their expertise to cover digital curation in order to preserve, promote, and provide long-term access to digital—that is, digitized and born-digital—collections as well as quantitative data for nonnarrative humanities projects. The debates around classifying Digital Humanities projects, including collection building, writing codes, and creating new digital artifacts necessarily bring the theoretical and scholarly scope of archival work into the discussion.

Chapter 2 investigates the relationship between archives and the digital humanities. It begins by establishing a conceptual framework through a review of definitions and discussions on Digital Humanities. A discussion on the various debates in the field also provides a theoretical framework for the role of archives as well as the scholarly orientation of Digital Humanities projects and work contributed by archivists and digital curators. The chapter then addresses what constitutes critical discourse in the field and the various positions stated in the *Digital Humanities Manifesto* (2008) followed by concerns from the archivist perspective. The range of Digital Humanities projects from the early CD-ROM initiatives to current interests in data visualizations, geospatial representations, and social curation underscore the continuing role of archives as leading heritage institutions. The chapter examines specific Digital Humanities projects that have involved archives and used digital content, data, or both.

Defining the digital humanities

In his Columbia University's Center for Digital Research and Scholarship presentation, Cohen (see Cohen, Frabetti, Buzzetti, & Rodriguez-Velasco, 2011) defined Digital Humanities as "the use of digital media and technology to advance the full range of thought and practice in the humanities, from the creation of scholarly resources, to research on those resources, to the communication of results to colleagues and students" ("Defining the Digital Humanities"). Cohen's reference to the role of archives, libraries, digital collections, and finding aids lays the foundation of a continuing

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long-term relationship between humanities scholars, archivists, and subject librarians. Digital Humanities represents the partnership of humanistic scholarship and computing as referenced in the much-cited definition from Wikipedia (2013):

> *The Digital Humanities are an area of research, teaching, and creation concerned with the intersection of computing and the disciplines of the humanities. Developing from the field of humanities computing, digital humanities embrace a variety of topics, from curating online collections to data mining large cultural data sets. Digital humanities (often abbreviated DH) currently incorporate both digitized and born-digital materials and combine the methodologies from traditional humanities disciplines (such as history, philosophy, linguistics, literature, art, archaeology, music, and cultural studies) and social sciences with tools provided by computing (such as data visualisation, information retrieval, data mining, statistics, text mining) and digital publishing.*
>
> *"Digital Humanities"*

In their introduction to the *Companion to Digital Humanities*, Schreibman, Siemens, and Unsworth (2004) address the history of close collaboration of disciplinary experts, technologists, librarians, and other information specialists such as archivists. They write,

> *[Humanities] remains deeply interested in text, but as advances in technology have made it first possible, then trivial to capture, manipulate, and process other media, the field has redefined itself to embrace the full range of multimedia. Especially since the 1990s, with the advent of the World Wide Web, digital humanities has broadened its reach, yet it has remained in touch with the goals that have animated it from the outset: using information technology to illuminate the human record, and bringing an understanding of the human record to bear on the development and use of information technology.*
>
> *"History," para. 1*

At the aforementioned Columbia University event, Frabetti (see Cohen, Frabetti, Buzzetti, & Rodriguez-Velasco, 2011) called the digital humanities simply as "Humanities in dialogue with digital technologies." While addressing the relationship between the humanities and technology however, she points out that the purpose of this relationship is not solely to use technology for the sake of technology but also to understand what implications digital technology has for the humanities. This rationale may also apply to the technorealist approach taken by Cohen and Rosenzweig (2006), which advocates a center-of-the road position between the extreme technoskeptics' and cyberenthusiasts' positions. In fact, digital curators will share an interest in applying technology toward investigating critical humanistic and historiographical questions that might not be possible without digital technology. Hall (2011) addresses a two-way relationship between the humanities and technology when she writes, "just as interesting as what computer science has to offer the humanities is the question of what the humanities…have to offer computer science; and, beyond that, what the humanities themselves can bring to the understanding of computing and the shaping of the digital" (p. 2). A frequently asked question has been whether

Digital Humanities scholars can present a humanistic—that is, nonquantifiable, nonmechanistic, and qualitative—perspective on digital technology and computing. Drucker (2012) takes this concern a step further by addressing the incompatibility of current interfaces used to create timelines, maps, and other data visualizations with the qualitative methodologies of the humanities since all retrieved data appear in abstraction and out of context as not to support the analysis and interpretations of texts. She writes,

> *The challenge is to shift humanistic study from attention to the* effects *of technology (from readings of social media, games, narrative, personae, digital texts, images, environments) to a humanistically informed theory of the* making *of technology (a humanistic computing at the level of design, modeling of information architecture, data types, interface, and protocols).*
>
> *p. 87*

Using quantitative data in visualizations, timelines, and maps may be more effective when used with specific addresses, dates, and names, but such technologies must—as Drucker notes—accommodate qualitative data that humanities scholars also use for analysis, interpretation, and writing. Drucker points to the incompatibility or incommensurability of mechanized spatial and temporal representations for works, ideas, and events that simply cannot be plotted along rigidly parameterized lines. According to Schmidt (2011), sufficient support from available data is necessary for theory and practice in Digital Humanities research, but the field can also engage experts from other fields to contribute. He writes,

> *At their core, the digital humanities are the practice of using technology to create new objects for humanistic interrogation…This has rightly led much of digital humanities' focus to lie in public humanities; there is enormous excitement about the potential of visualizations, exhibits, and tools to encourage non-humanists to think humanistically.*
>
> *para. 5*

In this spirit, Davidson's concerns (2012) should not be too surprising. She alludes to the growing gap between the sciences and humanities, which had flourished contemporaneously since the industrial revolution: wherever science made progress, humanistic scholarship provided the interpretive, analytic, and narrative perspectives that shaped culture. Davidson's vision for the digital humanities departs from diametrically opposed sciences and humanities in favor of an integrated intellectual landscape. She writes,

> *Perhaps we need to see technology and the humanities not as a binary but as two sides of a necessarily interdependent, conjoined, and mutually constitutive set of intellectual, educational, social, political, and economic practices. More to the point, we need to acknowledge how much the massive computational abilities that have transformed the sciences have also changed our field in ways large and small and hold possibilities for far greater transformation in the three areas—research, writing, and teaching—that matter most.*
>
> *p. 477*

Characteristics of Digital Humanities

While these and similar definitions have evolved around the relationship between the humanities and computing, characterizing the field involves identifying the deeper aspects of the discipline including its theoretical foundations and critical discourse. The *Digital Humanities Manifesto*, published in two versions in 2008 and as *Digital Humanities Manifesto 2.0* in 2009, is an aggregation of over 70 statements from the broader digital humanities community of practitioners, and it serves as an important framework for an evolving discourse with various definitions, descriptions, questions, and debates in the field. These statements represent the most commonly voiced positions in the *Manifesto* but they also characterize a new discipline that integrates technology, collaboration, multivocality, quantitative methodology, interdisciplinarity, and the decentralization of knowledge, all of which distinguish this field from the earlier humanities and other legacy disciplines of earlier scientific paradigms. The first version of the *Manifesto* (2008) describes this field as "not a unified field but an array of convergent practices that explore a universe in which print is no longer the exclusive or the normative medium in which knowledge is produced and/or disseminated" (para. 2). A review of statements in the *Manifesto* yields the following characterizations for Digital Humanities:

1. it aggregates practices from multiple fields with a focus on disseminated knowledge;
2. it immerses quantitative aspects of computing with qualitative and interpretive aspects of the humanities;
3. it emphasizes multivocality in humanistic knowledge; and
4. it emphasizes teamwork and the collaborative production and reproduction of humanistic knowledge, using teams committed to risk-taking, collaboration, and experimentation.

These statements represent a marked departure from practices of traditional Humanities scholarship, which has varied from monodisciplinary to various degrees of interdisciplinary approaches, focused on qualitative and narrative-driven scholarship, and solo production of knowledge with little or no space for intertextuality. The emerging model of humanities scholarship has been increasingly inter- and transdisciplinary, collaborative, culturally diverse, and intersubjective. Svensson's (2010) review of an emerging digital humanities landscape points to a multifaceted paradigm for relationships between scholarship and technology. Tara McPherson's typological approach identifies "computing humanities, blogging humanities, and multimodal humanities…[which correspond to] building tools, infrastructure, standards, and collections…the production of networked media and peer-to-peer writing…[and the aggregation of] scholarly tools, databases, networked writing and peer-to-peer commentary while also leveraging the potential of the visual and aural media that are part of contemporary life" (as cited in Svensson, 2010, Typologies of the Digital Humanities section, para. 14). Another paradigm shift occurred in the transition from Humanities 1.0 to 2.0 as discussed in Davidson's work (see Svensson, 2010), which addresses the transition from data-based projects to greater interactivity, openness, and interdisciplinarity. The decentered authorship parallels the shift from computing to multimodal humanities.

Discursive concerns in the digital humanities

Outside the scope of the two versions of the *Manifesto*, there are several other concerns related to critical discourse (or lack thereof) in the digital humanities. These concerns address the overuse of technology to the potential detriment of scholarly pursuits, the gap between the humanities and sciences, and the scholarly nature of digital humanities research and practice. Frederick Gibbs (2011) warns about the absence of critical discourse in the digital humanities, which in his view (1) "must be concerned with both interpretation and evaluation"; and (2) "is central to establishing the importance of the kind of scholarly and even cultural work that it does" ("II. The value of digital humanities criticism," para. 1). As for the current status of critical discourse, Gibbs presents three issues: (1) there is no effective critical discourse around digital humanities work; most energy is focused on peer review; (2) Digital Humanities work requires more practical and theoretical rubrics as evaluative criteria; and (3) criticism of Digital Humanities work needs to be more collaborative and multivocal. Gibbs's concerns reflect the emerging discursive differences between Humanities and Digital Humanities in that the latter has yet to establish a greater degree of autonomy from the humanities while integrating into a more interdisciplinary discourse marked by a greater degree of critical discourse, intersubjective readings, and collaboration. The historically solo-oriented work in the humanities is in contrast with the collaborative orientation of the digital humanities.

A broader and equally important question in the debates stirring in the digital humanities (Gold, 2012a) is whether building collections is theoretical in nature as to merit recognition as a scholarly activity for tenure and promotion of humanities professors. In fact, the same may apply to archivists and librarians in tenure-track positions. Gold (2012b) alludes to a presentation at the 2011 Modern Language Association's (MLA) "History and Future of Digital Humanities" panel in which Stephan Ramsay identified the mandate to build (including coding) things in order to be considered a digital humanist. According to Ramsay (see Gold, 2012b), IT skills to build repositories per se will not make one a digital humanist; however, collections and metadata both support searching, information retrieval, discovery, and writing in digital humanities scholarship constitutes scholarly work in the digital humanities. The role of archivists engaging in Digital Humanities research is vital, as they operate in a similar theoretical framework while developing the necessary metadata framework and the data.

Ramsey and Rockwell (2012) address the difficulty in interpreting and determining "what 'digital work' means in the humanities, and the context in which that term is being applied [which] can differ between scholarly and non-professorial positions, and the normative concerns of tenure and promotion" (75). According to Rucker (see Ramsey & Rockwell, 2012), digital artifacts are theoretical in nature because they reify and communicate knowledge. Lev Manovich (see Ramsey & Rockwell, 2012) classifies digital artifacts such as software, code, and platforms as prototypes that validate theories in digital humanities instead of predicting as theories do in the sciences. Likewise, digital artifacts such as research data, digital objects, metadata, and the digital repositories used for digital curation are also prototypes that explain observations through metadata. Whether they should be experimental or developed for the purpose of modeling, digital artifacts remain theoretical in nature since they must integrate into the

ontology of the discipline via the various subject fields in the metadata record. Hence, this level of relationship brings archivists in direct connection with digital humanists. If building things in Digital Humanities is equivalent or equal to communicating scholarship, then we must accept the argument that artifacts are also an act of scholarship.

Ramsey and Rockwell (2012) approach digital artifacts as theory in that they view digital artifacts as hermeneutical instruments used to interpret events, phenomena, or answer humanistic questions. Digital artifacts are suitable for hermeneutic activity since they present contextual data for intersubjective reading, analysis, discussion, and interpretation; thus, they become "theory frameworks" for interpreting. Visualization tools, they argue, may also be hermeneutical tools, and while digitized artifacts *per se* are not, the presentation of analytical and interpretive information in the metadata is scholarly activity. A significant part of metadata (such as provenance information) requires research on the curator's part, which is publishable content. Considering the digital artifact as a theoretical model, Ramsey and Rockwell (2012) turn to the basics of computation wherein computers are used to "transform information from one state to another" (81)—which is identical to reformatting in digital terms, and presenting descriptive information in the form of metadata—also a strictly electronic (and structured) form of data. Bauer (2012) recognizes the theoretical significance of databases that present a prototype of data deeply theoretical in scope and methodology. "When we create these systems we bring our theoretical understandings to bear on our digital projects including (but not limited to) decisions about: controlled vocabulary (or the lack thereof), search algorithms, interface design, color palettes, and data structure" (para. 7).

The debates and other communications about discursive concerns in the digital humanities lead to the growing argument that the digital projects undertaken in the digital humanities merit recognition as scholarly activity with theoretical foundations. A reasonable extension to these arguments is the inclusion of work by archivists, curators, and librarians who collaborate in such projects.

The role of archives in the digital humanities

The definitions and characterization of the digital humanities and the discussion of discursive concerns in the field can help in identifying the roles archives play in the digital humanities. This is because archives have systematically appraised and accessioned collections with relevance to institutional (including curricular) needs in mind, and carefully appraised and processed collections only benefit the field's recognition in scholarly communities. Integrating archival collections into the conceptual framework of digital humanities takes into consideration the theory and practice in archiving and digital curation. What are the roles of archives and digital curation in the digital humanities context, and to what extent does digital humanities discourse elevate or clarify these roles? The growing need for raw data in digital humanities scholarship communicates new expectations for archives and libraries, which already produce enough data as part of the metadata record to facilitate visualization. For instance, geospatial data on donors from the deed of gift or other donor records would not only support provenance research on donation history in specific geographical areas during a specific period but would also allow mapping the donors in a timeline, and allow

researchers to establish relationships among several donors by specific regions and periods.

This is where archives and special collection areas can make tremendous contributions to digital humanities scholarship, but at the same time retain nondigital research methods. As Schmidt (2011) writes, "The unreconstructed texts of the past make us think in old ways. Archives, libraries, censuses, atlases: all of these force us to read juxtapositions far more aligned with historical ways of thinking than the reconfigurations possible with digital texts" (para. 10). Archives have been, still are, and will be collecting materials from the past when historical records were produced in print, but at the same time, they can also participate in the transition to digitally oriented and reconfigured landscape.

The theoretical scope of digital projects in the digital humanities may scaffold a similar recognition of archival and librarian work associated with such projects. In their introduction to the *Companion to Digital Humanities*—which clearly speaks to this issue—Schreibman et al. (2004) write,

> *Widely spread through the digital humanities community is the notion that there is a clear and direct relationship between the interpretative strategies that humanists employ and the tools that facilitate exploration of original artifacts based on those interpretative strategies; or, more simply put, those working in the digital humanities have long held the view that application is as important as theory. Thus, exemplary tasks traditionally associated with humanities computing hold the digital representation of archival materials on a par with analysis or critical inquiry, as well as theories of analysis or critical inquiry originating in the study of those materials.*
>
> *"Principles, Applications, and Dissemination," para. 1*

The various debates in the digital humanities regarding the theoretical scope of digital projects include Bauer's work (2012), which brings into this discourse the archival and information science theory related to the ontological, epistemological, and methodological aspects of archival and digital curation practice. Bauer recognizes research databases and digital libraries as prototypes having theoretical significance because they support data modeling and interpretation, which may be equally theoretical in scope. Owens (2011) addresses the variety of data across the traditional and digital humanities practices. He writes,

> *We can choose to treat data as different kinds of things. First, as constructed things, data are a species of artifact. Second, as authored objects created for particular audiences, data can be interpreted as texts. Third, as computer-processable information, data can be computed in a whole host of ways to generate novel artifacts and texts which are then open to subsequent interpretation and analysis.*
>
> *"What is data," para. 1*

Owens recognizes that data in the digital repositories represent three dimensions of humanistic knowledge: First, the data itself represents a digital object that is either a digital reproduction of the physical original—one that humanists may or may not necessarily be interested in if the digital copy is reliable and certified. Second, the data itself supports a new narrative for some humanists; for instance, the provenance field may support meta-research or a completely new line of inquiry into the history

of donations. Finally, the data itself may be useful in data visualizations for additional interpretations with applicable technologies.

Herein rests the role of archivists and librarians with description and metadata skills to reinforce the semantic connections between object and context. If the building of collections, data analysis, and interpretation fall into the category of scholarship, will the work undertaken by archivists and digital curator also do? If digital preservation, metadata (and ontology) construction, information architecture, and digitization involve computing, then one must also ask if some of the work by digital curators also qualifies as modeling. Charles Isbell (see Ramsey & Rockwell, 2012) regards computing as modeling activity through which the modeler establishes correspondence between a phenomenon such as an event, place, or community, and the computer that contains the data describing such phenomena. The language associated with digital artifacts—that is, metadata—is a symbolic representation of the world captured in digital form. In digital preservation terms, therefore, it is equivalent to presenting an abstract (historical or imagined) world in the form of a digital object comprised of a combination of media, text, and data.

Owens and Bailey (2012) argue that using digital interfaces such as Viewshare to visualize data constitutes a mode of inquiry that is more than merely providing access to information. "Visualization can be thought of as part of a hermeneutic research process: 'generative and iterative, capable of producing new knowledge through the aesthetic provocation.' In short, the development of an interface to a collection is itself an interpretive act, which brings to light particular vectors for further exploration and interpretation" (para. 2). This understanding echoes a similar position by Ramsey and Rockwell (2012) who argue that digital artifacts are theoretical in scope. Digital humanities data curation brings actual data from archival materials into a new level of curation that focuses on annotations, visualizations, and other hermeneutic research activity. In a relationship between archivists and digital humanists, both are involved in scholarly activity even though the project workflow assigns researchers to specific activities. Furthermore, digital curation is scholarly pursuit if the collections that archives build, preserve, reformat, and promote meet established criteria for theory.

Digital collections, virtual exhibitions, virtual museums, and other digital artifacts must project beyond esthetic appreciation, frame the learning process (with a well-planned information architecture), and present theoretical problems to the researcher or to the curator or to both. Theoretical questions must include the following:

1. Does the collection contain information on organizational, community, or personal history to support critical historical writing?
2. If there is no such material (possibly because the donated material did not contain any), does the arrangement of historical records represent the internal structure of the organization?
3. Can that arrangement lead to or support new theories?
4. Does the metadata record present critical and verifiable information on provenance?
5. Does the metadata record contain subject terms that delineate and represent the domain within which the records were created?

Thus, if the information in the metadata record can drive new humanistic and historiographical questions for further analysis, such digital collections provide the intellectual framework for scholarly and theoretical work and therefore merit recognition

on similar grounds. Therefore, defining the digital humanities in terms of its relationship with archives and digital curation should take into consideration archivists' ability to produce knowledge of scholarly and theoretical value.

Archives and the linguistic turn

Along with the increased use of archives came a linguistic turn leading to a variety of new meanings associated with the term "archives" and to an extent that such references no longer associate the term with an archival facility, specialization, process, service, or materials kept for preservation. Cox (2005) underscores the importance of archives in the information age amidst the growing criticism of archives as "bureaucratic obstacles" (p. 209). With the vision of paperless offices come questions of accountability not backed up with commitments to preserve the digital records, which have evidently proliferated often beyond institutions' and corporations' control. Cox also addresses the trustworthiness of records with respect to authenticity and reliability. For the purposes of this book, these two aspects of digital records should be of concern to humanities scholars and historians regardless of the extent to which they are using (or even relying on) digital information in repositories and social media.

Theimer (2012) addresses the widespread misunderstanding, misuse of, or challenges to using "archives" in its historic context, and the lack of awareness about the fundamental differences between archives and manuscript collections, both of which remain vital primary sources for digital humanists. The divergence in the usage of the term "archive" is nevertheless important because many digital humanities projects—including digital historical representations (Sternfeld, 2011)—continue to apply this term in reference to materials from traditional, digital, and hybrid archives.

Theimer argues this point by pointing out the formal definition according to which archives are

> *Materials created or received by a person, family, or organization, public or private, in the conduct of their affairs and preserved because of the enduring value contained in the information they contain or as evidence of the functions and responsibilities of their creator, especially those materials maintained using the principles of provenance, original order, and collective control.*
>
> *Peace-Moses, Ed. 2005, "archives"*

The information on provenance, for instance, provides important historical context to scholars researching the history of objects and that may apply to born-digital content, assuming the increasing scarcity of such contents due to obsolete technologies (hardware, software, and file format). Although collective control *per se* may not be obvious in virtual exhibitions, the information architecture of virtual exhibitions can represent the physical–logical organization of the original collections. It is important to assume some level of organization—in the form of an organizational chart for instance—which may have been in place prior to transferring and accessioning the records, and such a visualization may frame the arrangement of records at the time of archival processing. Finally, the concept of original order may mean keeping together

records that otherwise have little to do with each other but are related through the historical context, such as letters of Civil War soldiers to their families, or a collector's subsequent activity. Thus, a broader thematic framework must be evident for such records to appear together.

Price (2009) questions the meanings of such terms as archives, databases, editions, projects, and thematic research collections. Do these have any different and significant implications for digital humanities scholarship, or are they merely the legacies of predigital institutions of humanities, archives, and scholarly communication in general? The legal and ethical framework within which archivists work may never exclude predigital records even as the momentum for digitization is still growing. What digital humanists should realize is that there is much more to archives than the instant access (and gratification) coming with digital collections; there are buried treasures in those archival boxes that are worth the travel and digging through old papers.

As for the use of "archives" in the digital humanities context, Price cites the definition by Peter Schillingsburg (see Price, 2009), which is noteworthy since it dramatically diverts from the historic mission and nature of archives. If, according to Schillingsburg, an archive is merely a "library of electronic texts, linked to explanations and parallels and histories" (as cited in Price, 2009, "Archives and Digital Thematic Research Collections," para. 3), the term implies a product evolving outside of the archival practice since archival processing does not involve interlinking selected individual records. Theimer (2012) points out that archivists prepare finding aids that help researchers navigate throughout processed—that is, formally accessioned, described, arranged, and catalogued—collections. Internal hyperlinks in HTML, PDF, and EAD (Encrypted Archival Description) finding aids enable researchers to navigate throughout very large finding aids and between key sections of finding aids. External links may take researchers to related collection finding aids or selections of materials digitized from the collections. Digital humanists can collaborate with archivists and digital curators to develop highly specialized digital archives that focus on a particular humanistic or historiographical question while take advantage of high-quality digital collections and finding aids to navigate knowledge in that field.

Digital humanities projects involving archives and libraries

Since the mid-1980s, there have been copious efforts to innovate scholarship and teaching through CD-ROM and early Web-based projects. Although archivists and librarians have contributed to these projects, there were no digital curation standards to ensure long-term preservation and access. Madden (2008) addresses lessons learned in the *American Memory* project, which started in 1998 with disadvantages of not having any standards for the creation and preservation of digital content. Archives and libraries played a visible role in developing digital curation standards from the earliest conversations on digital curation in the United Kingdom and United States. The development of metadata schemata (Dublin Core, Visual Resources Association, and Cultural Description for Works of Art, and others) and professional standards involving

archives and libraries has significantly improved access to digitized collections from archives, libraries, and museums. Archivists were active in the development of archival description and encoding standards such as the DACS (Describing Archives: Content Standard) and EAD, respectively.

The *Companion to Digital Humanities* (Schreibman et al., 2004) ascribes practical and theoretical significance to text encoding. Although textual analysis is not synonymous or otherwise equivalent to archival practices, the Text Encoding Initiative (TEI), with early beginnings in 1987, had interested archivists and librarians from the beginning. Mylonas and Renear write,

> *TEI is now itself a research community, connecting many professions, disciplines, and institutions in many countries and defining itself with its shared interests, concepts, tools, and techniques. Its subject matter is textual communication, with the principal goal of improving our general theoretical understanding of textual representation, and the auxiliary practical goal of using that improved understanding to develop methods, tools, and techniques that will be valuable to other fields and will support practical applications in publishing, previous hit archives, and libraries.*
>
> *As cited in Renear, 2004, Chapter 17, "Larger Significance of TEI," para. 4*

Not only does the TEI standard highlight structure, but within a text, it presents an ontology that, via proper mapping, corresponds to specific fields in Dublin Core, MARC (MAchine-Readable Cataloging), or other metadata schemata. While mapping at a low (item-) level of granularity may be laborious and cost-prohibitive, the strategy opens new research avenues for digital humanities scholars to linked data and, through that, new knowledge hitherto unassociated with known texts. For instance, mapping between geospatially tagged oral histories and other digital objects (images and hypertext) elsewhere is possible. If they all reference the same or nearby locations, a digital map may expose spatial and/or chronological proximity of hitherto unassociated people, events, and places for historians and other digital humanities researchers to consider further study. Metadata for oral history transcripts with geospatial data can help researchers establish thematic and logical connections to existing bodies of text proven incorrect, falsified, or fabricated, or may simply underscore the veracity of information in some texts.

TEI projects were among the earliest digital humanities projects involving archives and libraries sharing and curating metadata. The rationale behind text encoding was simple: to present text using computer technology and in machine-readable form using the XML schema, which evolved out of the earlier SGML schema. This approach has not only presented (digitally transcribed or born-digital) text but also the local structure of the material in a way that supported analysis (and it required textual analysis), interpretation, and organization.

Archives may also play an important role in verifying the transcription for accuracy and reliability for researchers to use as primary sources. While TEI texts are digital—and hence, suspect as inauthentic in some communities—PDF files may eventually allow access to layers of text produced by Optical Character Recognition. If the text in such layers were open to reading and editing (instead of the excruciatingly slow word-by-word inspection of suspects in currently available software),

it would allow archivists and historians to verify and tag such texts for analysis. The PDF/A file format enables archivists to add a digital signature attesting to the authenticity and reliability of the TEI representation of the original text, and a save action will prevent others from altering the document. This approach may not only support TEI tagging within the PDF file but allow digital humanists to view the original page image. This proposal has been submitted to the PDF Association's LinkedIn group for further discussion (see Sabharwal, 2012b), and the emerging PDF standards may one day support this application of the TEI schema. In addition, such data in TEI tags are stored in databases that can be queried and retrieved in archival repositories and library databases; thus, TEI offers continued support for digital curation in a linked data environment.

Among the various digital humanities projects involving archives and libraries are the thematic research collections, which Palmer (2004) calls "digital aggregations of primary sources" (Introduction, para. 1.) that digital humanities scholars can develop in collaboration with archivists and librarians. Thematic research collections are in hypertext format, containing digitized and heterogeneous primary sources. Therefore, archives may see an advantage in working with diverse technologies, data types, and file formats—a strategy using format-neutral, universal, and open metadata schema. The Dublin Core metadata schema, for instance, has been in use for digital collections since it supports interoperability across several platforms. Researchers' ability to search a wide array of digital resources and catalogs, retrieve the digital content, and discover related library holdings is a goal of archivists and librarians interested in digital humanities projects.

The challenges to libraries in this environment, according to Rydberg-Cox (2006), were not only to introduce digital resources as an extension to print material on which the traditional system of scholarly communication has evolved for centuries. If the digital projects replace print resources, then there will be questions as to whether these new digital corpora will compete with other commercially available products with much more robust citation analysis tools such as Thomson Reuters Web of Science. Besides, the preservation and curation of digital resources will add to the list of new challenges, which libraries with a solo curator or solo digital initiatives librarian will understand to be formidable. Rydberg-Cox also points to the implications these developments have on libraries, which have thus far focused much of their professional energies on measuring their impact on scholarship, readership, and teaching, as well as improving such areas as interlibrary loans, reference, instruction, and outreach. Working with digital humanities scholars will also mean collaborating on new and original archival and library resources that focus around specific humanistic and historiographical questions.

Digital humanities project descriptions

The projects described in this illustrate multiple scenarios for using archives and manuscript collections with varying extent of collaboration with archivists and librarians. In the scenario where the projects had direct access to primary source

collections, the collaboration must have benefited from the direct relationship with archivists and librarians who provided finding aids, metadata, and other bibliographical information. In cases where metadata was available via a subscription-based service, the project used such data for analysis, visualization, and interpretation. Elsewhere it is evident that first edition books or books out of print were the source of the information. Finally, there are such projects as Viewshare, that allow archivists and librarians to submit specific data for digital humanities curation in order to enable researchers to analyze, and interpret with the help of maps and timelines generated by the project. This chapter does not investigate the legalities of using those publications for digital reproduction; the discussions herein proceed with the assumptions that the project leaders have obtained the necessary permissions or used works in the public domain. In most cases, archives and libraries transfer the responsibility to identify copyright holders to the researchers who must make all efforts to locate the copyright holder, heir, or agent for permission. An analysis of bibliographic sources may also help researchers identify orphaned works.

The Perseus Project (1987)

The Perseus Project ("Perseus Digital Library," n.d.; see also Wiltshire, Pearcy, Hamilton, Eiteljorg, & O'Donnell, 1992) is an early digital humanities project to which archives, special collections, libraries, and museums have contributed with photographs and texts since its launch at Tufts in 1987. The project planning had begun in 1985, and the CD-ROM of Perseus 1.0 by Yale University Press came out in 1992, followed by version 2.0 separately for Macintosh users in 1996 ("Perseus Version History," n.d.). Because of the launch of the Web in 1994, 1.0 is of particular historical significance since it demonstrates the presence of hypertext structures and information architecture. Equally important are the sources of information since the Perseus Project focused on Greco-Roman art and architecture in digitized images and hypertext. The site was developed in HyperCard, which preceded the Web's hypertext environment. It featured guided tours using paths to the contents and the other CDs. The tours included guided, novice, expert, philological, art, and archaeology tours designated for beginning and advanced users. Maps were available to allow visually enhanced navigation commonly available on Google maps today. Additionally, search tools, direct and indirect links, and the lookup tools were also available as part of the site's information architecture.

The primary texts in Ancient Greek with translation into modern English featured 31 authors (dramatists, philosophers, poets, and historians) including Aeschylus, Aristotle, Euripides, Herodotus, Homer, Pindar, Plato, and Sophocles. The Perseus site (Tufts, n.d.) lists sources in excess of 2,500 works and over 3,500 encyclopedic entries from Loeb Classical Library. Images of vases come from Boston, Mississippi, and such institutions as Harvard University. A significant part of the collections comes from Tufts University Library and various digitized collections, which illustrates the role of archives and libraries during the early years of the digital humanities.

CD-ROM projects

Who built America? (1995)

Who built America? was an experiment (Thomas II, 2004) of the American Social History Project based at City University of New York, bringing social and labor history into the context of America's national history between 1876 and 1914. It was a project involving Roy Rosenzweig, Steve Brier, and Joshua Brown to introduce a new multimedia form of scholarship and teaching, but its stand-alone architecture had already presented a risk in an age that steadily and quickly moved toward hyperlinked and networked texts on the World Wide Web launched in 1994. It contained 450 pages accompanied by a CD-ROM containing multimedia histories, songs, recorded speeches, oral histories, and primary source materials in print form. The availability of archival materials in edited (i.e., typed) format has elicited criticism (see Darien, 1998) and praise (Saillant, 1994), depending on the audience: historians may have preferred to see the original texts perhaps side by side with the transcription while a general audience may appreciate the easy-to-read format and visual accessibility. The textual materials from archives appear in edited form, which demonstrates the role of archives in selecting source materials for the project.

Valley of the Shadow

Among the earliest digital humanities (and historical scholarship) project to appear on the Web was the *Valley of the Shadow* project undertaken by Edward L. Ayers at the University of Virginia. The project focuses on Civil War era records, which typically come from two sources: institutional manuscript collections and private collectors. The project was available in CD-ROM as well as Web version. Ayers (1999) referred to his project as "digital archive" and "interactive 'album' that organizes the story in thematic and chronological spaces, providing launching points into the archive and using multimedia as effectively as we can" (para. 6). Such a digital archive is also a critical resource for analysis and interpretation. This "digital archive" has extensively relied on manuscript collections at the University of Virginia, U.S. Army Military History Institute, Carlisle Barracks, Pennsylvania State Archives, National Archives and Records Administration, Virginia Military Institute Archives, Library of Virginia, and several other collections of regional and national significance. The current Web site provides access to transcribed texts, census records, maps, and images via an image map in the form of a floor map with an overview of a physical archive. Each level corresponds to large categories of records such as "The Eve of War," "The War Years," and "The Aftermath." While the arrangement of the archival material here fits the information architecture of the project site, the ontological structure of the archive is present through the indexing and topical arrangement of the material on the site.

Text Encoding Initiative projects

Aside from the continuous misuse of "archive" in the digital humanities context, Text Encoding Initiative (TEI) projects have involved archives and special collections

making their collections available for textual analysis and interpretation. Willett (2004) alludes to such projects as "textual archives" as envisioned for the works of Yeats, the *Canterbury Tales Project*, the Project Gutenberg, *Women's Writes Online*, and the works of Henrik Ibsen in Oslo, and Isaac Newton's manuscripts in London.

Binder's book

McCarthy, Welsh, and Wheale (2012) report on the basic application of TEI tagging in the case of Bodleian Library's *Binder's Book* (BB) at Oxford University, a 150-page record of seventeenth-century book binding orders. The authors considered the potential audience for such a resource to include rare book librarians, collectors, bibliographers, digital humanities scholars, archivists interested in such aspects as provenance, and graduate students interested in TEI encoding. From a digital humanities data curation perspective, this TEI project offers value to the library wishing to curate and share such data with researchers. The authors address the methodological conflict between the requirements of the EAD schema used for finding aids and the TEI used for textual analysis:

> *Text encoders should act with the awareness that they impose meaning upon texts, rather than merely presenting them for others (McGann, 2001). This can be problematic when the document expressed is technically archival in nature, as the BB. The archive world has developed Encoded Archival Description (EAD) for finding aids; it has its own system for allowing interpretation by the encoder, but the world of textual encoding has no similarly evidential language.*
>
> *McCarthy, Welsh, & Wheale, 2012, p. 564*

Aside from such methodological differences, there have been attempts to establish crosswalks between standards and schemas. The Optimizing Resources for Repositories and Archives working group (METS, 2009) has addressed integrating such metadata standards as TEI, EAD, DDI (Data Documentation Initiative), and Metadata Encoding and Transmission Standard (METS). Reconciling the domain-specific encoding and description standards of the archivist and digital humanities communities of practice will remove barriers and foster new projects.

Thematic research collections

The William Blake Archive (Eaves, Essick, & Viscomi, 2014) has been a freely available site since 1996 with a focus on William Blake's prints and poems. The illustrations are accessible on separate pages as to minimize obstructions to linear reading of the text. The intuitive arrangement of illustrations allows viewers to access the illustrations in a separate frame although this approach may not quite support assistive equipment and users relying on such support. The digital archive draws its sources from several contributing sources such as Auckland Art Museum, Bodleian Library, Glasgow University Library Special Collections Department, Library of Congress Rare Book and Special Collections Division, New York Public Library Rare books division, and many others. The metadata records with the contributed material contain such information as technical metadata, provenance, and other administrative metadata. The images have been digitized with a resolution of 300 ppi using TIFF file format

for "dark archive" preservation and detailed viewing. Additional storage options include DVD disk copies as well as storing raw images in a storage provided for the archive by the Carolina Digital Library and Archives. This project illustrates that the archives are no longer just providing data and content for viewers' and researcher's sake—they are also involved in the digital preservation of the digital archive contents.

The *Walt Whitman Archive* (Folsom & Price, 2011) began in 1995 with a focus on the life works of Walt Whitman who left behind, in addition to his poetry, notebooks, manuscript fragments, prose essays, correspondence, and journal articles, all of which present important contexts for analysis of his works. The original site also offered a biographical essay by Ed Folsom and Kenneth M. Price and supplementary biographical materials. The current site also provides a timeline. More importantly, researchers can find hypertext and facsimile versions of his works. The archive makes extensive use of TEI methods to reveal editing work in his manuscripts, which allows researchers to follow the changes to his work.

The presence of archival resources is evident: not only does this site offer its own library of TEI-encoded files, audio and image files, notebooks, manuscripts, and translations; it also presents alphabetically arranged finding aids to manuscripts at various repositories holding Walt Whitman's works. The listing of finding aids identifies various source repositories such as the American Antiquarian Society, British Library, and Harvard University. Manuscripts Department, Houghton Library, Huntington Public Library, Library of Congress, Liverpool Central Library, Musée de la Coopération Franco-Américaine, University of Pennsylvania's Walt Whitman Collection, Walt Whitman House in Camden, and many more. The structure of these integrated finding aids represents the arrangement of the original archives in series with links to digital images—an added benefit of EAD finding aids. Each series-level description provides the identification of the content in the home repository, which enables researchers to search for originals at the home repositories. As archival finding aids go through subsequent updates due to added items and background biographical information, one may see the benefit of links to those finding aids. In fact, there are no links to holding institutions in the Whitman Archive, just their contact information. These internal finding aids, however, lend a solid information architecture to the entire site. In some cases, the Web site of the individual repositories may not even have an online-accessible finding aid, so these integrated finding aids are very helpful. A quick comparison with the Whitman Archive finding aid for the Trent Collection of Whitmaniana at Duke University, for instance, reveals differences in detail (like in the Abstract) but the structural correspondence is reliable. The argument for the integrated finding aid approach is logical, however, given the migration of content from one infrastructure to another: despite some minor differences, the Whitman Archive provides the stability of information that researchers can highly appreciate. The archive demonstrates a close integration of archival role into this project.

Recent digital humanities projects

Digital Literary Atlas of Ireland

Recent innovative developments in the digital humanities involve the transdisciplinary use of geospatial data to emphasize the spatial, not just the temporal, dimensions

of digital humanities since narratives migrate when people do, which directly contributes to the diffusion of artistic, musical, literary, and other creative genre over extensive geographical spaces. The *Digital Literary Atlas of Ireland* (Travis, 2010) project at the Trinity College of Dublin combines biographical data with interactive timeline and geospatial technology, which present Irish literary history and biographical knowledge in a temporal–spatial context. The project focuses on the literary, historical, and cartographic perspectives on Ireland between 1922 and 1949 through the eyes of 14 Irish writers. The user interface presents three access points to biographical information:

1. featured authors' life paths with biographical narratives and references to sources;
2. timeline using Google Earth timeline outlining the life of these authors as they moved around. This data has significant bearing on their development as writer during these periods; and
3. vimeo presentation of maps, which begin with an overview of the terrain, gradually zoom in to the street-level view, though not quite as close as Google Street view but a Google Earth plug-in allows a significant level of interactivity for viewers interested in the geographical aspects of humanities.

The significance of these Irish writers in the context of Irish national history and cultural identity can be accurately established through authors whose works have not only shaped the lives of these authors, but may have had transformative effects on their literary works.

Mapping the Republic of Letters

The *Mapping the Republic of Letters* (2013) project seeks to visualize the international network of scientific academies through the networks of correspondences and patterns of travels followed in this project. The time frame for this project was from Erasmus to Benjamin Franklin, which spans the development of modern sciences and humanities from the Renaissance to Modernity. Given the geographic and date ranges, the project presents case studies as individual frameworks for data presentation. The results were manageable sets of visualized data for individual writers like Voltaire, Galileo, Locke, and many more. Based on the data that could be generated from the material located in archives and manuscript collections, the next step was to visualize such information as the number of letters sent by Galileo in a given year between 1588 and 1616, and their recipients. The geospatial visualization of Voltaire's and Locke's correspondences shows geographical overlap in the two individuals' scholarly networks although their lives have only had a 10-year overlap. The digital curation and visualization of such data may present surprising facts such as mutual acquaintances of historical personages who did not interact with each other. The network map reveals geographical proximities of famous people whose relationships otherwise might not have been known due to the lack of literary or historical study. This type of visualization provides new avenues for humanities scholars to follow.

Although the background information on the site does not present any direct involvement of archivists and librarians, the material for this project included 55,000 records and 6,400 correspondents ("Mapping," 2013) in the Electronic Enlightenment Project (EEP) database, which is available via subscription and contains nearly 64,000

historical records as of 2013 and a network of over 8,000 historical figures. The EEP is a product of the Bodleian Library at Oxford University with its own archives and manuscript collections, but the project sources include information on almost 60,000 manuscripts and over a 100,000 early edition sources.

Archives, ViewShare, and digital humanities data curation

Of all the digital humanities projects, perhaps this model illustrates the fullest involvement of archives in the digital humanities landscape. Archivists and digital curators collaborating on the preservation of content and metadata can extend access to the data in the metadata record in ways that enable researchers to use such data. For instance, timelines can use the values in the date fields to generate timelines, and of recent, there have been efforts to include geospatial data in designed fields to allow such services as ViewShare (n.d.) to generate maps. Viewshare requires either a metadata worksheet or an Internet address (URI) where it can harvest the metadata complying with the Open Archives Initiative Protocol for Metadata Harvesting (OAI-PMH) standard. As the discussion of the Republic of Letters has demonstrated, researchers can use these tools to research hitherto unrelated events and people in nearby places.

Windhager and Mayr (2011) have introduced two visualization models for the navigation of information in space, time, and ontology: geographic and topographical space-time-cubes. These methods facilitate the visual representation of time, space, and ontology in a single three-dimensional space, in which human and other historical relationships unfold over a period of time in specific geographical locations and in relation to various topics. Although these visualization models have presented in the context of museum exhibitions to orient visitors, their applications may extend to representing information visually in digital repositories, virtual museums, and in such visualization tools as ViewShare or EEP. This is definitely an opportunity for archivists and librarians to play a more visible role in humanities data curation since they already have all or most required metadata: date, temporal and spatial coverage, subject, names, and provenance.

Many archives have submitted data to Viewshare for data visualization. The *History of Fairfax County in Postcards* (Trow, 2012) illustrates an ideal use of Viewshare, as it plots the locations of the postcard sites on a map with pins. Clicking on the pin will open a small pop-up window with the thumbnail and description. Viewshare can also generate a pie chart, timeline, as well as gallery with the images residing on the collection's host server; Viewshare simply harvests the metadata and makes available any visualization with the help of code that archivists and technologists can embed on their institutional Web site.

Digital humanities curation in the classroom

Thus far, this chapter has focused on defining, characterizing, and discursive aspects of the digital humanities and the role of archives, but has not addressed pedagogy in the field. While this book does not focus on pedagogy and learning theories

(e.g., constructivism), it places archivists in traditional classrooms and e-learning environments with an important role to fill as teaching partners or even as instructors. The conceptual model in this chapter demonstrates this role as well as the relationship of interpretive layering and metadata enhancement, both of which comprise collaborative curation in the classroom. Panofsky's approach to iconographical study serves as a framework even though its original context for the methodology was art history and the interpretation of art. Applying some of the methodology to classroom curation of works of cultural, literary, and historical significance may present learning opportunities in current learning environments as well. More importantly, any advanced course may potentially turn into a thematic research collection or some digital archive described earlier in this chapter. Another chapter (on information architecture) will address the critical hypertextual and navigational relationships between the classroom and digital resources.

Recent professional literature notes the lack of coverage on pedagogy in Digital Humanities. Brier (2012) notes that the *Digital Humanities Quarterly* (at or near the publication of his work) contained 19 (out of the 90 plus) articles related to research and only two to pedagogy. Full-text searches on "research" received 81 hits whereas the figure for "teaching" and "learning" was at least 40 and nine for "pedagogy." In his calculations, Hirsch (2012) notes that while the *Companion to Digital Humanities* (published in 2004) contains 504 instances for "research," instances of "pedagogy" only appear eight times; "teaching" 60 times, "education" 30 times, "teach" 7 times, and so on. These numbers indicate the visibility of such topics in the Digital Humanities research community, and may be of concern to archivists and librarians collaborating educators in the field.

With advanced classroom technologies and learning management systems (such as Blackboard, Moodle, Sakai, or others), educators can integrate digital collections and digital humanities sites into their curricula and specific coursework. The student learning objectives in such courses may include learning about ancient civilizations and Western traditions (as covered in most humanities courses in the United States) to developing digital humanities curricula (for education majors), developing digital humanities projects (for humanities majors), and developing digital archives (for library and archives students). In most cases, online courses include lectures with materials from digital archives, digital libraries, and virtual museums. The role with instructors is to relate that information to the course material while students will analyze and interpret what they select for their assignments. Kapelos and Patrick (2012) report on an architectural course engaging special collections and subject librarians in the process at Ryerson University. The archival materials included architectural photographs, rare texts, and other materials in a physical classroom setting. A curator's role may range anywhere from selecting collections to support specific coursework to teaching an entire course for humanities and history students in using primary sources (in print, analog, and digital forms). The curation process may involve writing annotations to enhancing the metadata through interpretive layering (Flanders & Muñoz, 2011).

This last section focuses on the relationship of interpretive layering, metadata enhancement, and the application of Edwin Panofsky's iconological framework

(Panofsky, 1962). Although designed specifically for the interpretation of art, Panofsky's three-level iconological approach is ideal for the study of humanities whereby the interpretive layering process begins with the identification of the artifact or document discovered and located through research. Reliable descriptive metadata is vital as it can considerably improve the identification process, which is essential to the selection process in order to lend a project much-needed focus. This approach enables students to enhance the existing metadata record (supplied by the curator or subject librarian) with missing data and information discovered through research.

At the initial stage called preiconographical description, "the objects and events whose representation by lines, colors, and volumes constitutes the works of *motifs* can be identified…on the basis of our practical experience" (Panofsky, 1962, p. 9). In curation terms, all metadata contain simple data and information to aid the identification of author or artist, title, style, geographic location, language, provenance, time period, cultural context, and others. This metadata must support historiographical or hermeneutical analysis at the subsequent stages.

The next stage (called iconographic analysis) focuses on contexts such as themes, concepts, stories, and allegories, which require greater familiarity with such foundational elements as objects and events. Students begin working with information specific to the artifact and beyond the general information that appears in the initial metadata record, and this stage requires some research of related sources. They can enhance fields like dc.description.abstract or dc.description.notes, which provide adequate space for analytic annotations. These repeatable fields also present space for critical notes by curators and faculty for future reference. Panofsky (1962) ascribes great importance to iconographic analysis, which requires correct preiconographical description.

The last stage (called iconographic interpretation) focuses on the interpretation of deeper, intrinsic meanings, deeper meanings of objects, events, content, and symbolical values. After the three-tiered process, the effects of interpretive layering on the curation process show extensively in the metadata records. Curators will notice the difference in annotations from history students, compared to those from other areas of the humanities, which is due to the significant differences in methodologies across the humanities. Historiographical studies will produce different kinds of interpretative annotations in comparison to those produced by students in courses focusing on arts and letters and inviting more subjective interpretations. As a result, the metadata record will contain progressively deeper and more introspective keywords available for analysis and interpretations in future courses. Figure 2.1 below demonstrates this layering effect in the curation process.

The role of an archivist or curator selecting the course material and providing the metadata can significantly improve the learning experiences (and outcomes) of students having the benefit of reliable metadata and high-quality digital material from a well-curated digital collection. The enhanced information developed and added to a virtual museum or digital collection can support future interpretive and analytic studies in digital humanities coursework and professional research.

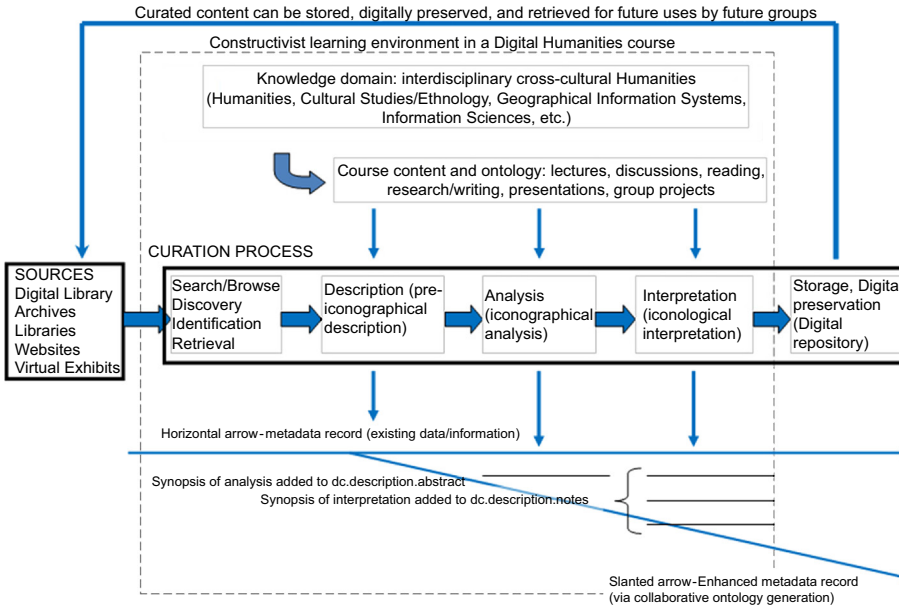


Figure 2.1 Interpretive layering on the curation process.

Conclusion

Archives and special collections play a vital role in the digital humanities through participation in digital humanities projects and teaching. As the digital humanities field expands, questions about scholarship and the theoretical nature of projects will emerge and require discussion. The debates in the digital humanities provide a framework for arguing that digital curation activities may also be just as theoretical in scope and depth, and the various projects demonstrate validity to such arguments. As digital humanities pedagogy gains greater ground in higher education, so will digital curation in order to provide the critical information students need in their coursework.

Digital history, archives, and curating digital cultural heritage

3

Within the broader context of Humanities, History represents a specialized field where archivists have worked particularly closely with researchers, educators, and students. In fact, archives have evolved over the century around this nexus focused on historical questions and primary sources. The scope of preservation issues has gradually expanded from paper (or earlier media) toward film and audiovisual collections in analog and digital formats. Today, digital history presents a new chapter for collaboration among historians, archivists, librarians, and technologists, as it focuses on using digital technologies while archives expand their preservation and curatorial work to digital materials.

Historical computing has roots in the 1960s New Social and New Political History wave with historians using computers and databases for quantitative analysis. The 1980s introduced quantitative analysis of Census data using computers, and in the 1990s, historians also began to use the Internet and the World Wide Web for teaching and research (Burton, 2005). For archivists and digital curators, this stage also presents new possibilities, questions, and concerns about preserving, promoting, and providing long-term access to born-digital, digitized, and hypertextual content. Throughout modern history, the relationship of archives and historians has evolved along on the availability of documentary evidence preserved for research in archival facilities. The need for evidence, however, is equally important in the digital environment today, as the volume of born-digital materials accessioned by archives increases along with the growing backlog of existing collections awaiting digitization. Due to the ephemeral nature of digital material and the concerns about their preservation and long-term access, archives will continue to play a crucial role in providing evidential information to historians. A digital curation program for digital history will need to consider the disciplinary focus of collections and encourage collaboration between historians and archivists.

For a century now, various paradigm shifts in academia, society, politics, economics, and technology have transformed historiography and, with it, the curatorial agency shaping public memory and community identity. Interdisciplinary research, postmodernist and postcolonial, Marxist, and feminist methods of representation, and the growing corpora of born-digital materials have informed historical analysis and writing, which in turn inform academic and public discourse on history, including local history and cultural heritage. Almost parallel to these developments in history were paradigm shifts in archival practice with respect to the curatorial role and visibility of archivists who have emerged from the arcane institutional repositories of European nations after the Enlightenment era and the French Revolution to occupy prestigious public spaces. Cook's chronology (2013) of this paradigm shift traces the evolving role of archivists from two centuries ago through changes after the two

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world wars. The recent decades have brought even greater visibility and public role to archivists whose work can shape community memory and identity through more creative outreach practices. In terms of technology changes, digital historians followed trends similar to those by digital humanists: the trends toward mobile technology, open digital standards, Web 2.0 platforms, and new file formats have had significant implications for digital curation and history. The convergence of these trends will have varying effects on the directions taken by historians and digital curators, their working relationships, and collaborative scholarly output.

This chapter focuses on the relationship between digital history and archives, addressing also the implications of digital history and historiography for digital curation and vice versa. A review of definitions presents a general scope for digital history and the effect of technology on history. The chapter then presents a brief chronology of developments in the archival profession from a records keeper to an activist-curator model following Cook's analysis (2013), which fits well into the evolving digital landscape where archivists are creating blogs and engage in activity outside the traditional frameworks of the profession. This is followed by a discussion on digital historiography and the role of archives in digital historiography and shaping digital historical representations. The next section focuses on historical hypertext and the structuring of historical texts in the Web environment, and addresses the challenges of preserving both traditional Web and TEI-XML documents. Next, the chapter addresses digital history data curation, following the flowcharts developed by Lord and Macdonald (2003), which present the comprehensive digital landscape of digital history, taking into consideration the emerging digital publications, data curation mandates, and the role of researchers, archives, libraries, and publishers in this curation ecosystem. Digital history is an important context for digital curation as presented by Lord and Macdonald's framework, which brings together archival resources, publication, and data into a three-level curation workflow.

Defining digital history

As seen with Digital Humanities in the previous chapter, digital history is also a binary concept with two dimensions: digital (technical dimension) and history (disciplinary dimension). Dictionary definitions for "history" usually include generic entries such as "the study of past events," but for deeper meanings of history and its relationship to archives, definitions need to address specific aspects of history. The Oxford Dictionary (2013) provides two such definitions: "The whole series of past events connected with someone or something" and "a continuous, typically chronological, *record* [emphasis added] of important or public events or of a particular trend or institution" ("History"). The former definition addresses the series aspect as well as associations with people who could fall anywhere on the spectrum of historical significance (in spite of the political prestige they may have enjoyed); the second one speaks to history as a record of events, trends, or institution. Both definitions provide important elements of such relationships: series appear in the structure of archival finding aids, and each item in a manuscript collection or institutional archive constitutes a record.

"Digital history" first appeared in the name of the "Virginia Center for Digital History" established in 1997, and it was the subject of an online roundtable "Interchange: The Promise of Digital History" (2008) involving Daniel Cohen, Michael Frisch, Patrick Gallagher, Steven Mintz, Kirsten Sword, Amu Murrell Taylor, William G. Thomas II, and William J. Turkel. From this discussion came an early definition for digital history:

> *Digital history is an approach to examining and representing the past that works with the new communication technologies of the computer, the Internet network, and software systems. On one level, digital history is an open area of scholarly production and communication, encompassing the development of new course materials and scholarly data collections. On another, it is a methodological approach framed by the hypertextual power of these technologies to make, define, query, and annotate associations in the human record of the past.*
>
> *Cohen et al., 2008, p. 454*

While the association of digital history with digital technology is commonplace, in Burton's view, digital history is not merely about the digital reformatting of content; it is about using the computer for historical analysis that is impossible without computers (Burton, 2005).

Rosenzweig (2003), however, had warned about multiple unexpected effects of digital technology on the historical record when he wrote, "Historians, in fact, may be facing a fundamental paradigm shift from a culture of scarcity to a culture of abundance" (p. 739). The world of limited surviving sources, small audiences, and limited scholarly output has exploded into a world of abundance and digital ephemera. The number of digital publications has grown due to improved access as well as the steady growth of digital publications; moreover, improving classroom technologies and e-learning environments only add to this already expanding dynamic.

The volume of digital heritage materials has also increased with significant implications for archives and libraries warning about data loss due to crises, lack of control, or emerging political landscapes intolerant of open knowledge. Even with what may survive, the chances of accessing that material comes into question due to obsolescence. The short life span of digital media (compared to film and acid-free paper) adds to the problem of preserving interactive hypertextual content, as seems to be the case with some dynamic Web sites in the Internet Archive: links to background images and external content disappear, rendering the "archived" site functionally and aesthetically undesirable, useless, and meaningless. Rosenzweig (2003) also addresses the authenticity of digital texts as well as the rise of born-digital texts still subject to suspicion by historians. Digital curation plays a critical role in mitigating the effects of this abundance, but selection criteria will evolve into a critical process in archival as well as digital historiographical practice.

Paradigm shifts in archival curation

Digital history and its present relationship to archives lend archivists significantly greater visibility than what they have had in the past. Terry Cook (2013) traces these

developments in the archival profession through the centuries following the Enlightenment and the French Revolution. He writes, "the archivist has been transformed…from passive curator to active appraiser to societal mediator to community facilitator" (p. 95), which brings archivists into the present as mediators between history and the public. This shift also has implications for curatorial agency in the archives, which also shifted from the centrally positioned archivist to the community, and as the focus of curation moved from evidence (historical records) to community memory. As archival identity has significantly changed along the sweeping changes in the sciences and historical study since the late nineteenth century, Cook examines four paradigms (or frameworks) for curatorial practice: evidence, memory, identity, and community. They are not mutually exclusive but overlapping; for instance, the legacy of the "evidence" paradigm still lingers in some institutions' special libraries or dedicated archives. How archives curate collections (including digital ones) is defined and communicated through these paradigms reviewed in this section. There have been a number of works (see Greene, 2002; Nesmith, 2002) addressing the influence of postmodernism on the archival profession and the record, which affects historiography and historical representation equally influenced by similar paradigm shifts.

Paradigm 1: evidence (premodern archiving)

European national archives emerging after the French Revolution were accessible to scholarly elites and general citizenry for research. Archivists were custodians of official records of the state (and later, other privileged institutions) and curated these records as evidence. Principles as provenance and original order have emerged during this period, replacing a practice of arrangement by subject or other authorities. It is interesting to note that library catalogs have adopted this method, which is helpful to researchers gathering sources by subjects. While these methods validated records as evidence, this appears to be a model of compromised curation practice. Cook (2013) notes that institutions did not involve their archivists in the appraisal and selection of records—a practice that has not only survived into the present but has also become a preferred method at some institutions even today. The resulting compromised historical representation not only favored ruling (and official) elites but also supported a national scope for such practices. This paradigm did not address or prioritize the collection of private records and personal archives and there was no regional, local, or social focus either. Instead, collectors and donors sought libraries to acquire private records. These collectors and donors were systematically regarded as dilettantes, and such collection efforts were fragmented and discontinued after deaths of the donors and collectors. During this period, however, archivists oversaw descriptive practices aiming to "illuminate the contextual origin of records" (p. 106). The importance of contextualization has remained an important aspect of archival curation.

Paradigm 2: memory (modern archiving)

The period following the two world wars, falling between the 1930s and 1970s, saw a sharp increase in the volume of state records. This explosion has mandated swift

and sweeping changes in archiving practices, especially in appraisal and selection by archivists making critical decisions. The stronger affiliation with historians has altered appraisal focus to support historiographical practices. The top-down model of the previous paradigm gave way to the bottom-up model with a new focus on everyday people working in factories and farms, also focusing on family and community history, and providing greater details about the "social, cultural, economic, and scientific activities of the state as much as its constitutional, legal, military, and foreign relations spheres" (Cook, 2013, p. 108). The elitist focus of archival work gave way to one on the community, everyday people, and public memory. Description practices have also adjusted to the reality of high-volume records, as archivists have added collection summaries and focused on collection and series-level descriptions rather than individual records. Archivists have also begun to conduct historical research to verify biographical and historical information, provenance, and other pieces of information needed for accurate contextualization. Cook notes that while the focus on subjectively created cultural memory resource replaced the focus on evidence in the previous paradigm, this period also provides important historical data on families, communities, private individuals, and the underprivileged. Digital cultural heritage representing this period is possibly due to the emergence of interpretive approaches to history and the records/archives kept because of archivists' mediation.

Paradigm 3: identity (postmodern archiving)

The 1970s saw the emergence of the professional archivist with deep roots in history and historiographical methods while expanding interdisciplinary and multicultural interests. Due to the postmodernist challenges to knowledge, society, and culture, the monolithic understanding of a universal "Truth" gave way to plural truths embedded in multiple and diverse perspectives and narratives. Archivists, historians, and archivists as historians themselves have shifted their focus to fragmented, diverse, and pluricultural societies. As a result, the collecting focus began including private citizens and the "broad spectrum of human experience" (Cook, 2013, p. 110). The arrival of personal computing (including mobile) devices presented challenges to legacy methods of archiving, as digital records painted a much more fragmented world, disrupting the more traditional practices such as arrangement according to original order. In fact, the historic one-to-one creator-record relationships replaced by one-to-many, many-to-one, and many-to-many provenance models have significantly transformed the approach to documenting provenance. Early digital archival standards took centerstage in the curation process. Archivists in this paradigm have become advocates for various social issues and causes, thus speaking for issues through archival curation. Exhibitions may speak volumes about the preferred selection process. Archives became a societal resource, not just a cultural and heritage resource. Archival curation in this paradigm was framed by the archivist's own identity (gender, political, professional, religious, race, ethnic, etc.), which helped archivists to mediate between record and societal identity.

In the inaugural issue of *Archival Science*, Cook (2001) addressed the effects of postmodernism on archival science and the nature of the archival record. In this issue, he writes

> *At the heart of the new paradigm is a shift away from viewing records as static physical objects, and towards understanding them as dynamic virtual concepts; a shift away from looking at records as the passive products of human and administrative activity and towards considering records as active agents themselves in the formation of human and organizational memory; a shift equally away from seeing the context or record creation resting within stable hierarchical organization to situating records within fluid horizontal networks of workflow functionality.*
>
> *p. 4*

In the same article, Cook summarizes key changes—or "new formulations"—affecting archival science as well as practice. A shift from focus on product to process has transformed key archival concepts:

1. function, not organizational structure, is the new framework for provenance;
2. original order no longer adheres to the initial placement in a registry but can correspond to multiple functional and intellectual contexts;
3. records no longer have fixed structures, contents, and contexts but are fluid in terms of their relationship to contexts over time;
4. fonds expand from single to multiple creatorships based on new contextuality related to function and activity;
5. arrangement and description move away from a focus on entities and groups toward representing multiple interrelationships responsible for record creation;
6. appraisal will shift away from expressing potential research value to the record creators' place in a social or professional network regardless of status;
7. preservation will shift away from conservation and maintenance toward a digital curation approach involving migration and emulation; and
8. archives will be more publicly accessible instead of remaining storage or dark archives with no public access. These immense changes taking place for the past 12 and more years have begun to reshape digital curation, digital historiography, and the place of archives in the digital humanities.

Paradigm 4: community (participatory archiving)

This paradigm represents a curation model in its current formative stage, as it evolves around such platforms as the Internet, social networks, communication technologies, and Web 2.0 platforms. The archival process experienced significant fragmentation as each person, community, institution, and other units build archives separately (see Hawkins, 2013; Lee, 2011a). Personal archiving now happens not only on shared platforms like Internet Archive but also in social networking sites despite some disadvantages over designated digital repository platforms that institutions use. The hypertext connections across these personal archives are also arbitrary, inconsistent, and unstable. Cook (2013) adds,

> *There is simply too much evidence, too much memory, too much identity, to acquire more than a mere fragment of it in our established archives. Furthermore, removing such archives, such memory, such evidence, from the originating communities to our archives may be problematic and undesirable for several reasons.*
>
> *p. 113*

The fourth paradigm focuses on community-based archiving that focuses on identity and memory as well, and introduces a high level of cultural sensitivity to the curation process. It represents a significant shift from exclusive custodianship to a shared one and collaboration. The collective aspect of and collaborative approach to this model presents an intersubjective approach informed by multiple identities, memories, and perspectives. The democratization of archives in this paradigm liberates it from the former limitations of archives and curator models.

Digital historiography and archives

The paradigm shift from a tacit recordkeeping role to a highly visible community participant presents a long trend leading to the direct relationship between digital historiography and digital curation today. Sternfeld (2011) defines digital historiography as "the interdisciplinary study of the interaction of digital technology with historical practice" (p. 550), resulting from the combination of historiography, archival theory, and various computational and technical standards. He also regards historiography as the epistemological basis for accessing historical knowledge, which—in digital context—means using any digital historical representation. A more general definition for historiography is "the writing of history based on the critical examination of sources, the selection of particulars from the authentic materials, and the synthesis of particulars into a narrative that will stand the test of critical methods" (*Merriam-Webster Dictionary*, 2014, "historiography"). Both The Merriam-Webster and Sternfeld's definitions speak to the relationship of digital historiography to archives and digital curation, respectively. The definition of digital curation in Lee and Tibbo (2007) communicates the need to develop "trustworthy and durable digital repositories" ("Opportunities and challenges," para. 3) as a requirement for digital historiography. The trustworthiness, durability, and authenticity of contents are among archivists' and curators' concern when preserving digital collections, and this is where digital curation offers historians long-term partnership for collaboration.

The role of metadata standards and schemata underscores the mutual interest in digital preservation and long-term accessibility of digital material. Thus, digital curation can reinforce greater confidence in digital technology among skeptics of what Sternfeld (2011) calls "digital historical representations" (p. 547) in reference to digital collections, Web sites (such as virtual exhibitions and virtual museums), research databases (with electronic journals), mobile applications, and visualizations.

The preservation of Web sites comes with significant challenges for archivists and digital curators, which arises from the need to preserve such content in their original hypertextual context. Because virtual exhibitions and museums feature several

links to external materials, those hyperlinks will eventually lead to deleted, moved, or otherwise rendered inaccessible sites. The purpose of the Internet Archive Wayback Machine is to preserve Web content along with their innate structures, but while backing up a site in the first era of Web sites (Web 1.0) was as easy as copying the entire folder to a storage medium, the Web 2.0 environment has considerably complicated that process through a very different structure. It requires backing up entire platforms, not just self-identifiable PDF files and images. Preserving the full-text (the historical narrative) is impossible since the text resides in the platform's complicated database, which also requires a backup. Thus, curating such collections translates into greater attention to (and need for) detailed structural, descriptive, technical, and administrative metadata. The following examples of digital historical representations illustrate how hypertext narratives, descriptive metadata, and various applications present contextual data to researchers.

Digital historical representations

Historical representation is integral to historical writing and historiography. Ankersmit (2001) writes, "we may 'represent' something by presenting a substitute of this thing in its absence. The real thing is not, or is no longer available to us, and something else is given to us in order to replace it. In this sense, it can be said that we have historical writing in order to compensate for the absence of the past itself" (p. 11). Historians present text and nontextual materials to substitute things or people no longer in the present, and much of that material comes from archives. However, there are fundamental differences in what historians and archivists do, what archivists do as authors of histories, or what historians do as archivists: historical representation versus description.

Ankersmit treats description and representation as opposites: whereas description assumes fixed and tangible relationships between a subject and its property, historical representation recognizes unfixed and intangible relationships between representation and the represented. What is absent in the present cannot be described but can be represented in historical discourse. Description, in contrast, requires tangible relationships (fixed in the present) between objects and their properties, both of which represent the past. While historical representation brings events, people, and places into the present through writing, archivists rely on description of objects used by historians (many of whom are actively publishing archivists) for writing. Descriptions can take the form of label descriptions in exhibitions, series- to item-level descriptions in archival finding aids, and data added to metadata records in the various fields of a schema such as Dublin Core. Therefore, historical representation and archival description are interchangeable but are mutually informative.

According to Sternfeld (2011), digital technologies use digital historical representations in reference to digital archives, collections, research databases, Web sites (which could be digital archives, thematic research collections, virtual museums, or virtual exhibitions), mobile applications, and visualizations. All contain text and media with data, information, and nontextual objects that represent a person, event,

place, or some organization that may no longer exist. In fact, Roegiers and Truyen (2008) address "visual historiography," which is a method integrating nontextual materials into historical representation and historical hypertext with social, spatial, and chronological perspectives. They write, "The strength of digital media is that one is able to represent the complexity of a historical subject, without having to fill out the gaps, or having to choose between different interpretations, but using an architecture that places the subject in its context(s)" (p. 70). Historical representation, therefore, may rely on both text and nontextual media to represent people, places, events, and others that can no longer be present in the Present. Digital historical representations, therefore, are tools that enable historical representation through a combination of text and nontextual media.

Virtual exhibitions, virtual museums, and such visually oriented Web sites as *Castles of Historical Hungary* (Szabó, 2001) demonstrate the value of visual historiography in which it can engage researchers, students, and teachers. In this case, the interactive images teach about the architecture of Medieval and Renaissance Hungarian fortresses and their role in the region's defense system throughout several centuries. Each castle—standing or in ruins—comes with a historical narrative (in Hungarian), map, and a collection of recently produced photographs showing the structure from multiple angles. The hotspotted (hyperlinked) arrows serve as navigation points between photographs, indicating the current state of these fortifications. The photographs and narratives can not only assist in structuring a historiography but can also assist historical preservationists in documenting the building. Code-embedded documentation may also be necessary from a digital curation perspective in the event that the decade-old digital images do not survive the many-centuries-old historical structures. The images and sequences thereof deliver the historical lesson through a combination of images and texts. Historians will still need to analyze images—print or digital—and integrate information in nontextual format into a linear or hypertextual narrative. Digital historical representations described by Sternfeld can contain critically selected images, recordings, interactive media, and historical text to accompany visual historiography. The descriptions available through the metadata record will not be, in Ankersmit's framework, representations but will directly support historical analysis, interpretation, and writing, which constitute historical representations.

Digital collections

Archives and libraries have been digitizing cultural heritage collections for open access to researchers and the public. Depending on the metadata schema (Dublin Core, VRA Core, and others), the fields provide descriptive information at varying granularity. At the very basic level, title, description, creator, and date have been available, but many collections may also include such contextual information as the collection name, provenance, and others to enable researchers to continue. Table 3.1 shows the metadata for a collection in the UTOPIA (The University of Toledo OPen Institutional Archive) repository where the provenance field provides textual information on the provenance and previous ownership history and where the Source Collection field (dc:relation.ispartof) provides the physical context for the collection image. The LC subjects may

Table 3.1 Metadata in a digital collection

Object description	
Title	1862 Diary
Description	The diary started on July 11, 1862 and ended on June 1, 1863
Creator	Hussey, Cyrus (1838–1926)
Date created	1863
Document type	Image
File type	Image/jp2
Source collection	MSS-017: Cyrus Hussey diaries, 1862–1864
Extent	12 images
Time period	1862–1864
Geographical area	United States
Provenance	Received from Donald D. Duhaime, 1970
Institution	University of Toledo
Repository	Ward M. Canaday Center for Special Collections
Digital publisher	Digital initiatives
Date digitized	April 20, 2009
Subject	War, soldiers, and veterans
LC subject	United States – History – Civil War, 1861–1865 – Personal narratives Vicksburg (Miss.) – History – Siege, 1863

help researchers locate related information in a library using the Library of Congress Subject Heading. While dates are important to historical research, they can also be useful in chronological visualizations (not available for this particular collection).

Historical hypertext

Ayers (1999) writes about historical hypertext, "Hypertextual history promises to be a tool that lets us think more rigorously. We might be able to imagine ways to write that let us deal more effectively with multiple sequences, multiple voices, multiple outcomes, multiple implications" (para. 18). In the "Interchange" roundtable (Cohen et al., 2008), Thomas addresses the hypertextual capabilities of the emerging digital technologies, which offer readers multiple ways to navigate historical texts by following preferred pathways through the Web of hyperlinked texts. This is very different from reading linear texts of works in print formats commonly found in archived manuscript collections, with which historians are familiar. However, not all historical texts or media require hypertextual rendering, but including such documents in a Web presentation will enhance contextualization of digitally processed texts and media. It is also possible to transcribe series of related historical texts—such as correspondence, diaries, memoranda, newspaper clippings—and present them in hypertext format for improved navigation. The information architecture of such Web sites must completely support the thematic structure of such sites therefore.

Historical hypertext may well have roots in the Memex project of Vannevar Bush, Director of the Office of Scientific Research and Development in the Library of Congress, who had envisioned the use of computing using the Memex program by historians accessing historical information through small-scale high-quality microfilms located inside the unit, which was a highly sophisticated desktop user interface with controls to access media. The Memex breaks away from the model of ontologically structured data (common in library catalogs) requiring researchers to navigate the classification system's classes and subclasses, and reinforces mental associations, which, in Bush's view, demonstrates how the human mind works (Bush, 1945). In the case of history, such associations among people, events, locations, dates, and various social, political, and economical contexts are the backbone of the Memex. Bush writes,

> *The historian, with a vast chronological account of a people, parallels it with a skip trail [or paths], which stops only on the salient items, and can follow at any time contemporary trails which lead him all over civilization at a particular epoch. There is a new profession of trail blazers, those who find delight in the task of establishing useful trails through the enormous mass of the common record.*
>
> *p. 8*

Bernstein (1999) demonstrates further that hypertextuality had already been present in the linear forms of historical narratives even as early as the fifth century BCE when Herodotus wrote *Histories*. Constructing the linear historical texts known to generations of historians antedating the arrival of computers, thus, did not require hyperlinks common in Web sites; according to Bernstein, the reader's mind shapes those paths based on associations that readers establish during reading. Definitions, clarifications, references, linkages to other events (covered in more or less detail than those in the main narrative) already provide a structural framework for historical hypertext, enabling researchers to put the text aside, consult references, and verify names, locations, and dates. Such diversions should be voluntary on the readers' end; however, skeptics may view such hyperlinks as prescriptive and programmatic as to instruct readers where to wander away from the main narrative. The associations that Bush writes about are subjective in nature; all hyperlinks can communicate is the availability of related texts at specific points throughout the texts. Virtual museums, exhibitions, and history Web sites fit this model best as long as their information architecture supports spontaneous and subjective reading.

TEI-encoded historical texts represent another significant category of historical hypertext that is both human- and machine-readable. In contrast to manually designed Web sites, TEI documents feature tagged data corresponding to specific taxonomies such as name and subject authorities, geographical names, and document and file types in an XML database. This allows project teams to link documents to indefinite sets of TEI documents featuring the same tags. A Web archive of transcripts from multiple archives can utilize such a method because they can also share the same database. TEI coding can also improve oral history transcriptions with important geographical locations, organization names, personal names, events, and other data. The tags are intentionally prescriptive as multiple documents may use a shared index to enhance

navigating considerably larger corpora of historical texts, but will not prevent readers to divert away from the hypertext as they like, which is what Bernstein is demonstrating. TEI projects also represent some of the earlier digital humanities data curation initiatives in archives and libraries participating in TEI projects.

Preserving TEI projects will also present challenges for archivists not familiar with the schema and database structure. Although it appears that the full text has been tagged with key tag elements of the system, digital curators must ensure the long-term preservation of not just the full text but the database and its structure and relationship of the vocabularies (or ontologies) used in the text.

Information architecture, hypertextuality, and cybertext—all discussed at length in the next chapter—have significant implications for digital historiography and historical hypertext, and it may not be too far-fetched to classify such corpora of texts as ergodic literature (Aarseth, 1997). Today's historians will plow through large numbers of primary and secondary sources in print and digital form to locate answers for historiographical questions, and research is well past any trivial reading activity. In fact, in digital historiography, researchers will navigate multiple cybertexts—as Bernstein (1999) illustrates—and rely on writing skills to represent the past through digital sources. Digital historiography, nonlinearity, hypertext, and the experience with ergodic literature represent a continuity of research process from earlier, predigital times in human history.

Digital history data curation

Digital curation and humanities data curation are integral to the work of archives pursuing new strategies for preserving data for humanities and history use. While data in TEI databases represent a sizable body of data for preservation and curation, archivists will also need to plan on preserving and curating metadata because they contain rich contextual data such as provenance, creation dates, geographical locations, and internal annotations. Historians have also begun working with larger volumes of historical data for analysis, interpretation, and visualization, and the metadata associated with such collections as historical letters, diaries, and other primary source material will support ensuing analysis and interpretation. Additionally, the onslaught of born-digital content, digital data, and digitized collections presents new mandates for archives to mitigate the effects of abundance resulting from the information explosion in the historical field.

The eScience Curation report by Lord and Macdonald (2003) also discussed in Chapter 1 of this book presents a starting point for a model of digital history data curation. The discussion in the "Interchange" (Cohen et al., 2008) points to the use of technologies to create, enrich, disseminate, and collaborate on historical scholarship digital form; thus digital history and data curation involve researchers, publishers, archivists, libraries, and educators alike, as presented in the flowchart in the eScience report. Among the reasons for curating primary research data (as well as primary sources as historians use) are the retention of data (as in archival records), validation of research results, use in teaching, and for public good. Contents of digital

repositories include text, images, and other media requiring metadata and standards for preservation and interoperability. Long-term preservation strategies include migration, emulation, formal descriptions, data reformatting, digital archaeology (as well as digital forensics), and the use of computer museums. Universally used metadata schemata like Qualified Dublin Core enable institutions to share and harvest metadata across network boundaries in an Open Archives Initiative framework.

The eScience data curation model by Lord and Macdonald presents three levels of curation, which applies to a comprehensive digital history data curation framework. At Level 1, which represents the traditional research process, raw data resulting from research go through analysis producing secondary data ultimately published through scholarly communication channels such as journals, proceedings, and—of more recent—open-access publication sites. These works eventually reach libraries, archives, the public, and the industry, which the researchers represent. This framework is similar to predigital scenarios. Level 1 curation is simple:

> *Research process → Level 1 curation → Publication process → Library/peers/public/industry (feedback) → Research process (starting over again).*

Level 2 involves the addition of a data archivist who makes raw (primary) data available for new discoveries and community and industry feedback. The archivist uses metadata to enable searching, discovery, and information retrieval of such data. The framework for digital historiography described by Sternfeld (2010) builds on three archival processes—selection (or appraisal), search, and metadata—all of which require the role of a digital archivist. Digital historians will utilize one of the multiple forms of digital historical representations (digital archives, virtual museums, etc.) to obtain primary data and sources (prepared by the digital archivists) for analysis. Level 2 curation:

> *Research process coupled with data archiving and Web presentation → Level 2 curation → Publication process → Library/peers/public/industry (feedback) → Research process (starting over again).*

Level 3 represents a more completely developed curation process outside the research and publication contexts. In the previous (Level 2) model, the archivist worked in close relationship with the research process, but in Level 3, the curation process becomes a separate process preceding the research process wherein the curator works directly with primary data and data repository functions. The repository does not only contain primary (raw) data but the secondary (derived) data as well, and leverages metadata for greater interoperability between various data types. Because research in the humanities (more so than in history) can build knowledge on interpretations—not just the primary sources and data—the additional loop can be equally valuable to humanities researchers. For historians, however, primary data remain essential for accurate historiography and historical representation. Level 3 curation:

> *Curation process → Research process coupled with data archiving and Web presentation → Level 3 curation → Publication process → Library/peers/public/industry (feedback) → Research process (starting over again).*

The eScience report presents a useful model for a comprehensive metadata framework for digital history at each curation level. At Level 1, researchers access primary sources and/or data from archives and digital repositories using the Encoded Archival Description and Dublin Core, respectively. The resulting publications go into periodicals and monographs cataloged using MARC (US and Canada), UNIMARC (Europe), and similar schemata elsewhere. Crosswalking between Dublin Core and MARC, and EAD and MARC has advantages where digital collections and finding aids have no corresponding MARC records in institutional catalogs; otherwise, there must be an agreement regarding metadata practices in order to avoid duplicate records.

Level 2 accommodates additional metadata schemata, given the presence of a data archivist who catalogs primary data to support searching, discovery, and retrieval of primary data. For digital humanities and history, TEI data in databases may support analysis in ways traditional textual documents do not. Where there are hypertextual relationships between digitally analyzed texts and nontextual digital contents, crosswalking between Dublin Core and TEI may be desirable and practical, especially if there are shared controlled vocabularies in place. Finally, Level 3 adds the digital curator and another level of data curation process, but the metadata crosswalking possibilities are similar to Level 2. Researchers at archives and libraries should be able to use a single resource, such as the library catalog, to access institutional data unless the support for the integration of institutional repositories into normal channels of information access (such as discovery layers) is absent. The benefits of such an integration are clear in a scenario where researchers interested in locating a particular faculty author's scholarly output are able to access that information from a single interface—a strategy that should gain unanimous support from stakeholders. Such decisions, however, are just as political as they are logistical.

Digital historiography and digital curation

The framework that Lord and Macdonald (2003) have introduced can work for curating historical data. Historians will continue to use archives and manuscript collections, publish in peer-reviewed journals, and their scholarship will reach various outlets including archives and manuscript collections (typically at the researchers' home institutions).

In his analysis, Sternfeld focuses on three archival processes—selection, search, and metadata—as crucial for the contextualization of raw data, which highlight the differences between just a document and the record that is inseparable from the historical, social, political, and cultural contexts leading to its creation. This connection with the context—as documented in the provenance field of digital libraries—is what validates sources for authenticity and evidence in historians' work. One of digital curation's principal efforts in this relationship is to preserve the record as well as provenancial properties coded in various forms of metadata such as descriptive, administrative, technical, preservation, and copyright metadata. Elsewhere, Sternfeld (2010) identifies search and metadata as building blocks of digital historiography, but metadata is essential for both information retrieval and information organization.

Sternfeld (2010) presents a theory for digital historiography closely associated with three areas of archival theory—selection (appraisal), search, and metadata—related to evidence. It is ideal for this type of data curation, as it helps in determining what materials to select, preserve, and curate through various communication channels. Because the information in the metadata record supports the contextualization of information, the availability of such information, according to Thibodeau and Duranti, distinguishes records from documents (see Sternfeld, 2010). The preservation of provenancial properties of such records is particularly important because the data and information in the metadata record bring digital content into a contextual relationship with the record-creating organization or person. The ensuing discussion considers Sternfeld's theoretical framework but draws from other areas of expertise.

Selection

The practice of selection involves a complex maze of professional, ethical, sometimes personal, but often political perspectives at individual and collective levels. In archival context, selection constitutes a significant part of appraisal, which—as Cox (2000) writes—"has been defined through a delineation of values such as evidential and informational, as well as through the development of techniques such as sampling and institutional collection analysis" (p. 215). The collecting focus of a special collections unit, usually aligned with institutional curricula, may determine what collections archives can or cannot accept or solicit. The notion that such collections will contain the universe of knowledge on the subject may be far-fetched therefore. Cox addresses appraisal theory with 12 principles with varying relevance to selection in digital historical representation. In summary, these principles address the long-term systematic preservation of selected records based on a common set of criteria or standard in order to prevent haphazard accumulation or reduction of records, which affects the repository, record creators, and possibly society due to inadvertent loss, destruction, misplacement, or accumulation of records. It is also important to consider records to be an organic part of the record-creating institution, which underscores contextualization. The increase in the volume of nontextual—that is, photographic and audiovisual—records also presents new challenges to appraisal including the trend that not all materials (such as popular culture collections) arrive at the archives. Personal digital archives (Hawkins, 2013; Lee, 2011a) are more common today on social networking sites than they are in archives. Today, this is a significant preservation challenge due to the obsolescence of old media, the deteriorating condition of older media devices, and the relative instability of digital media and technology. Finally, Cox (2000) addresses the new role of archivists who are more actively involved in the selection process and the documentation of underdocumented and underrepresented communities and individuals.

Selection in digital historiography is an extension of archival appraisal: archivists and historians must consider the long-term historical value of the records digitized and made accessible in a digital historical representation. There is an interpretive and subjective (or perhaps intersubjective) dimension in selection, which—as Sternfeld (2011) points out—requires interpretation, placing both archivists and historians in

the role of the interpreter, mediator, or facilitator between knowledge and researchers. This is a dramatic shift considering earlier focus on objectivity among American archivists (Cox, 2000). As curators, archivists at some institutions have frequently planned exhibitions to display and promote special collections. They have developed expertise on various topics through primary research and interpreted each exhibit (a record) on the merit of its relevance to the exhibition theme. Thematic research collections and virtual exhibitions demonstrate this role with archivists selecting the materials. In both cases, as Sternfeld writes in reference to *The Valley of the Shadow*, "we see selection criteria enforced at every level of the site…We can also begin to see how an assessment of a digital historical representation's selection can lead to larger questions about the representativeness of historical content, which ultimately point to the archival matter of trust. A representation is only as effective as the extent to which the user trusts that the creators have exhaustively considered available historical content" (p. 556).

The selection process and criteria affect the curation process and the digital content lifecycle. The volume of selected contents will have a direct effect on the volume of contents preserved throughout the cycle. In the trilevel curation model of Lord and Macdonald (2003), the selection process pervades all levels. Authors select and evaluate their primary and secondary sources for their scholarly publications. Journal and book publishers accept, review, and select submissions on the merit of their scholarly potential. Finally, archives appraise collections while libraries select and acquire publications with the help of published professional reviews. Selection is a cyclical process, which iteratively intersects with the curation process.

Search

The design behind intuitive search interfaces is well outside the scope of archival theory and digital historiography but is an appropriate concern of information architecture (Rosenfeld & Morville, 2002) interested in the navigation, search, and labeling systems of Web sites, information portals, digital libraries, catalogs, and other user interfaces. Despite the natural association of search interfaces with computer science, algorithms, and artificial intelligence, some systems rely heavily on ontologies, taxonomies, and controlled vocabulary lists. This is where such archival skills as indexing and cataloging join computer science. Historiography and archival science, however, serve as conceptual frameworks for evaluating a user interface's ability to present data and records (with meaningful context), suggest related topics, or provide users with other guides for historical analysis. Sternfeld (2011) compares various search systems using formal lists (for guided searches), tag clouds (indicating the frequency of cited terms), and user tags in order to demonstrate how search interfaces assist in the contextualization of retrieved information.

Outside the digital history and archival communities, there have been debates around formal professionally developed classification systems (ontologies) versus social tagging approaches (folksonomies). The line of opposition between those preferring ontologies and taxonomies versus those preferring social tagging is clear since proponents of formal classification view social tags as unsystematic, subjective, and

open-ended (Peterson, 2006) while those in the latter camp view formal classification systems and ontologies as rigid and counterintuitive. Lee and Neal (2010) point out that subjectivity in social tagging impedes information retrieval since—compared to professional indexes and taxonomies—there is no central oversight behind folksonomies, which leads to isolation among similar and potentially related contents. Social tags such as "awesome" and "decay" in reference to images of neglected historical structures circulating in the Instagram user community further illustrate the inadequacies of social tagging. A counterargument in support of social tagging comes from Avery (2010) who connects social tagging with collaborative indexing and group intelligence, which underscores the value behind collaborative (and distributed) curation practices common to social networking.

Information architects regard formal classification systems as "top-down" whereas social tagging (combined with database) approaches as "bottom-up" organization structures (Rosenfeld & Morville, 2002), but from a curator's vantage point, there may be a benefit to considering both approaches. Classification systems identify established (but often rigid) disciplinary domains that are going through interdisciplinary changes while social tags may represent emerging trends, subcultural or highly localized concepts, and other community (local) knowledge that merit analysis and evaluation before indexers select and integrate them into formal ontologies. Ding et al. (2010) indicate that their results strongly favor a structural (i.e., ontological) rather than semantic organizational (i.e., tag-based) paradigm, but proposed the use of upper tag ontologies (UTO) to integrate social ontologies, social tagging, and structural organizational methods. "While other ontologies developed to represent folksonomies tend to focus on the meanings of tags, UTO is designed to capture the structure of social tagging behavior" (Ding et al., 2010, p. 509). Such a hybrid approach may provide extensive support for digital historiography since such sites would benefit from a combination of full-text searching, guided searching, and browsing. Given the interdisciplinary scope of digital historiography, historians and archivists may consider several taxonomy approaches such as lists, trees, hierarchies, polyhierarchies, and matrices (Lambe, 2007) to accommodate complex conceptual frameworks.

Metadata

At the most fundamental level, metadata is data about data; in the present context defined by archiving and digital historical representation, metadata describes the digital content and the information needed for contextualization. Thus, an archival record's trustworthiness directly translates into the reliability of the digital content and the metadata that describes it (Sternfeld, 2011). Skepticism about digital material among researchers favoring print format rests on the widespread impression that just about anyone can publish anything on the Internet, and when those files turn up in Internet searches, they typically appear decontextualized with no access to the standard metadata to inform researchers of the context. Contextualization does not merely place a photograph or historical text into historical context; archived items have provenancial properties informing researchers about the history of ownership, spatial and temporal coverage, and subjects representing formal ontologies.

The framework of digital historiography brings historical scholarship, digital preservation, and curation into direct contact. Taking the historiographical perspective, Sternfeld (2011) writes,

> *In developing a set of evaluative criteria, we must consider the association between a representation's form and content, which together comprise the representation's overall historical argument. While there are numerous components worthy of consideration, two in particular—search and metadata—determine to a large extent how a representation organizes its historical information. Without a robust search engine the user cannot access historical data; similarly, without quality metadata, a strong search engine is rendered ineffective.*
>
> *p. 2*

The relationship of metadata to search represents an entirely different, albeit related, set of concerns among archivists and librarians. As seen under Search, the use of subject keywords, taxonomies, ontologies, and social tagging aims to represent historical information and knowledge. The information provided or discovered here appears in the subject field, which usually corresponds to the subject headings. Digital historiography, therefore, relies extensively on historical knowledge representation.

Metadata is also a requirement at all three levels of data curation, as historians and humanists will search for primary sources, scholarly publications, and quantitative data. Lord and Macdonald (2003) list metadata as one of the several strategically important components of data curation: metadata quality, authenticity, and provenance ensure trust in digital repositories and the curation program. An internationally accepted metadata schema is the Qualified Dublin Core, which complies with the Open Archives Initiative Protocol for Metadata Harvesting (OAI-PMH) for improved interoperability as well as ISO standard requirements. Considering the extent of the eScience data curation framework (Lord & Macdonald, 2003), crosswalking between Dublin Core and various metadata standards (MARC, EAD, and TEI most notably for digital historical representations) requires collaboration among archivists, humanists, catalogers, and technology staff configuring the digital repository. For bibliographic management software such as EndNote, there must also be support for the Z39.50 protocol, which enables researchers to download metadata from library catalogs and research databases.

Conclusion

The role of archives and archivists is crucial in the digital history landscape. While the traditional relationship between the discipline and archival practice will not transform into all-digital-no-paper models, digital history will develop new expectations that digital curation will meet. As the volume of digital—digitized and born-digital—material arriving at archives increases and as technology shapes new historiographical practices, the emphasis on digital technology will sharply increase. Metadata as a building block for digital historiography will not only serve database users but will also provide

the critical link between collection and context in such fields as date, location, creator, and provenance. Archivists will continue to transform their role in this environment by combining and transforming traditional roles where they attain more visibility in the community while still focusing on preserving the historical record. Finally, digital curation processes will involve archivists and facilities at all levels of curation.

Information architecture and hypertextuality: concerns for digital curation

4

Information architecture, hypertextuality, and digital curation are related concepts of interest to archivists preserving and curating complex Web sites, text-encoded (TEI) projects, EAD finding aids, and other networked hypertextual collections of cultural heritage. Although information architecture is not a substitute for curation or archiving, the field can effectively inform curators about the internal and extended structures of Web content requiring preservation and documentation. Unlike stand-alone digital files, multiple networked files contain one or more hyperlinks to other interrelated digital pages. Such networks of such hyperlinked documents have comprised the World Wide Web, but the fragility of files and the links among them have significant implications for historians producing historical hypertext and curators preserving the integrity of information. There is also a direct relationship between information architecture and hypertextuality, which represent several forms of hypertext in digital history projects. Understanding these forms will help curators assess the structure of large-scale projects and prepare proper documentation. Information architecture, therefore, may interest curators in assessing the structure of digital files, multipage Web sites, and even entire systems of information architectures. Digital curation spans several levels of representation for more complex file structures, aggregations, and networks (Lee, 2011b) where understanding information architecture on various scales can benefit digital curators.

The importance of information architecture to some areas of archival practice had been evident prior to the arrival of the Web and current digital technologies. The Working Group on Standards for Archival Description (1989) addressed the importance of information architecture in standardizing description for finding aids, which serve as a map to the described collections. Through finding aids, archivists present the physical and logical arrangements of collection contents as well as the relationships of such contents in order to preserve the historical context according to provenance or original order. Very large collections usually end up getting very long finding aids, which will benefit from having anchors and links throughout the document. This is information architecture at a basic level but digital (EAD) finding aids usually involve much more intricate structures discussed later in this chapter.

Weissman (1994) predicted that in comparison to computing technologies (such as mainframes) available to archives in the early 1990s, the model for information processing would be more object-oriented and document-centered in 2000 with improved user interfaces, software architectures, and information retrieval and collaboration tools. In 1994, he wrote,

Digital Curation in the Digital Humanities. http://dx.doi.org/10.1016/B978-0-08-100143-1.00004-0
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> *Tomorrow's architecture will be fully client/server based…[and these] personal devices will be networked—sometimes physically and sometimes via wireless technology—to powerful servers containing trillions of characters and millions of pages of information stored locally. These servers will create a distributed information architecture throughout an organization, linked by high-speed networks.*
>
> *p. 21*

These predictions were quite accurate since they had forecasted the broader application of hypertext structures over an intranet. There is no reference to cloud-based storage yet, which has since emerged as storage for many consortial and institutional repositories. Archives and libraries had explored and adopted digital technologies to develop digital collections sometime before the World Wide Web in 1994 however. For instance, the HyperCard and CD-ROM projects of the 1980s featured digital versions of classical texts, encyclopedias, and other humanities text corpora, and underscored the suitability of hypertextual adaptation of historical texts despite skepticism about the reliability and authenticity of such sources for purposes of humanities or historical research.

More recent literature addresses the value of information architecture in archives (Yakel & Kim, 2003; see also Bbmarchives2011, 2011) and libraries (see Batley, 2007; Duncan & Holliday, 2008), which allows these institutions to organize, present, and provide long-term access to their archives and special collections. Archives and libraries have also participated in developing digital material for thematic research collections, which were essentially Web "archives" of classical knowledge. Information architecture played a vital role in planning these Web archives as well as designing the sites for intuitive navigation and visitor experience. Their preservation and curation will require understanding their unique information architectures, which evolved independently for different collections at different institutions.

This chapter focuses on the place of information architecture in digital curation where it helps curators assess the diverse structures of information spaces and the relationships of interrelated resources. The chapter will not delve into the technical details of specific curation tools (such as BitCurator) or digital repositories (e.g., DSpace, CONTENTdm, Omeka, Joomla). *From Bitstreams to Heritage* by Lee, Woods, Kirschenbaum, and Chassanoff (2013) provides details of working with BitCurator while technical information on the repositories is available on the respective developers' (i.e., DuraSpace, OCLC, CHNM, and Joomla) Web sites. The chapter will discuss information architectures and the hypertextual relationships between two or more resources with navigation scenarios in mind. Current and recent work at the Ward M. Canaday Center for Special Collections at the University of Toledo illustrates the structural, textual, ontological, spatial, temporal, and thematic relationships of archival information resources.

Defining information architecture

Rosenfeld and Morville (2002) present several definitions for information architecture, all of which are important from the vantage point of digital curation:

1. "The combination of organization, labeling, and navigation schemes within an information system";

2. "The structural design of an information space to facilitate task completion and intuitive access to content";
3. "The art and science of structuring and classifying web sites and intranets to help people find and manage information"; and
4. "An emerging discipline and community of practice focused on bringing principles of design and architecture to the digital landscape" (p. 4).

While most—if not all—definitions may well apply to designing good virtual exhibitions and other historical hypertext content, the application of these concepts to digital curation will add value to the curation and informed documentation of Web contents, much of which may fall well beyond the control of curators.

As community and other collaborative curation practices reach beyond the archives—as it is with the use of social media and other Web 2.0 platforms preferred by digital humanists, historians, and personal archives—the expansion of the old information space into a system or network thereof and the proliferation of digital content within this expanding space present new opportunities and challenges for curators. In an editorial of the *Journal of Information Architecture*, Morville (2011) writes, "A system of information architecture [speaks to a] design for experiences across channels. Mobiles and social are just the beginning. Our future-friendly, cross-channel information architectures need to address the full spectrum of platforms, devices, and media…it's no longer okay to build a website and then talk about mobile, social, and search engine optimization. These systems are all intertwingled" (p. 3). The digital curation of content in such "intertwingled" systems will greatly benefit from information architecture skills, if not solely for purposes of designing but for documentation also.

Arango (2011) takes a social-structural approach to information architecture, which considers how physical buildings frame daily interactions among people. He views digital spaces as "container…of our day-to-day interactions with our institutions and with other people" (41), and compares information architecture to buildings as settings for social interactions, turning information architecture into a metaphor for social and textual spaces for knowledge creation. Metaphorical associations of information architecture can significantly help Web users relate to physical places such as car dealerships or hardware stores (Rosenfeld & Morville, 2002); similarly, virtual exhibitions can evoke similar experiences with exhibition spaces, such as galleries or libraries, where visitors interact directly with digital texts and media. In digital context, this translates into leveraging information architecture and hypertextuality in a framework to support learning in virtual spaces with hopes of inspiring new ideas and research projects. After all, Arango's aforementioned building can be a metaphor for an archive, library, museum, or an entire institution functioning as the framework for interaction among people who like to use digital technology.

Digital curation of hypertextual content

While information architecture is not archiving or digital curation, it offers significant advantages to digital curators engaged in creating and preserving hypertext collections on the Web. Digital curators have built, or at least collaborated with

historians on building, historical hypertext sites. There is an information architecture in those historical hypertext projects, which holds the site and its content together via hyperlink relationships and enables users to navigate and search the site to locate historical information. This architecture must remain intact in order for the site to remain functional throughout its entire curation lifecycle; however, the constantly changing nature of the Web and emerging digital technologies present challenges—in addition to the opportunities—for curators. Web sites appear and disappear, rendering hyperlinks on those sites useless, which affects the readability (and therefore the usability) of the site.

A sad reminder by Cohen and Rosenzweig (2006)—that even such valuable Web sites as *My History is America's History* can fall victim to economic, political, and cultural forces—reminds curators to face the ultimate demise of cultural heritage collections and plan for preserving such collections with or without the interactivity of the content. For example, creating and saving high-quality screenshots of digital history Web sites as PDF/A (an ISO-created PDF file format for archiving purposes) files will extend the life of the information well past the usual lifecycle of most Web sites because all information is self-contained in this format.

Curation should include the practice of documenting the Web site among historians and curators, and such documentation should be present in a metadata record separate from the site as well as—as Cohen and Rosenzweig (2006) indicate—embedded in the HTML code of the Web site as `<!–comment-->`; therefore, they are not visible on the Web site. Likewise, a Web page's header provides additional space to place metadata for additional document as well as improved discovery. Such tags as `<meta name, title>` and others also provide context and other related information to curators' successors. Code-embedded annotations with dates, initials, instructions, and helpful details will help curators identify the resources and linking sites that may have been available at some time but no longer are. Additionally, the use of ALT or TITLE tags, which cause pop-up texts to open upon hovering a mouse over links, will also inform curators and users about the linked site.

The Internet Archive has become a popular and reputable place for archiving various hypertext documents, but there are widespread instances of continual deterioration—especially in the case of link relationships to external sites. Ximm (2013) discusses the use of the Wayback Machine to archive over 150 billion Web pages. The growth rate 50 TB per week for additional billions of pages comes with challenges to archiving. "The Machine uniquely affords users the opportunity to browse the web backward through time [with the help of calendars], providing a reasonably representative, if admittedly incomplete, view of otherwise vanished digital heritage" (Ximm, 2013, p. 191). Archiving entire Web sites such as virtual exhibitions requires using internal, relative, static links across the entire site, but the links to contents external to the exhibition will eventually break if those external resources migrate, cease to exist, or if the domain is no longer accessible to Web crawling by the Wayback Machine. Hence, the documentation of all related external sites and the hypertextual relationships is necessary, and some knowledge of information architecture will provide benefits in this process.

Lee (2011, 2012) presents a framework with several levels of representation with each corresponding to a specific curation activity around digital objects or information artifacts.

In this scheme, each layer builds on the one below it and represents a progressively growing microarchitecture. Objects on each level have unique information architectures but only a few (at the top, with more complex architectures) are meaningful to human users. While interactions taking place at levels 1 and 2 (bitstreams, subfile data structures) are physical, those at levels 3 and above are increasingly more intellectual since users can derive meanings from files. Files such as finding aids at these levels can be stand-alone (axial) or networked (Landow, 1994) with their own intrinsic information architectures. Levels 6 and 7 represent complex objects in Web sites, databases, and institutional repositories where curation activities must take the entire Web site or the platform into account. While the first decade of the Web relied on manually coded and created HTML pages, the subsequent developments have led to Web 2.0 platforms without self-identifiable pages. In some cases, the backup procedure includes the entire platform. With respect to the former model Lee (2012) writes,

> *The rendering of a web page in a browser, for example, does not reflect any comments that are within the text of the HTML file. Viewing the HTML file through a text editor would reveal the comments (and other properties of the HTML markup such as whitespace and style sheet references), but it would not reveal specifically how the page is presented to users who visit the site through a browser. The developer of a web site who is trying to fix a "broken" page will routinely shift between these two levels of representation, in order to see specifically how the HTML code is expressed and how changes to the code affect the appearance and behavior of the page in one or more web browsers.*
>
> *p. 516*

The role of information architecture in digital curation is not merely for design purposes but also for preservational and curational ones. Annotations added to the code may have important information for curators developing structural metadata, which will help in transferring contents and representations in space and time.

Information architecture and hypertextuality

Information architecture and hypertextuality are closely related concepts: the former presents a tangible structure to dispersed and electronically interlinked texts—or cybertext (Aarseth, 1997)—while the latter establishes narrative fluidity between those interlinked pages and texts. This fluidity requires writing skills, however; an information architect's contribution is to represent the narrative through the design of the information space occupied by hypertext. As discussed earlier in this chapter, Rosenfeld and Morville (2002) define information architecture as the "structural design of an information space" (p. 4). Hypertextuality refers to the dispersal, decentralization, delinearization, and de-sequentialization of interrelated texts allowing readers to define their own paths (Aarseth, 1994, 1997; Bolter, 1991; Bush, 1945; Landow, 1992, 1994, 2006; Nelson, 1992). Architectures of networked and interlinked information spaces present a structural framework for hypertextuality and may span over a single device (such as a server), an intranet of an institution (on several network drives),

or the Web of networked computers across the globe (on several independent servers). Landow (1994) addresses some fundamental differences in axial (stand-alone) and networked textual structures where the former emulates linear text such as a book or other print document (such as finding aids) and the latter exists in a hypertext network with links to other decentralized texts.

Aarseth (1997) addresses the place of cybertext in what he calls ergodic literature. He writes, "The concept of cybertext focuses on the mechanical organization of the text, by positing the intricacies of the medium as an integral part of the literary exchange" (p. 1). The etymological basis of "ergodic" (a union of Greek works *ergon* and *hodos* for work and path, respectively) signifies what Aarseth describes as nontrivial efforts to reading with "extranoematic responsibilities placed on the reader" (p. 2), which requires considerably more effort than eye movement and automatic page turning as routines in reading print materials. In this context, cybertext is a metaphorical labyrinth that readers must make considerable effort to navigate as though in a computer game placing readers at the center of the literary experience. For purposes of this book, cybertext presents the framework for the intellectual experience associated with scholarly research and the construction of new knowledge resulting from research using cybertext. Information architecture, then, provides the structure of the information space within which such sophisticated reading of ergodic literature can take place, but also shares that "extranoematic responsibility" by providing a design that enables readers to occupy the center of their reading experience.

Elsewhere, Landow (2006) presents hypertext theory, which underscores the textual relationship between information architecture and digital curation. A highly developed hypertext environment in archives can connect finding aids to special collections, Web sites, virtual exhibitions, and digital libraries with digitized historical records, and social networking sites with discussions on the topic. Textual connections enable researchers to navigate the Web of links between textual and nontext content. Landow identified several forms of linking: *lexia* to *lexia* (unidirectional); *lexia* to *lexia* (bidirectional); string to *lexia* (unidirectional); string to *lexia* (bidirectional); string to string (unidirectional); string to string (bidirectional); one-to-many; many-to-one; and typed links (hyperlinked words or phrases). This node–link system of relationships between textual units appears in social networks (as addressed in a later chapter in this book) where groups of people produce interrelated texts.

Information architecture in finding aids

Finding aids are essential tools to facilitate the discovery of information within archival records and manuscript collection (Pearce-Moses, Ed., 2012) and are vital to the lifecycle of records and curation in archival practice. The Working Group on Standards for Archival Description (1989) addressed the importance of information architecture in the construction of archival finding aids among the 17 recommendations it made. Recommendation 13 treats information architecture as "a model identifying the sources and users of information; the processes by which it is collected and used; and the structures within which it resides" (p. 473). Viewing the finding aids as a part of larger information systems underscores the more recent definition for systems of

information architecture by Morville (2011), both of which require a systems approach to constructing and placing finding aids.

The internal design of finding aids communicates two critical aspects about the described records: first, it serves as a map to the processed collection; and second, it represents the work of the record creating organization as well as its structure, thus supporting the documentation of provenance at departmental levels, not just institutional. Large and complex collections will require lengthy and complex finding aids with extensive information in its collection summary, biographical/historical overview, scope and content notes, and series and folder listings. For instance, a collection 200 linear feet (more or less 200 Hollinger boxes) in size may take up as many pages in the finding aid. In contrast, a small manuscript collection containing a scrapbook or just a handful of items may be as short as two pages. Thus, there is not necessarily a close relationship between a collection and organizational size and structure. A blog posting in the Bebe Miller Archives (2011) explains that finding aids can "let a researcher understand a lot about company organization and logic, how they categorize information." Thus, finding aids present a unique architecture reflecting the internal logic of the collections and the ontological framework of the information prior to the transfer of described collections to the archives for preservation and curation.

In hypertextual terms (Landow, 1994), finding aids fall into two distinct groups: stand-alone (or axial) and networked. Print and legacy finding aids without hyperlinks fall into the former category whereas new HTML, PDF, and EAD finding aids feature internal navigational links as well as external links to subject guides and related finding aids. Figure 4.1 illustrates the structural differences between legacy (stand-alone) and digital (networked) finding aids. In both types of finding aids, information architecture improves navigation within the hierarchical structure of long and complex documents and preserves the relationship of information on different levels. In the case of EAD finding aids, more hyperlinks allow connecting to external documents, collections, and other EAD finding aids in the same repository, usually via subject links (not shown here).

EAD finding aids in repositories and discovery systems (such as ArchiveGrid) present yet another advantage as access points to finding aids of archival institutions using ArchiveGrid. The ArchiveSpace project builds on the Archivist Toolkit and Archon platforms for the management of digital collections and finding aids at institutions using ArchiveSpace. Finding aids demonstrate a two-way relationship between information architecture and hypertextuality: Hyperlinks may bring archival records and manuscript collections into relationships via hyperlinks, but in such cases, information architecture provides the necessary structural foundation curators and researchers to identify such relationships and discover more. The benefit to converting legacy finding aids in the stand-alone category into networked ones (HTML, EAD, new PDFs) is that researchers can focus on reading through the most relevant collections and records identified through preliminary research.

Hypertextuality is ubiquitous in EAD finding aids as the SGML (Structured Generalized Markup Language; precursor to the XML) schema presents extensive navigational paths within and beyond the digital finding aid, most specifically to related digital collections, but also to finding aids via subject, location, date, and other

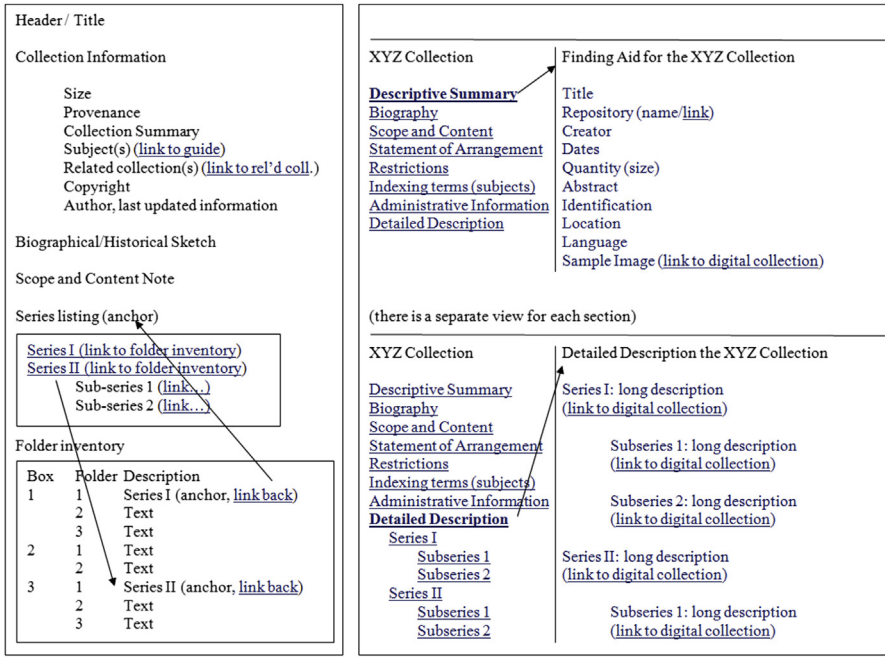


Figure 4.1 Information architecture and hypertextuality in legacy (stand-alone, left) and EAD (networked, right) finding aids.

authority field links. Ruth (2001) traces the development of EAD finding aids from the birth of the MARC schema in 1966 through the birth of the SGML as part of the Bentley Finding Aid Project in 1987, first Gopher (ASCII-text-based) finding aids in the early 1990s, to the release of Encoded Archival Description Document Type Definition in 1998. In reference to the improvements from the nonsearchable early versions toward the fully navigable and searchable EAD, Ruth emphasized that while technological advances have led to the development of the EAD, the new hypertextual standard directly speaks to archivists' "longtime interests in access, descriptive standards, and information storage and retrieval" (pp. 27–59). EAD finding aids are highly structured to represent the structure of manuscript collections and archival records described within. Some frame-based finding aid repositories provide easy navigation throughout the finding aids and the descriptions at record-group, series, subseries, folder, and item levels. The General International Standard Archival Description, or ISAD(G), has prescribed these levels for purposes of representing relationships and contexts for records. Haworth (2001) points out that another important advantage of this multilevel description approach is that information on the higher levels does not repeat at lower levels, which is also a principal in information architecture to achieve clarity. Figure 4.2 illustrates nonprescriptive linking relationships among digital collections in an archival repository, which outlines a typical hypertextual and information architecture network in archival practice. Bidirectional arrows indicate two-way linking relationships where supported whereas unidirectional links identify one-way

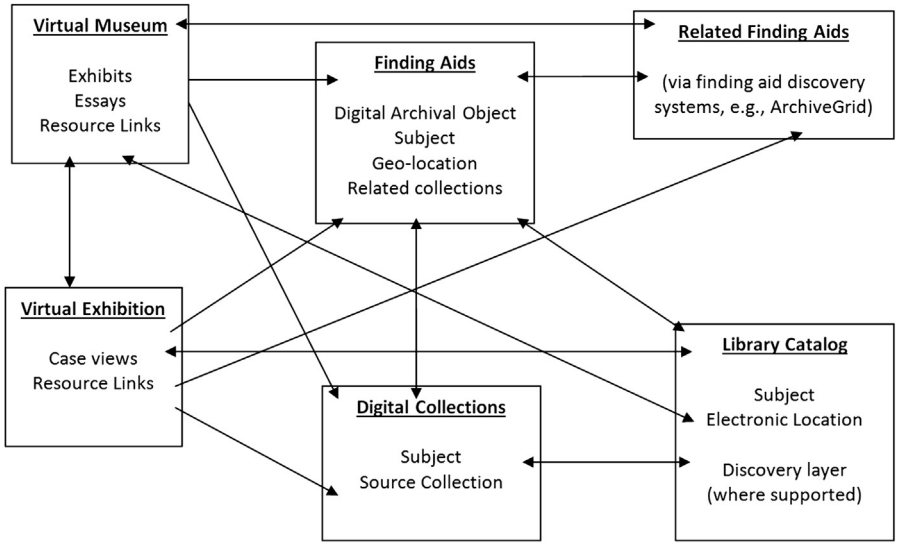


Figure 4.2 A system of information architecture and hypertextuality.

linking. The textual relationships are important where two related resources (historical texts in digital and print collections) appear in finding aids, catalogs, virtual exhibitions, or digital collections. Extending the information architecture, therefore, builds a bridge between digital and nondigital (but digitally described) collections with both finding aids and catalogs playing varying roles in digital curation.

Virtual exhibitions

Exhibitions fall within the scope of curation and outreach; virtual exhibitions do the same but benefit from Web design skills in addition to the core archival competencies. Effective virtual exhibitions not only improve the virtual visitors' experiences with virtual exhibitions but also represent the theme, context, and narrative behind the original exhibition through information architecture and other Web-specific coding. Furthermore, virtual exhibitions with highly developed information architectures can facilitate navigating across multiple domains housing digital finding aids, digital repositories, virtual museums, library catalogs, and other sites with their own architectures in place.

Dallas (2007) discusses spatialized knowledge in museum exhibitions through the arrangement of exhibits and related information as "mapping of conceptual relationships underlying the intrinsic and extrinsic properties of artefacts (e.g., a typological order, a historic sequence) onto exhibit arrangements in gallery space" ("Curatorial Agency in Museum Practice," para. 6). Spatializing knowledge in virtual museums and exhibitions is equally effective, and information architecture significantly contributes to the value of some curatorial work.

Information architecture is vital to the navigation and thematic coherence virtual exhibitions. It builds on metaphors—or mental models—that are vital as well as

meaningful to planning Web sites. Rosenfeld and Morville (2002) view buildings as metaphors for "the complex, multidimensional nature of information spaces. Like buildings, web sites—that is, information spaces—have architectures that cause us to react" (3). Arango (2011) has also used buildings as a metaphor for relationships between physical and virtual structures. Likewise, museum, art galleries, or even very small exhibition spaces can serve as metaphors for virtual exhibitions that can emulate the exhibition environment and evoke visitors' experiences with exhibits. In both cases—of physical and virtual exhibitions—the architecture of the building and the Web site can enhance visitors' ability to navigate the exhibition space. They can follow the narrative as published in well-researched exhibition catalogs or proceed by skipping some exhibits, returning to them later, or switching orders. Virtual exhibitions can enable visitors to follow the narrative as presented in the exhibition catalog or use hyperlinks to divert, return, and change the direction in which they are viewing the exhibits or access related resources. Virtual exhibitions can also present interactive media, such as panoramas with embedded links, in the pages as additional navigational devices. A combination of these approaches can help virtual exhibits emulate the physical exhibition. In this respect, virtual exhibitions present some advantage over physical ones in that they are remotely accessible and can also lead visitors to digital resources in the library such as digital finding aids, digital collections, virtual museums, and other information. Experiences with real exhibitions dramatically differ from those with Web sites; hence, the designers of virtual exhibition can close the experiential gap with other features. Exhibitions featuring three-dimensional artifacts (as in art, science, natural history) are different from those with mainly two-dimensional exhibits such as photographs, textual material, and moving images. There are no prescriptive rules for developing virtual exhibitions around archival resources, and no expectations for metaphorical relationships between the physical exhibition space and the virtual exhibition.

Compared to lecture series, presentations, and publications, exhibitions represent a different approach in archival outreach: archivists can select a theme based on multiple manuscript collections and archival records and organize an exhibition of unique primary source materials that document the history behind the theme. While the range of topics may be limited to the archives' collecting scope, combining the topics from multiple collections may leave outreach archivist and curators with few limits on creativity. Several archives may collaborate and contribute collections toward very specialized topics. Archival exhibitions usually feature correspondence, photography, films, diaries, organizational records, press releases, ephemera, and three-dimensional artifacts in the featured collections. Virtual exhibitions usually succeed the physical ones after they end, but they require a laborious process involving the scanning, photographing, and planning the Web site. This is where information architecture starts to play a more important and visible role.

The information architecture of a virtual exhibition takes the following elements into consideration: layout of the original (physical) exhibition; the narrative in the exhibition catalog; metadata (subjects, source collections, dates, spatial coverage, etc.) in the digital library and library catalog; links to the archival finding aids; links to media; and links to related social media for curation efforts. The structure of virtual

exhibition often represents the order of exhibits in the physical structure in order to reinforce the narrative it follows. The exhibit-to-exhibit design represents the order of exhibits in a gallery as well as the structure of the narrative in the exhibition catalog. Each exhibit case view in the virtual exhibition provides information on items displayed in the exhibit cases, which includes exhibit labels—helpful sources of descriptive metadata (location, creator, date, source collection, etc.). Visitors can use the back and forward arrows to navigate the site as though they are walking from case to case, but can also change their path through the exhibition space as they like (Landow, 1992, 1994, 2006). Figure 4.3 represents these two scenarios with solid and dashed arrows, respectively, which also demonstrates that viewing behavior in hypertext environments mimics human behavior in physical spaces.

In virtual exhibitions, however, there are additional navigational channels such as ordered lists of cases with descriptions, carousels with thumbnail images representing case themes, interactive panoramas with hotspots linked to corresponding cases, and slideshows linking to specific case views. Taking a template approach to designing the case-level pages ensures the uniformity of layouts across multiple pages. Each case view page, for instance, contains an image of the case in the physical exhibition with hyperlinks to shelf-level views. Each shelf-level view provides some descriptive overview of exhibits on that shelf, along with a link to thumbnails leading to item-level pages. The information on the case view level provides researchers with descriptive information and links to related digital finding aids, digital collections, and catalog records. This approach turns virtual exhibits into access points to the archival and library resources.

Figure 4.4 illustrates an approach where each case view presents viewers with links to images and related information. For example, the case view page with items from specific collections can also provide links to the corresponding finding aids, specific records in the library's collections (with the help of prepared searches on subjects, geographical locations, titles, authors, etc.), specific digital collections (related to subject, spatial and temporal coverage, provenance, creator, etc.), and various Web sites (virtual exhibitions, digital libraries, etc.).

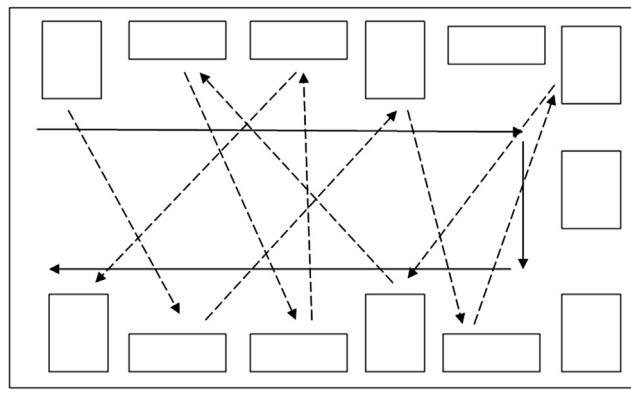


Figure 4.3 Straight and alternate view paths in exhibition areas.

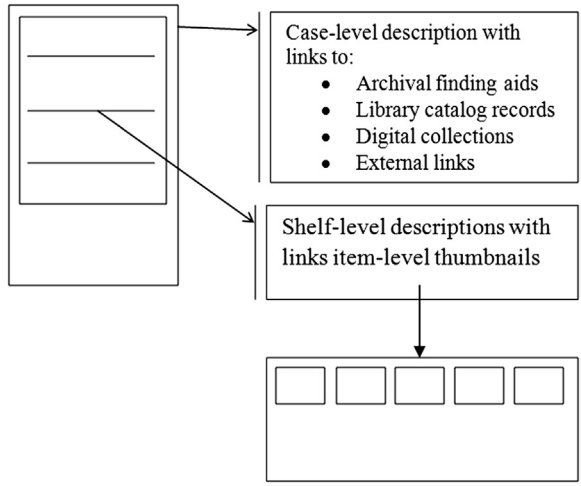


Figure 4.4 Interlinked case-, shelf-, and item-level views.

Linking at this level of granularity, however, requires preparing the target sites. Library catalogs may provide persistent links to search results, but it is necessary for curators to ensure that adequate metadata is available in essential fields such as subject, geographical location, author, and others as needed. If these fields contain no data, curators should work with catalogers to enhance the metadata record. The same approach applies to digital collections where specific indexed metadata fields must contain information to enable searching or browsing. Links to finding aids and external sites are relatively simple, as most have specific Web addresses (URLs) while those in digital (EAD) finding aid repositories have URLs assigned by the system. Alternatively, virtual exhibitions may also provide designated pages to digital collections and finding aids to corresponding manuscript collections. Links to social media channels used for social curation may help visitors engage in (hopefully) meaningful discussions on the exhibit and to generate reference interactions for archivists and librarians. These conveniently placed links from the virtual exhibitions to related resources can initiate the research process, and perhaps this advantage can set virtual and physical exhibitions. Museums and archives wishing to emphasize the social, aesthetic, and intellectual qualities of physical exhibitions may designate separate spaces for such technologies as mobile devices, QR codes, or social software, which may interfere with the museum experience. Others (HASTAC, 2012) may argue, however, that social media and mobile technologies facilitate networking and meaning construction through crowd-sourced curation. Museums and archives may, in fact, designate times for such crowds and benefit from the enriched discourse.

Virtual museums

Virtual museums differ from virtual exhibitions in scope and content, and unlike digital repositories, they have no robust digital preservation and metadata processing

capabilities. Virtual museums may support limited indexing, but do not always peruse controlled vocabularies. Like virtual exhibitions, virtual museums can reside in traditional Web sites for which developers must write the code for full functionality. However, Web 2.0 platforms such as Joomla, Drupal, Omeka, and WordPress continue to offer greater flexibility over the control of the design process and the overall information architecture with customizable navigation, labeling, and search systems. Joomla, Drupal, and WordPress also offer rich libraries to extend the capabilities of the platform, which can include metadata functions similar to those of the digital repositories. A recent addition of Dublin Core metadata handling capability to Joomla, and Drupal's ability to harvest descriptive metadata from DSpace repositories presents archives, libraries, and museums with an opportunity to utilize these platforms for outreach and curation purposes.

The information architecture of these platforms includes full-text searching, labeling, and navigation systems including top, main, and footer navigation as well as breadcrumb trailers. Each content requires creating an article to which curators can create links in the menus, submenus, internal article texts, and external sites. The information architecture for an entire museum requires thorough planning, as menu labels need to be clear and signify mutually exclusive content areas. A virtual museum of local history and cultural heritage may offer separate areas for essay and image collection, virtual tours, interactive media, research resources, social media, external links, and information about the virtual museum. As with the virtual exhibition, links to related resources such as library catalogs, archival finding aids, and digital collections should be available to local historians and citizen researchers interested in community history, genealogy, and interacting with others in social networking sites. Social media significantly contributes to the networked information architecture and intertextuality while enhancing visitors' experiences with the content. Fiske (2001) writes, "[the] theory of intertextuality proposes that any one text is necessarily read in relationship to others and that a range of textual knowledges is brought to bear upon it" (p. 219). In this context, postings on local historical topics in blogs and other social media may contain data, information, and knowledge to inform other postings or even shape current discourse on local history, cultural heritage, memory, and identity. Cook (2013; see also Greene, 2002) presents an overview of the paradigm changes in archival work from the eighteenth century to the present, and places archivists in this discursive milieu.

Figure 4.5 shows a part of a virtual museum's navigation system as it comprises the larger system (or network) of information architecture connecting various archival resources. Here, the navigation scheme not only maintains internal relationships but provides links to other important archival resources such as finding aids, virtual exhibitions, digital collections, library catalogs, and other important resources. The information architecture of the virtual museum serves as a framework for hypertextuality—as well as intertextuality—which enables visitors to access texts and related media within the virtual museum as well as across multiple Web sites, social networks, and domains. Perhaps it is the virtual museum where the role of information architecture to digital curation is more evident than in other resources. Not only does information architecture serve as a framework for spatializing

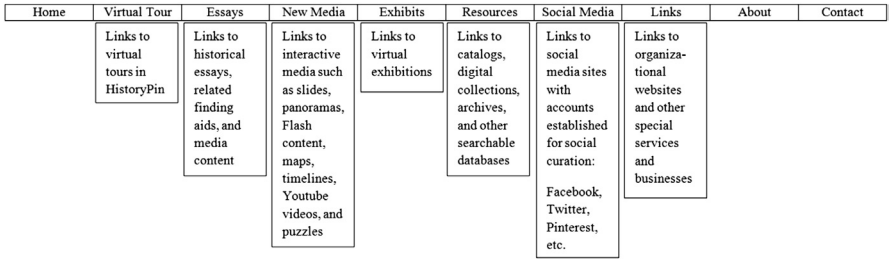


Figure 4.5 Navigation bar and menus in a virtual museum.

knowledge (Dallas, 2007) but it also presents a structure for hypertextuality, and through that, new pathways to reading and constructing new discourse.

Digital libraries

Compared to freely created Web sites such as virtual exhibitions, digital repositories reside on Web 2.0 platforms such as DSpace, CONTENTdm, Omeka, Fedora Commons, and other platforms. Web 2.0 platforms enable curators to focus on developing digital collections without any coding skills, but this obvious advantage comes with the price of losing control over the information architecture at the macro (platform) level, which includes the search, navigation, and labeling systems. However, curators may still make micro- (collection-) level decisions to facilitate or improve navigation within repositories, which varies from one platform to the next. For instance, both Omeka (open-source) and CONTENTdm (commercial) present curators with ways to develop virtual exhibitions, but the platform provides different resources. The primary function of these repositories is to provide access to digital collections and metadata to enable searching and browsing. This section does not cover the internal workings of a digital repository and neither does it engage in comparing platforms; rather, it focuses on strategies to improve access to information about the digital collections.

Most digital repositories allow the upload of digital collections and metadata into the repository with additional indexing, advanced search, commenting, rating, and RSS capabilities. There is a greater variation in the user interfaces and curator's control over the design of the user interface. The arrangement of collections also varies by platform; for instance, DSpace allows organizing the collections into three distinct levels: communities, subcommunities, and collections, which is an intuitive approach because researchers can easily identify archival collections by the manuscript number and title in an organized approach. This is not the case in CONTENTdm: it offers a flattened model of content organization, which may suit most nonarchival needs, but it is somewhat counterintuitive from an archival curation perspective. The platform provides some space for adding a limited number of custom pages, which may be enough for a few additional pages and navigation labels. Ambitious curation approaches involving enhanced information architectures for digital collections, however, will require Web coding skills and creative strategy.

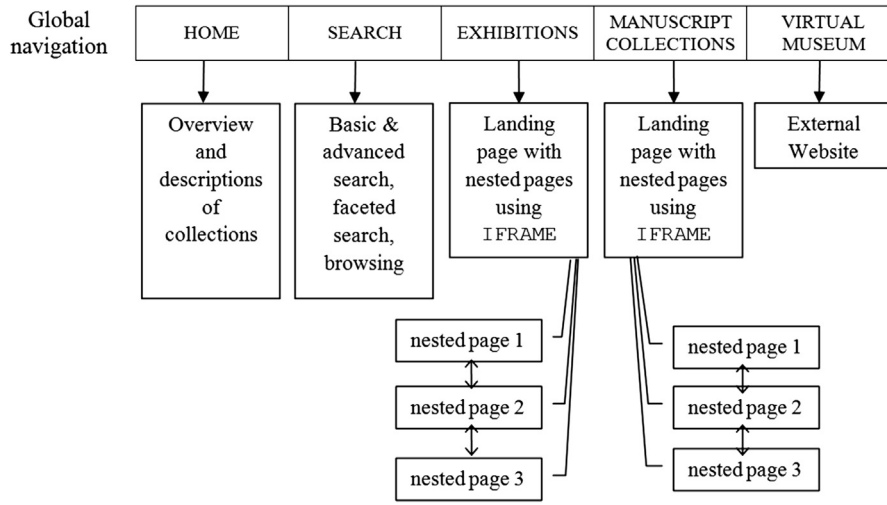


Figure 4.6 Nested information architecture in the digital repository.

All of these repositories allow indexing selected fields, which supports enhanced browsing by those fields. CONTENTdm supports the use of several controlled vocabularies, which turns the associated fields into hyperlinked fields and enhances the navigation possibilities. For curation purposes, the indexing of such fields as source collection, creator, spatial and temporal coverage, provenance, and date presents additional value because these fields support metadata harvesting for visualization, analysis, and hermeneutic uses by researchers. For instance, use the information in the provenance fields to research donors in specific time periods and/or geographical locations. The indexed fields can also support queries (e.g., by subject, spatial and temporal coverage) for which permanent links can be added to virtual exhibitions for easy access. To provide easier visual access to archival collections, curators will need to query the system for source collections (which would retrieve all records with a corresponding manuscript number and title in that field) and use the resulting URL as a link in the prepared list of manuscript collections in that system. The same link will also work from an external domain where the virtual exhibition resides. Figure 4.6 provides a schematic of an existing approach in CONTENTdm, which demonstrates the use of a nested information architecture.

While there are limits to adding custom pages and navigation labels in CONTENTdm, it is possible to nest an external site along with its native navigation scheme. Such an approach will create a nested information architecture, and may work well for a virtual exhibition using the `<iframe>` method to bring external sites with their native information architectures into the framework of the digital library. The principle of nested information architecture illustrates the expanded capability of designers to embed foreign information architectures along with their navigation and labeling schemes within the host environment. Elsewhere, the use of nested information architecture has accommodated the expansion of subject categories in the Dryad Database of Biological Data (Chapman, 2013). Users can navigate pages external to

the repository as long as the window behavior settings force windows to open in the same window using the <target="_self"> method. Window behavior must be consistent throughout the site so that appears virtually inside the repository. Depending on the host system, the nested information architecture may (or may not) work in the host information architecture. Nested information architectures may also work toward solutions to integrate timelines, maps, and other visualizations generated by tools that operate outside the Web site's or digital repository's domain.

Spatial, temporal, and ontological dimensions in information architecture

In addition to the structural and textual aspects of information architecture, there are spatial, temporal, and ontological aspects as well—each representing information on maps, timelines, or subject trees. Gruber (2007a, 2009) defines ontology as the description of concepts and their relationships for purposes of knowledge representation, sharing, and reuse. Ontologies are essential to organizing and visualizing geospatial, temporal, and other data in the metadata record, and as such can help curators preserve the content, information representation, and the internal structures of digital collections, Web sites, catalogs, and other resources.

This three-dimensional representation of information architecture is present in geographic and topographical space-time-cubes described by Windhager and Mayr (2011) who use "time geography…visualize temporal processes in geographical space…[where] two visualization methods became conceptually intertwined: cartography as method to map arrangements on geographic surfaces and chronography as method to map arrangements of events along timelines…[resulting in] space-time-cubes" (3). As advance orientation devices in virtual exhibitions, these cubes present information architecture along its three dimensions (space, time, and topic) which visitors can navigate separately via maps, timelines, and subject trees. HistoryPin offers visualization along spatial and temporal dimensions while ViewShare allows visitors to discover related topics through the subject field as well.

Johnsen's work (2010) on topic maps as information architecture devices presents yet another approach. He writes,

> *In a topic map, topics may be given one or more names; they may be categorized in types, subtypes and instances and they may be connected to internal content (information within the topic map itself—descriptions, data values, etc.) as well as external content (resources outside the topic map itself—web pages, files, etc.). Topics may be related in typed associations in which they are assigned semantic roles and they may be linked to external descriptors.*
>
> *"Topic Maps," para. 3*

With their support for multiple subject descriptors, topic maps offer a significant advantage over some taxonomy forms. In archival context, they facilitate the discovery of older collections with obsolete subject terms, which young users may not succeed

while using current subjects or the popular social tags. For instance, the collection of the Toledo Woman's Suffrage Association may have "woman's suffrage" in the subject field of the metadata record, someone searching with more current terms like "gender" or "gender equality" may not find this collection. However, mapping the topics may help in two ways: first, it will help researchers locate more records under a wider range of related subjects; and second; it will help younger researchers explore archival collections under obsolete subject terms. By revisiting the subject fields in the MARC, Dublin Core, and EAD records, the information architecture remains up-to-date and more robust.

The spatial, temporal, and ontological dimensions of information architecture present curators additional possibilities to organize and visualize their materials using various emerging Web 2.0 technologies. As all of metadata schema (Dublin Core, EAD, and MARC) in archival use include fields for spatial (location), temporal (date), and ontological (subject) entries, data in each of these fields can appear on maps, timelines, and subject trees, respectively, supporting discovery of digital collections from corresponding areas, date ranges, and subject areas. Because of disparate metadata and description standards and cultural interpretations, the description issues may hinder efforts of digital curators to find a single platform for all purposes.

Localized approaches at the Ward M. Canaday Center for special collections

The Ward M. Canaday Center for Special Collections at the University of Toledo houses the university's archival records, manuscript collections, rare books, and digital collections. The center's participation in the Toledo's Attic virtual museum project since 2001 and in the OhioLINK Digital Resource Commons (DRC) project between 2007 and 2013 has moved the center toward implementing digital curation of the university's digital collections. While using EAD finding aids have come into consideration, most finding aids are in HTML and PDF/A formats accessible on the center's Web site, via the library catalog, and ArchiveGrid. The Center has also converted exhibitions into virtual ones, which remain accessible to the public after the physical exhibition ends. There was no extensive implementation of digital curation to the extent described in recent publications however. The OhioLINK Digital Resource Management Committee (Newman et al., 2012) has investigated using DSpace, DuraCloud, and other repositories for purposes of dark archiving and long-term digital preservation and presented a detailed report with recommendations based on its findings. Discussions on digital curation standards worldwide only began in 2001 although there had been such projects as the American Memory, which began in 1995 without the benefits of approved standards in place (Madden, 2008).

The University of Toledo's bifurcated curation plan utilizes UTOPIA (The University of Toledo OPen Institutional Archive) using OCLC's CONTENTdm repository to

house all digitized heritage collections and the University of Toledo Digital Repository (hosted by Bepress) as the designated repository for born-digital materials, open access scholarly publications, departmental reports, and various data curation initiatives. Both are considered trustworthy repositories that meet digital curation standards (Lee & Tibbo, 2007); however, not only is this approach costly but it presents a bifurcated curation model with some disadvantages over using a single repository, metadata standard, and a unified workflow process. Perhaps the mitigating factor in this strategy is that the library can keep its heterogeneous digital collections in distinct platforms, and focus on two distinct content categories: cultural heritage in UTOPIA and Open Access Scholarly Communication/data curation in the University of Toledo Digital Commons. The Canaday Center, which has predominantly contributed to digital heritage materials (in UTOPIA) from its manuscript collections, may eventually develop greater interest in Digital Commons as a repository for humanities data, electronic records, and audiovisual materials. However, this chapter will not cover Digital Commons, as its implementation is still in its infancy.

The Canaday Center has organized medium- to large-scale exhibitions almost annually. The archival records and manuscript collections provided most of the exhibits, but some exhibits were on loan from nearby museums, libraries, historical societies, and private collectors. Since 2000, the Center has also been interested in virtual exhibitions that came online after the physical exhibition ended. These virtual exhibitions, which offered continuous access to the exhibits (mainly the images or cases or the artifacts) afterward, have ranged from simple to complex Web sites. Adding resource links to the virtual exhibition at the case and shelf-levels ensured that visitors have complete information on the exhibits and could explore the archival and library collections. This strategy aimed to promote the collections to faculty, students, and other interested researchers after viewing the virtual exhibitions, and in a broader framework of curation, it aimed to bring historical context and cultural meaning to the exhibits and inspire new scholarship in the areas of history and the humanities. Virtual exhibitions contain primary source material in some thematic context that digital historians and humanities scholars can use.

Improving the information architecture across all digital domains of the Canaday Center became an important curatorial interest in the past four to five years. The management of digital collections in the DRC, virtual exhibitions, Toledo's Attic virtual museum, and the digital finding aids came under the purview of the digital initiatives program, and it was important to improve access to all digital resources. The improvements in the virtual exhibitions, finding aids, virtual museum, and digital collections involved providing links from one resource to the other. Working with catalogers established a new linking relationship between the library catalog, and the DRC's platform enabled developers to turn the Library of Congress subjects into a link leading to the subjects in the University of Toledo library catalog. The concept behind this strategy was that each resource could serve as an access point to the other digital resources, and this strategy would enable users to discover resources from any direction. For instance, visitors of the virtual museum and virtual exhibitions would use the links to related materials in the catalog, finding aids, and digital collections. In a

different scenario, researchers using the catalog or the digital repository may discover the virtual exhibitions. The recent addition of social media for social curation purposes has expanded some possibilities for the Canaday Center and Toledo's Attic to reach new audiences and extend access to the archives and special collections. Using such customized strategies in the long run, however, will only work if the virtual exhibitions, virtual museums, and digital repositories reside on servers directly accessible to curators. The DRC experience may be all too familiar to curators at institutions with shaky commitment to digital heritage curation; and the consequence is all too real: as soon as one element disappears, the system is broken.

Figure 4.7 is the digital curation map showing the node–link relationships in this improved information architecture, which became the framework for hypertextual relationships across the various channels. Each node represents archival content, programs, and people connected by links with the action words describing the nature of uni- or bidirectional relationships between notes. The relationships also outline the workflow associated with digital curation; for instance, the workflow from archives to social media involves collection development and setting up exhibitions, which implies managing metadata, preserving the digital content, and developing an outreach program: ARCHIVES – develop – COLLECTION – organized into – VIRTUAL EXHIBITIONS – link to – SOCIAL MEDIA. Virtual exhibitions play an important role in that part of the center's digital curation program. The exhibits add meaning and context to the information in the exhibits, and therefore virtual exhibitions begin to function as hubs in the system of information architecture and hypertextuality.

The virtual exhibitions developed in the last four years feature improved navigation systems (navigation bars, sitemaps, breadcrumb trains) and access to the exhibits and related resources. Since the metaphor for the design was a museum, the

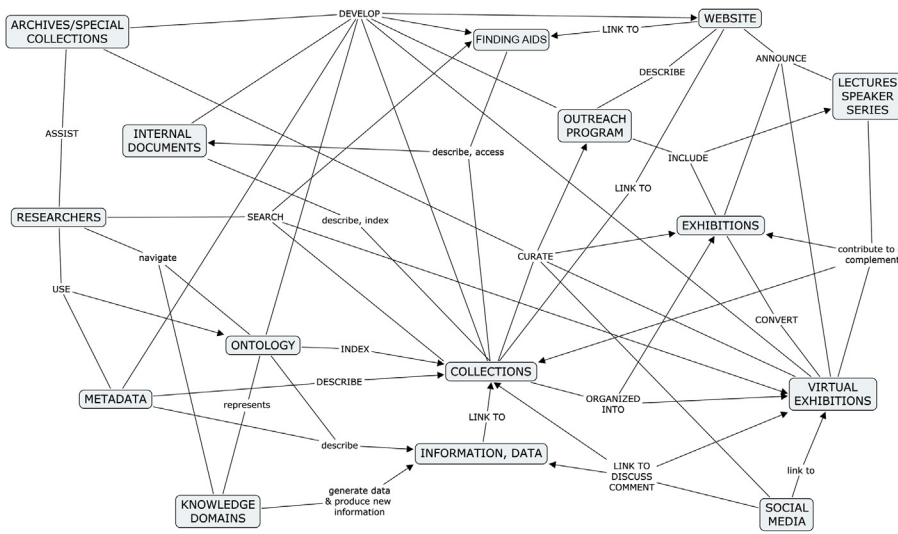


Figure 4.7 The Canaday Center's curation map.

idea behind the design approach was to emulate the experience of walking through the exhibit space (Sabharwal, 2012a). Virtual exhibitions may never compete with the social, spatial, and tactile aspects of physical exhibits; however, well-planned sites can ameliorate that in other ways. For instance, visitors can read the complete document, inspect images by zooming in much closer than possible in most physical exhibitions, download and print the documents, and gain instant access to related resources from the archives and libraries available online. Researchers interested in the entire featured collections could access the linked finding aids and request specific boxes or folders.

Visitors of virtual exhibitions can usually navigate sites with the use of navigation bars, sitemaps, and breadcrumb trails; however, slideshows and interactive panoramas can be equally effective and may evoke an almost equally strong sensation of being very close to the exhibits. In their work on historical representation and "visual historiography," Roegiers and Truyen (2008) write, "one distinguishing feature of digital media is the fact that it is able to integrate many modes of communication: static and moving imagery, sound and text. The digital medium does not simply allow integrating all three modes into one presentation; it is also able to reproduce characteristics of older media like print, photography, and film." Interacting with the images in such a direct manner underscores what Bolter and Grusin (1999) call immediacy in the context of virtual reality, transparent technology, and remediation. While archives and libraries have not relied on immersive technologies to represent the past through historical texts and media, the paradoxical relationship of immediacy and hypermediacy within the context of remediation may interest digital curators and digital humanists. Immediacy aims to connect viewers with digital images as though no user interface mediated between user and content. This notion may contradict what Sandweiss (2007) attributes to the relationship between photographer and subject; however, while immediacy is evident between the two prior to taking (and posing for) the photo, the printed photo and the digital copy rule out immediacy since the photo and the media are now inseparable. Old photos need paper or digital media just as much as virtual reality media needs required devices. Hypermediacy, therefore, is that unavoidable dimension of remediation, which reinforces the presence of media equipment and renders technology not so transparent. Information architecture facilitates remediation through both immediacy and hypermediacy. The hypertext structure (via well-planned information architecture) can enhance visitors' experiences with Web sites using embedded media players, widgets, and other applications. This requires collaboration among archivists, humanists, and information architects.

The home page in virtual exhibitions is the most important starting point for visitors to consider a path. Most visitors start by viewing the exhibits right away while others may be interested in exploring the timeline, reading the exhibition catalog (see Floyd, Brownlee, Jones, Free, & Chelminski, 2010; Floyd, Russ, Uppal, Stevens, Jones, Miller, 2012), or searching for information in the virtual exhibition or the archival and library resources. Figure 4.8 presents the Wholly Toledo virtual exhibition in which the vertical navigation bar and some text links in the main content area presented easy access to visitors of the site. Figure 4.9 in the "Medicine on the Maumee" virtual exhibition illustrates a horizontal navigation bar. Some designers

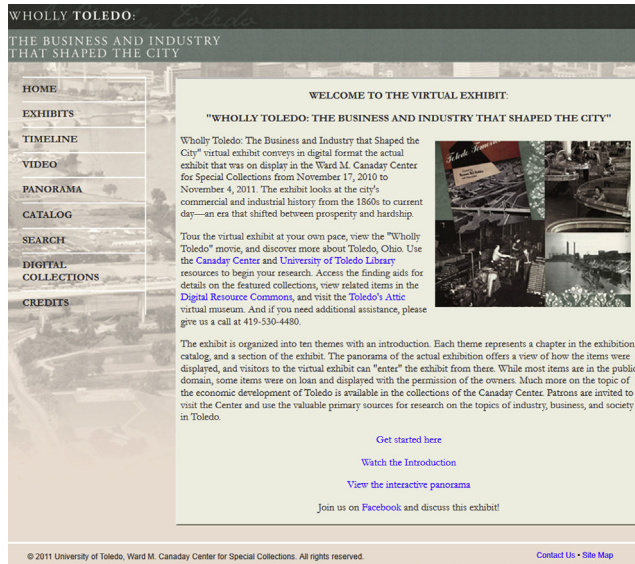


Figure 4.8 Information architecture using a left navigation bar.



Figure 4.9 Information architecture with horizontal navigation.

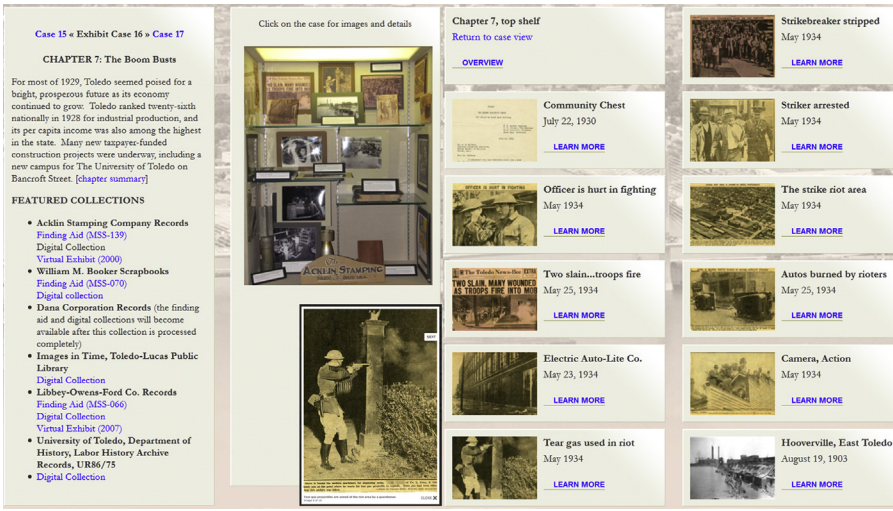


Figure 4.10 Moving between case-, shelf-, and item-level views.

prefer right-side navigation systems (e.g., online article submission sites). Global navigation systems appearing throughout a site must be clear and represent mutually exclusive content (Rosenfeld & Morville, 2002).

Virtual exhibitions must also translate the spatialization of knowledge (Dallas, 2007) into concrete hypertextual relationships throughout the site, and this is where information architecture requires careful planning. If the aim is to emulate experience at exhibitions, there must be a way to navigate from case to case via textual or graphical links or both. Figure 4.10 illustrates this approach. Each case view (left) should present shelf-level descriptions and link to the individual exhibit accompanied by the text appearing on the original exhibit label (right). Navigation should be possible between shelf-level views and individual item views. To this end, the Lightbox script for Web pages has been very useful because it presents the label description, navigation paths (next, previous), and darkens the background (right bottom). Each case view comes with a list of resource links that takes visitors to the finding aid, digital collection, library collection, or some related item in the virtual museum.

Getting the exhibit gallery "feel" through Web sites does not always come with expected results, and the capabilities of mobile devices may enhance or hinder what a site is trying to accomplish. Figure 4.11 illustrates the use of an interactive panorama with hotspots—or linkable regions on an image—leading to the corresponding case views. This is different from the familiar hypertext linking method in that the image (or selected regions thereof) becomes the link. Not showing in the illustrations, but desktop and laptop users will see a pop-up window with a short description of the hotspotted region. Clicking inside that region will take visitors to the selected case view. The panorama is available in both Flash and QuickTime versions to accommodate a larger community of technology users.

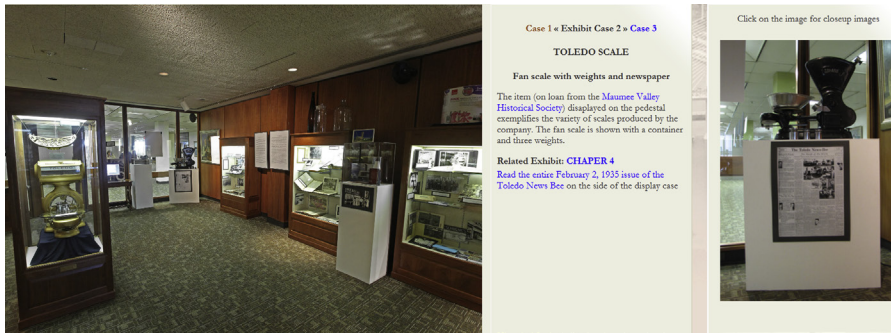


Figure 4.11 Accessing case views via an interactive panorama.

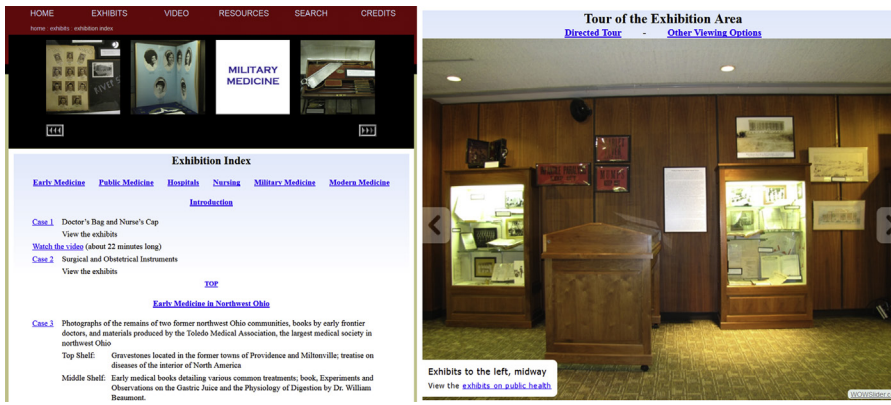


Figure 4.12 Carousel (top left), textual navigation (bottom left), and slideshow (right) methods.

Figure 4.12 illustrate the use of an interactive carousel and slideshow as another method of graphic linking where each thumbnail image functions as a link to the corresponding case view, and the slideshows combine image and text linking in one view. In addition to graphical navigation, the exhibition index provides textual navigation as may be needed for some visually impaired computer users.

Information architecture for online and hybrid courses in digital humanities

The growing interest for online courses in higher education has affected humanities faculty and academic libraries to varying degrees. Archives have remained somewhat external to this trend since a large percentage of their collections are in noncirculating predigital media. The development of digital finding aids and digitized collections can significantly reduce research time in the reading room of archival facilities, but the majority of records are still in their original format for legal,

logistical, or other reasons. The availability of digital history projects has opened the doors to designing and offering online digital humanities courses in institutions of higher and secondary education. Chapter 2 of this book discusses the state of digital humanities pedagogy (compared to research or technology), but the book as a whole does not explore pedagogy, teaching methodology, and learning theories although some may explore the relationship between cognitivist learning and collaborative curation in the classroom. This chapter focuses on the role of information architecture in online and hybrid courses.

Instructors experienced with learning management systems such as Blackboard, Moodle, or Sakai will distinguish between passive and active methods whereby they integrate Web resources on different levels. Shank and Dewald (2003) describe macro- and microlevel integration of library resources. In the former case, the online courses provide links to Web pages of library services and resources. The microlevel strategies included the development of course-specific resources such as resource guides, tutorials, citation style guides, and links to course-specific resources in the research databases. An even greater integration at a "nano-level" (Sabharwal, 2005) would result in improved navigation between the online course and resources such as research databases, resource guides, digital collections, virtual exhibitions, and historical hypertext material. This strategy would require improvements to the *coursescape* with customizations of labeling and navigation schemes, thus affecting only a part of the online course's information architecture, but there are several possibilities for collaborating faculty, archivists, digital curators, instructional technologists, and metadata librarians.

The results of integrations at the "nano"-level may produce two new architectures: First, nested information architectures, which involve weaving foreign HTML codes into the course structure, but this will utilize the limited visual space of the online course, giving the false impression that it is actually a part of the course. Because of copyright concerns, instructors should only consider content that they created for this approach. Second, juxtaposed information architectures involve two or more information architectures with intricate connections using dynamic links and scripts that integrate contents across the two domains although the linked pages may open outside of each other's domains. A historian may deliver a special topics course in an online or hybrid course, provide links to specific contents on a digital history project site, and even involve an archivist or digital curator to teach sections of the course, select specific collections (with enhanced metadata), and respond to reference questions. This arrangement may still leave the ontological gap between course and resource sites wide open, but there have been successful attempts at even further integration involving ontology-level crosswalking between course structure and a digital library.

Gasevic and Hattala (2006) have addressed semantic interoperability of Web resources and have discussed ontology mapping between an online course and the ACM Digital Library. These authors report on their solution to connect online courses to contents in remote repositories by mapping the course syllabus that lists the precise topics to be covered throughout the course with the subject headings in the linked database— the ACM Digital Library. This crosswalking approach not only narrows the searching field but also presents the results in the context of the course expectations. The authors describe the method involving a source (i.e., syllabus/curriculum)

ontology, a target ontology (i.e., the ontology in the repository), and a mapping ontology that operates behind the crosswalking process.

Although this chapter—or the book for that matter—does not explore any engineering behind digital curation or e-learning platforms, it presents a case for a similar ontologically framed approach in digital humanities and history courses offered online or in hybrid courses. An initiative at this level will bring archivists, librarians, digital curators, historians, digital humanists (mostly faculty), information architects, and instructional designers to the table. This is also an opportunity to present a case for recognizing the scholarly aspect of digital humanities projects as discussed in Chapter 7 in reference to the debates. A collection building activity that reifies the underlying theory is scholarly since it serves to prove points made through the collection, associated narratives, and metadata. A college course can use collections developed by digital humanities scholars to create new knowledge or build new collections taking a scholarly approach. Collaboration with archivists, digital curators, and subject librarians will be necessary in order to identify digital collections, supply the finding aids, and enhance the metadata to correspond to the curricular requirements of the course or degree program. The metadata record (representing the target ontology) will require enhancement to correspond to the course (source) ontology representing the syllabus and other curricular requirements. When the mapping ontology (with the necessary crosswalks) is in place, students can start accessing the resource from within the course management system and focus working with digital resources in the specific thematic context of the course. Since digital humanities and history courses will also need access to data and scholarly publication, technologists may be able to create mappings between the syllabus and multiple digital resources.

Conclusion

Information architecture brings value to digitally curated collections, as it frames not only the hypertextual relationships throughout an archive's information space but also the learning process (leading to new knowledge) through improved navigation. The navigation systems allow visitors to connect and synthesize information that they discover in a virtual exhibition and the various archival and library resources. In itself, information architecture is not an archival skill and nor is it equivalent to digital curation; however, it is important to recognize its role in archival public services, outreach, and digital curation at various levels. In his presentation on levels of representation, Lee (2011) points to levels 4 through 7 corresponding to the aggregation and relationship of multiple files. There is an information architecture of some type present at these levels, which highlight the hypertextual, ontological, spatial, and social relationships among files. Finally, information architecture can significantly improve accessibility to digital collections through learning management systems used to teach history and humanities.

Digital curation lifecycle in practice

5

The curation of digital heritage collections entails a sequence of related activities throughout the digital content lifecycle, which brings various related curation activities into the workflow involving various specialists in the archives, libraries, academic departments, and administrative units. The scope and extent of curation programs vary from organization to organization and range from extensive curation initiatives at large institutions designating entire departments for digital curation to small-scale curation efforts at smaller institutions with a solo curator or none, with most curation initiatives falling between these extremes. Each curation initiative also varies by the individual organization's structure, type of collections (humanities and scientific data, cultural heritage, electronic records, open access publications), the makeup of stakeholders, institutional mandates for open access and data curation, curricular requirements, and community outreach interests. These variations have a direct influence on the lifecycle of digital contents and related curation programs. The pioneering publications on digital curation (Beagrie, 2001; Lee, 2011b; Lee & Tibbo, 2007; Lord & Macdonald, 2003; Yakel, 2007) have addressed the need for digital preservation standards and curation tools, but the best solution remains the one that works within the curators' community where stakeholder buy-in and collegial support are equally important.

This chapter will use the Digital Curation Centre's Lifecycle model as the framework for implementing digital curation programs from conceptualization activities to assessments. The discussions will focus on the activities included in the lifecycle model and their relationships across the different levels of curation activities in this model. What the lifecycle model did not include is assessment—that is, measuring the impact of digital curation programs on the environment (such as parent institutions, historical societies, and education programs). Such assessments can lead to the redesign of curation programs and workflows. Assessment provides data helpful for evaluating access and use as well as appraisal and selection, which directly inform collection development. For instance, increase in the use of digital heritage collections in National History Day competitions will validate the argument (supported by assessments and empirical data) that more staff time and support for digitization equipment will translate into more successful presentations by participating students. If measurement tools such as Google Analytics report a high number of hits on these collections then expansion of such collections will continue to receive both institutional and community support. The chapter will also consider a broader conceptualization of curation; digital humanities curation and social curation play a crucial role in the construction of meanings around heritage collections, which positively impacts the usage statistics—therefore the relevance—of heritage collections. Since "data" in the lifecycle model may represent a multitude of heterogeneous collections (research data sets, administrative reports, open access publications, heritage material, and

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more) from various departments, this chapter will focus on heritage collections containing digital objects, humanistic data, and publications in the humanities.

Overview of the DCC curation lifecycle model

Higgins (2008) presents the Digital Curation Centre's Lifecycle model as a nonprescriptive framework for training and practice, although it may continually evolve as to make adjustments as digital preservation standards emerge or change. The model is also an excellent framework for conceptualizing, planning, implementing, and sustaining digital curation initiatives. Implementation teams may anticipate seeing variations to this model due to the evolving scope of their planned preservation programs, changes in size of the record-creating organization, and the types of collections and data sets planned for curation. For instance, cultural heritage collections, institutional research data, electronic records (reports, minutes, agendas, and memos), humanities data, or data required by grant agencies will inevitably expand the model and change the relationships within this model.

Figure 5.1 presents the lifecycle model in the shape of concentric circles representing a key element of the model (Digital Curation Centre, 2010): data, full lifecycle actions, sequential actions, and occasional actions. Data covers more than data sets in a database; it includes digital objects of textual and nontextual nature. Data is what the curation and

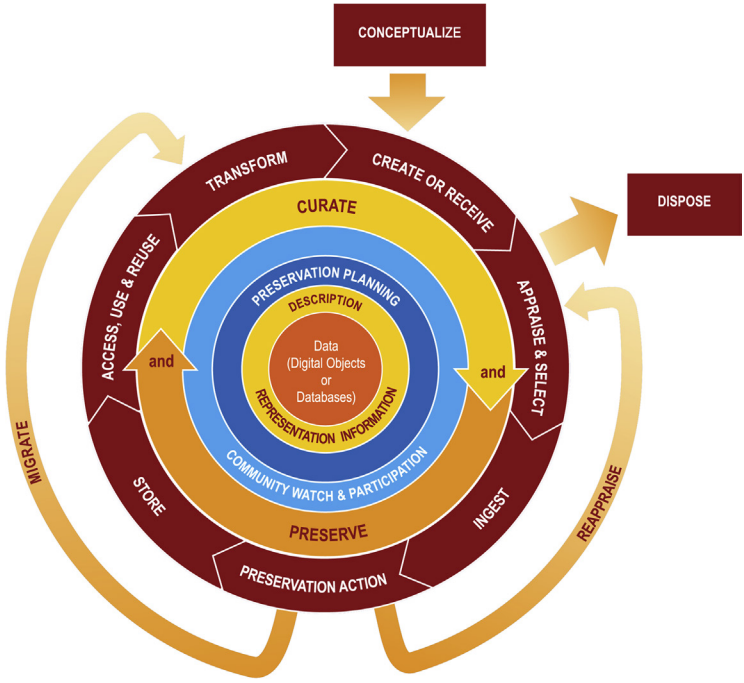


Figure 5.1 The DCC curation lifecycle model.
Credit: Digital Curation Centre.

preservation activities in the other region of the model focus on, but each region also identifies related activities and stakeholders vital to an institutional data curation. The outermost circle represents the preservation workflow with the circles between it and data presenting an organizational and practical framework for curating data. Thus, the entire model presents the necessary knowledge architecture—the people, content, and technology (Applehans, Globe, & Laugero, 1999)—needed for a curation initiative.

The lifecycle model uses arrows pointing toward the model (conceptualize, dispose, migrate, and reappraise) to represent or indicate activities tangential to activities and relationships in this model; for instance, conceptualizing takes place well before the process of creating content for curation. Likewise, the arrow representing migration arches back toward the model and represents digital reformatting into a new format, but it may also represent migration away from one curation process toward another one. Similarly, reappraise will cause a delay in the curation cycle of a particular collection or item before it reenters the curation lifecycle. It may be helpful to identify other relationships such as those between description/representation information and data (a link between data, metadata, and the intellectual context defined by academic disciplines or administrative practice) or between data and community watch (the link between data and the community that contributes and evaluates that data).

Conceptualization and the master plan

The lifecycle model begins with conceptualization; however, a critical discussion on institutional mandates must precede conceptualization because of compliance requirements (e.g., to develop a data curation mechanism) or interest in community outreach (e.g., develop cultural heritage collections), other reporting requirements (e.g., accreditation reports, administrative documents, etc.), collecting scope, and branding (as required by an institutional marketing office and implemented by the university's Web design team), all of which must inform curation planning and implementation. There are legal aspects also, which this chapter will not discuss in depth, but the online publication of any intellectual property must assume prior discussions of copyright, permissions, deeds of gift (and restrictions stated within), creative commons licenses, and materials in the public domain. A master plan is necessary to address these questions and bring departments and experts into long-term working relationships in order to complete the implementation process, and the first step in this direction is establishing an investigating team (a steering committee or task force) to complete and submit a needs assessment and feasibility report, involving the archives, library, and participating academic and administrative units as necessary, whose roles will inform the workflow in many cases. In most cases, such a team consists mostly of library staff, but ideally, it should also include faculty, administrators, and technologists.

The planning process considers the institution's knowledge architecture: its components are people, content, and technology. This architecture varies by organization, so for the present purposes, the organization is a university: "people" include stakeholders such as archivists, librarians, faculty, researchers, administrators, technologists, and perhaps a variety of representatives from the community surrounding the university; "content" includes digital objects in cultural heritage collections, electronic records, and

research data; and "technology" includes the repository platform, computers, cameras, digitization equipment, software, storage mechanism, and storage media.

"People" play a crucial role in this architecture since they not only create but also select the material for curation. The team, therefore, identifies the role of archives and libraries in the process because most digital collections and virtual exhibitions will feature collections from the archives and use metadata developed in collaboration with catalogers. The digitization of archived records of the parent institution will also involve the archives, which has received those records following an established retention schedule. The university archivist (who is often also the records manager) will also participate in—if not oversee—electronic records management as a process involving a single institution-wide records retention plan and standard in the digital curation program. The archives and libraries constitute a significant part of this knowledge architecture because of their commitment to preservation, information organization, knowledge representation, and continued curation. Researchers, teaching faculty, and administrators are also important since they will contribute data documenting their work at the institution. Researchers using government funds will follow mandates to curate their research data in repositories. Digital humanists may publish and seek research data for scholarly purposes such as analysis, visualization, and related hermeneutic activities.

Given the obsolescence of most predigital and a growing number of digital media formats, the team must have a good understanding of capture methods, digital media, metadata quality, various digital preservation options, and industry standards for long-term preservation of digital content. There is prolific literature on high-end curation tools, methods, and commercial services, which require serious investments in open-source operating systems (e.g., Linux), repository platforms (e.g., DSpace, Omeka, Hydra), and curation tools such as BitCurator (see Lee, Kirschenbaum, Chassanoff, Olsen, & Woods, 2012). Although the software is available free, their installation, configuration, upgrades, and maintenance require highly competent technologists, staff time, and institutional commitment. Commercial products (e.g., CONTENTdm, Digital Commons, Luna) are available for a fee, and while they take a great deal of work off the curators' shoulders, these platforms usually serve specific clienteles, leaving some institutions to develop bifurcated curation strategies, which are not efficient and may cost more than adding a dedicated trained technologist to handle routine tasks. Conceptualization also involved long-range planning for digital repositories, open access publication outlets, virtual exhibitions, and resource guides. Conceptualization is complete when there is a nearly mutual agreement among the stakeholders on the critical components about content, structures, and workflows; when there is an implementation of timelines according to the lifecycle model; and when the steering committee or task force has a master plan for implementation.

Curation of data sets and digital objects

Institutional mandates, academic agendas, and community relationships can have direct impact on the collections selected for digital curation. Such selected and curated data occupies the center of the lifecycle model and contains data sets (in databases) and digital objects that are either born-digital or digitized materials from the archives and libraries.

Archives have historically accessioned manuscript collections containing textual and photograph collections, but throughout the twentieth century, film-based, audio and video materials have also reached the archives. Born-digital materials have begun to arrive in archives in recent decades, containing textual and audiovisual materials in various proprietary (and often obsolete) formats. Historians and other humanities scholars have historically utilized manuscript and photographic collections in archives, but digitally accessible data has opened up new avenues for digital humanities and history scholarship. Digital humanists also contribute with data for open access publications, so data sets today are valuable to researchers in the humanities and sciences pursuing interdisciplinary scholarship. Additional data from scientists may include data required by funding agencies. Over the past century, therefore, this core area of the curation model has dramatically shifted from the homogeneous (text-based) collections to heterogeneous (text, media, analog, digital, scientific, and humanistic) ones. The selection of collections for digital collection will take into consideration such factors as curriculum, institutional mandates for open access, community relations, and legal requirements for electronic records management. These commitments, however, will evolve and change over time.

Full lifecycle actions

The curation lifecycle includes a number of interrelated activities and processes spanning the entire lifecycle of digital contents: preservation planning, preservation and curation, description and representation information, and community watch and participation. This practical and social framework for digital curation ensures the physical survival as well as social–cultural–intellectual relevance of digital content in the community. However, the lifecycle omits such important processes as conservation—a process closely related to preservation. Lord and Macdonald (2003) place three related activities—curation, archiving, and preservation—in a hierarchical order: preservation falls under archiving, which then falls under curation. Within the larger framework of curation practices and workflow, there is a different order to these activities: the first is archiving to ensure the proper selection and secure storage of data, so that curators and researchers can access and retrieve them. Second, preservation is an archiving activity aiming to maintain specific data and considering technology changes. Finally, curation entails the management and promotion of data in order that it meets current and anticipated scholarly, administrative, and collective needs. At this point, the role of the digital humanities curation as well as social curation is important: the life of data extends through analytic, hermeneutic, and social processes.

Preservation planning

Preservation planning is crucial to archives, which have, in recent decades, accepted and accessioned an increasing number of born-digital material coming with larger (print and digital) manuscript as well as digital-only collections. The implementation team will need to collect input from the stakeholders and the larger community

it represents, and then formulate a collection development and curation policy. This group can also study institutional mandates for open access in order to advise one for the parent institution if there is none at place.

Managing the digital collections involves documentation, staffing, equipment, budget, and workflow throughout the lifetime of those collections. Where a solo curator manages the process, staffing may often require temporary employees, student assistants, and volunteers. Workflow management requires a calendar and setting timelines for project management purposes. Reporting work in progress, challenges, and opportunities to department leadership and stakeholders opens the floor for discussions and constructive input useful for future projects. Therefore, project management skills are useful in a collaborative environment.

Curators may work with multiple projects simultaneously, each requiring separate workflows and relationships. One scenario is developing the collections within the archives with the help of assigned staff with some knowledge of the collections requiring digitization. This knowledge is useful for developing accurate metadata. Another scenario is to employ student assistants who may need to rely on background reading (of finding aids and institutional history) in order to provide context for the information on the digitized materials. A third scenario is collecting the material from the record-creating department, which has knowledge of the material selected for digital curation. This scenario requires a good relationship between the curator and that department, but a documentation of the process and communication will help in future projects.

Preservation and conservation in the curation lifecycle

While technologies to digitize texts and photographs and the quality of such archival-quality file formats such as archival PDF (PDF/A) and TIFF have significantly improved, there are formidable challenges to accessing obsolete media content and formats: accessing the source (original) media and using the equipment to access such media format. Obsolete source media today can be all: print, film, and various audiovisual media in analog and digital formats. Archives are not only facing preservation challenges but also conservation in order to prolong the life of the original material and access to the information within. Conservation is the "repair or stabilization of materials through chemical or physical treatment to ensure that they survive in their original form as long as possible" (Pearce-Moses, Ed., 2012, "Conservation"), while preservation involves "protecting materials by minimizing chemical and physical deterioration and damage to minimize the loss of information and to extend the life of cultural property" (Pearce-Moses, Ed., 2012, "Preservation"). While preservation refers to systematic efforts to prolong the life of digital content in this case, conservation refers to specific treatments to the original media. The conservation of digital media is an important concern, but as a preservation measure, content owners often migrate their contents to newer and commercially more accessible media. Creating the preservation (digital) copies, however, leads to new challenges.

Archives accepting media collections almost often do not own, receive, or have access to old equipment to access information in various obsolete media formats.

While film-based formats were easier to handle, there are significantly greater challenges with accessing the audio–video materials in analog and earlier digital formats. The existing and working equipment to play VHS videotape and audiocassette formats enables archives to transfer those materials to optical discs, but older formats like Betamax, U-Matic, Super 8, wax cylinders, wire recording, and open reel tapes (much more on the Museum of Obsolete Media, n.d.) present conservation, preservation, and access challenges. Equipment to play these older media formats is becoming rare and difficult to repair. Most universities have disposed of such equipment or transferred them to their archives in some cases, but the quality of the media and their physical condition will affect the quality of the digital copy.

Obsolete digital media presents yet another layer of challenges since it requires software to access the information on old digital media like old floppy drives, laser discs, Winchester drives, and other waning disc and tape-based formats. Locating obsolete software is only a part of the challenge; installing them on a computer with the necessary operating system is necessary to access the old files, assuming that the integrity of the physical media and the file structure do not present further issues. Computer museums own and can operate obsolete equipment, and many presidential libraries in the US National Archives and Records Administration system also maintain the players needed to access original footage associated with the administration whose records they preserve.

If the popular assumption about everything being on the Web holds, there must be another set of assumptions that institutional archives, citizens, and corporations have diligently and meticulously transferred their records to modern media every few years and that a copy of everything worthy of knowing is available somewhere on the Earth in digital form, in print, or as microfilm. Investigating whether that assumption is correct is simply beyond the scope of this chapter, but a safe advice for uncompromising researchers is that while the Web has grown considerably into a place for shared knowledge and memories they should continue to use archives for some time into the future.

Description and representation information

The development and management of metadata evolves from the beginning of the lifecycle with the creation of digital collections and data. Before developing metadata records for large collections and long-term projects, curators should collaborate with catalogers and possibly other technologists to obtain information for controlled vocabularies for use in the subjects and authors fields. For electronic records management projects, using the existing structures may help, but because of the parallel structures between paper-based and digital records, additional metadata is necessary to represent the digital continuation in the retention schedule. This is useful where the institution replaces print records with digital ones. Thus, using the existing authorities for departments, subjects, disciplines, and other critical fields will help curators establishing the relationships between paper and digital records. In addition to that, finding aids to those record groups will serve as a bridge between the transformed records.

Descriptive metadata requires collaboration; therefore, in one scenario, the record-creating department has the adequate knowledge to describe what is in the collections and who the author is. This approach will also work with data and digital objects contributed by researchers from the humanities and science fields. Within the curator's department, staff processing the collections can significantly contribute to descriptive metadata since they have developed the finding aids to the digitized collections. Elsewhere, student assistants and volunteers may be available for preparing the metadata sheet (usually a Microsoft Excel or OpenOffice spreadsheet) with item- or series-level metadata. In a second phase, the curator may enhance the rest of the collection-level metadata alone or in collaboration with catalogers. The latter approach may ensure a greater ontological integrity of digital records across an expanding digital repository, as it will help curators, researchers, stakeholders, and casual users establish critical relationships among the records. In addition to descriptive metadata, curators will need to manage technical, structural, copyright, and administrative metadata for purposes of documenting provenance and preservation actions throughout the lifecycle, some of which is an automatic process.

Institutional repositories contain heterogeneous collections requiring different metadata profiles or even schemas. While most repositories use Qualified Dublin Core, art and architecture collections have been relying on the Visual Resources Association (VRA Core) metadata schema specific to visual media. However, there is great variation of metadata use even within the Dublin Core camp: faculty and student publications, data sets, institutional archives, and heritage collections will vary significantly in their metadata profile and the fields required for these categories. For instance, student research papers, dissertations, and theses will require advisor fields. Data sets from grant-receiving departments will require extensive copyright and administrative metadata fields to document the lifecycle of the data set. Finally, materials from the archives will include provenance and other contextual information used in humanities research. Publications from open access repositories such as Open Journal Systems and Digital Commons will contain various metadata values developed during the submission and review processes (author-supplied subject keywords, various dates—submitted, approved, etc.). Thus, during the planning phase of the repository, the team must address metadata schemas and profiles before inviting submissions.

Community watch and participation

Community watch and participation are also vital parts of the lifecycle model, as the curation process involves stakeholders who use, evaluate, and comment on the collections. Russo and Watkins (2007) address an audience-focused approach to curating cultural heritage. Digital heritage collections and virtual museums, for instance, represent community memory, identity, and knowledge that citizen scholars can access and use. This is where the community can contribute to the discourse on community history and identity through discussions, item submissions, and other feedback to the curator. Social media presents an excellent channel for communication. Archivists and curators, in turn, can use their expertise and interests to shape community memory and history. This two-way approach to curation underscores the third and fourth paradigms in Cook's article

(2013) about the changing nature of curatorial agency: through the digital repository, archives expend their reach beyond the physical borders and take on new leadership roles to shape public memory through digital curation. Finally, the role of scholars in curation is also changing the landscape of digital curation. Digital humanists and historians interested in contributing data, digital objects, and publications to an institutional repository will ultimately enrich the repository for other scholars in the field.

Sequential actions

The lifecycle's outermost circle represents sequential and occasional actions. Sequential actions form the routine operations around collections or items, while occasional ones occur as necessary. The aforementioned master plan will govern these two throughout the lifecycle of digital collections. Conceptualizing may initiate these processes on two levels: the first—as addressed at the beginning of this chapter—starts the curation process, and that is where discussions begin to start the curation process; and second, the conceptualization may initiate a new collection, record group of data curation project. In this case, critical discussions must precede conceptualization in order to decide what the development of such collections will accomplish. Stakeholders and project participants will decide on the workflow, lines of communication, and data capture methods governed by established standards for digitization and metadata.

Conceptualize

The master plan assumes that the implementation team has used stakeholder feedback and other information to conceptualize the institutional repository. Conceptualization at the collection level implies working with the record-creating departments, special collections, archives, faculty, and possibly community members interested in developing cultural heritage collections. Capture methods, metadata schemata, and classification schemes will vary by the collections curated and, hence, the audiences in mind. Media formats will vary by intended use; therefore, if researchers want to access the original files or the best-quality digital copy, they will seek specific file formats, but for Web access, the teams must prepare surrogate sets accessible on the Web and in social networks. The use of social tags may be necessary while using social networks because these platforms do not necessarily support standard classifications.

Whether collections are originally created or received products, there must be questions about copyright, intellectual property, work for hire, creative commons license, and possible restrictions in the deed of gift. Due to the heterogeneous nature of the collections, a boiler-place creative commons license will not be appropriate. Digitizing and publishing oral histories, for instance, will require permission from all copyright holders before the editing process. Even with those permissions in place, such recordings must go through editing to remove information that may violate someone's privacy. Research papers in the medical field must comply with the Health Insurance Portability and Accountability Act of 1996 (HIPAA) and comparable international laws to protect patient privacy. Faculty and staff publications must remain the intellectual property of the faculty unless some prior

agreement makes them works for hire. Faculty may submit prepress versions of their articles for viewing by a limited number of people (such as departmental colleagues) during an embargo period. Consulting the Sherpa/Romeo database can help authors determine the length of an embargo period and restricting public access to the article in the repository. Articles submitted to an open access publication will not be accessible until after the formal publication of the volume in which that article appears. While archives prepare deeds of gift prior to accessioning manuscript collections, the creators of donated records retain the rights to the intellectual property in those collections. Researchers must locate them or their heirs to obtain permission to publish materials from those collections. Various laws apply to posting data on federally funded research and patent applications. Developing some collections, therefore, must await consultation with the university's legal counsel and attorneys representing faculty authors in contentious cases. Similar requirements may apply to research data curated in repositories. In all, the conceptualization as well as the sequential action must pass through legal scrutiny for some types of content.

Create or receive

Creating and receiving collections represent two distinct actions on digital collections with regard to lifecycle: the former assumes that the digital collection has not existed and is in the process of being created anew. Digitally produced oral history interviews illustrate this class since they may be a product of historians, archivists, and community volunteers interested in interviewing war veterans, government officials, or other eyewitnesses with a critical first-person account of events. Digital photographs of institutional events may also fall into this first class because there are no predigital originals. Curators will create administrative, descriptive, structural, technical, and preservation metadata that documents the creation, provenance, and other important data pertaining to its lifecycle. Identifying the audience is an important part at this stage, as the relevance of the collection will be measured through usage and input from the user community. This is a part of the master plan and collection development plan where curators can communicate with the audience about the relevance of the collection.

With regards to receiving collections from donors, provenance is an important concept in archival practice as well as digital curation. The archivist will document the history of ownership and transfers throughout the lifecycle of received manuscript collections. Ownership as a historical context can validate the contents of the manuscript collection, and associate records with various organizations and individuals who may have altered the collection intentionally or by accident. The documentation of such changes appears in the provenance section of the finding aid after an interview with the donor. In digital curation context, the ownership history automatically appears in the provenance field of the technical metadata. The integrity of this data lends reliability and trustworthiness of the collection and the repository. These provenancial properties are different from those of created collections since the provenance field documents the lineage of collection ownership. Received collections in born-digital formats must also comply with the collecting scope stated in the master plan and go through the same documentation process. With some differences (such as item-level metadata) a single metadata template may be sufficient, allowing researchers using the digital files to locate records submitted in print form and vice versa–to locate the digital files with the help of print records. Archivists

and digital curators can collaboratively develop enhanced finding aids as epistemic gateways in order to make connections between print and digital records and minimize the silo effect common in some institutions' libraries.

Appraise and select

The concept of archival appraisal is important in digital curation as well: appraisal helps archivists determine the evidential and informational value of collections (Cox, 2000). In archival theory, records have evidential value—or "the quality of records that provides information about the origins, functions, and activities of their creator" (Pearce-Moses, Ed., 2012, "evidential value,")—while they still serve an important purpose in the record-creating organization as they are still being produced by the organization. Informational value, or "the usefulness or significance of materials based on their content, independent of any intrinsic or evidential value" (Pearce-Moses, Ed., 2012, "informational value,"; see also Pearce-Moses, Ed., 2012, "evidential value," "intrinsic value") is the research value after the records are no longer a direct product of the organization.

The intrinsic value of collections—especially in archival practice—presents a unique set of rationale to preserve collections, but in the digital environment, the justification to do so must have a rationale. Intrinsic value is "The usefulness or significance of an item derived from its physical or associational qualities, inherent in its original form and generally independent of its content, that are integral to its material nature and would be lost in reproduction" (Pearce-Moses, Ed., 2012, "intrinsic value"). Digital repositories usually represent the latest technologies in preservation, and users will most likely interact with such content, using the latest technology. However, as digital collections reach a stage where they become obsolete or incompatible with emerging technologies (e.g., Flash files to handheld devices, older DOS computer games, or old Access database files that no longer open in more recent versions of Access), accessing them will be challenging without converting them into open-standard formats and making them accessible in these repositories at the risk of data loss. While their retention may consume space, it may also provide data historians and forensics experts an opportunity to analyze the inherent structures of the old files. Selection is the decision that reflects the collecting scope and policy. Historians may also select data sets for curation to support historiographical work using digital data, while other humanities scholars may require such data for hermeneutic purposes.

Ingest

Ingest involves the transfer of digital content from the record-creating department (archives, library, academic department, administrative office, or some committee that has produced publishable reports) to the curator of the digital repository. Laws govern the transfer of intellectual property and the protection of privacy, and no popular demand or other interests may supersede these laws. Although donors sign a deed of gift agreement with the archives when transferring donated collections to the archives, it is highly possible that a similar agreement will apply to digitization of donated materials. Some deeds of gift may already contain language to address digitization as a means of giving access to the information in those collections for research purposes, but limitations apply, affecting open access to such collections.

There are also technical aspects of the ingest process whereby data sets, digital objects, and descriptive metadata are added to the system. Ingest involves several methods, depending on the repository in use. DSpace, CONTENTdm, Digital Commons, EPrints, and others use different architectures and process for single-item and bulk import ingest methods. In most cases, the ingest process requires metadata spreadsheets with specific fields necessary to describe collections and items in each category. The completed spreadsheet generates XML data or a tab-delimited text file for the repository system, which parses the data into the fields included in the metadata sheet. The data becomes searchable and browsable through those designated fields. The ingest process, therefore, involves legal, intellectual, and technical processes.

Preservation action

Archives and libraries have digitized special collections as a preservation method to minimize manual handling of rare original manuscripts, photographs, and artifacts since most researchers were interested in the information in those collections rather than the medium. There is an ill-informed and widespread notion that the originals could be discarded because the digital copies are so much easier to access. Digital media is ephemeral; therefore, it will not outlast such physical media as film or paper and periodic content migrations will be inevitable. In fact, the digital copy requires more rigorous preservation actions over its lifecycle because the hardware, software, operating systems, and file formats become obsolete within a few years and inaccessible within a decade, after which accessing the information on old media becomes a preservation and conservation challenge.

There are short- and long-term preservation measures to prevent data loss in digital repositories and servers: short-term strategies usually include tape backups done overnight, which generates multiple copies of the data available for several months. A backup routine may not necessarily check the structural integrity of digital collections; thus, the disc image may retain structural flaws. Checksum reporting may be a useful technique to identify bit-level problems in the digital content and the associated metadata, but Newman et al. (2012) report that some repositories such as DSpace do not produce a complete report and point to the affected files. Long-term preservation is possible with dark archives in such platforms and services as Chronopolis, Duracloud, and Portico. Active preservation is a long-term strategy, which involves several routines such as format migration and data transformation in order to maintain access to the information, although this approach may have some unintended consequences.

Storage

The selection and use of storage methods will depend on the institution's technology and financial resources. There are several methods of storing digital content, but not all methods are identical to preservation or dark archiving, and not all are equally stable. At the time of creating the content, the immediate storage methods include a computer's hard drive, other temporary storage devices (such as Flash drives), or networked drives on the institution's intranet. For long-term storage, hard drives, network drives, and cloud-based storage are commonplace, but these are not equivalent to preservation

despite the periodic backups. Finally, some recordable and rewritable optical discs (CD-R, DVD-R, DVD+R, DVD-RW, and DVD+RW) are suitable for long-term storage but require ideal temperature and relative humidity levels to increase longevity of the media (NIST/LC, 2007; see also OSTA, 2001).

Each digital repository comes with storage capability, but there are differences among these platforms. There are content management systems such as Drupal and Joomla, which have no preservation functions but provide easy access to the files and turn into a Web site management tool. Repository platforms fall into two categories—open source and commercial—and can operate in in-house or hosted environments. In both cases, the storage may be physical, virtual, or cloud based, which raises questions about the quality of storage media, integrity of files and directories, and the frequency of backups. Operating systems are also important: DSpace, Fedora, and Omeka are open-source products that install only on Unix/Linux-based environments, while products such as Greenstone can use both Linux and Windows. Prior to launching the ArchiveSpace project, Archivist Toolkit had worked in both operating systems using either the MySQL or MSSQL database model; the decision to go with MySQL in the ArchiveSpace environment indicates a stronger support for Linux however. Curation tools such as BitCurator also use Linux as the ideal operating system, but it can run in a Virtual Machine (e.g., VirtualBox) environment. Open standards are crucial in long-term preservation because they are less susceptible to commercial trends than proprietary products are.

Transform

Hedstrom (2001) regarded temporal interoperability as the incompatibility of file formats with the changing technologies over a period of time, and this is what transforming resolves. Curators may be in a position to transform data, databases, files, and directory structures when the technologies used for creating the content are no longer accessible. For instance, databases created in Access 1997 required conversion into more recent database format in order to open in Access 2003. A simple patch in Word enabled users to open old Word (1998) files in Word 2007. Transforming such files as simple PDF, Word, Excel, PowerPoint, and image files into PDF/A have advantages because the PDF/A file is self-contained and sealed as to retain its original character by not allowing any changes. This strategy will allow archives to create archival-quality digital documents and will enable historians to verify the authenticity of digital files against a print or digital original.

Transforming metadata is also necessary from time to time because metadata interoperability standards, Open Archives Initiative Protocol for Metadata Harvesting, and best practices change with time. Such transformations entail changes in the metadata field, data entry methods, or identifiers, and may be necessary even if there is no migration. However, the transformation is more pervasive when collections must move between different repositories with different data structures, metadata requirements, and file types. In such cases—even if both repositories use the same metadata schema (e.g., Dublin Core)—the changes to headers, elements, data types, identifiers, file names, and other details will require significant changes to the metadata. An example of this level of transformation is when collections move from DSpace to CONTENTdm where

the former system requires submission packages in XML format while the latter, tab-delimited text files. The metadata elements and qualifiers are specified in DSpace but not in CONTENTdm. Even if DSpace exports the metadata file in a CSV format, the metadata file requires significant cleaning before conversion into other formats and templates compatible with CONTENTdm, Bepress, or other repositories. Such conversions can be minimal (e.g., changing headers) to extensive, affecting the name format, for instance.

Occasional actions

Reappraisal, disposal, and migration are related activities in both archival and digital curation practice. Reappraisal follows an assessment activity that identifies certain records, manuscript collections, and digital collections that do not merit further preservation. Although this is a rare action for cultural heritage collections, it is possible, and there are required procedures for the withdrawal of deaccessioned collections from public access. In archival practice, deaccessioning is "the process by which an archives, museum, or library permanently removes accessioned materials from its holdings" (Pearce-Moses, Ed., 2012, "deaccessioning"). Archivists withdraw the finding aid from public access, and remove any digital collection with any associated metadata in a digital repository. What follows is either the disposal (and subsequent destruction) of the records or their migration to a new repository. For digital collections, migration may not only mean a simple transfer of collections to a new repository; it may also involve transformation of the metadata record, migration to a new file format, and possible reorganization to fit the architecture of the new repository. Such a reorganization is almost necessary for collections migrated from DSpace to CONTENTdm because the latter cannot accommodate the hierarchical structure of DSpace collection; the structure requires flattening and simplifying the metadata record.

Conclusion

The DCC lifecycle model presents a useful framework for the conceptualization, implementation, and evaluation of a digital curation place. This chapter presented a modified curation model with all the stakeholders in place and analyzed the requirements at each stage. From the discussions, it is evident that while conceptualization only occupies the space of an arrow in the DCC lifecycle model, it is in fact one of the most formidable tasks before implementation. It must consider all the institutional factors such as curricula, data curation requirements, archival records management, scholarly communication, and community expectations. Deciding on the metadata schema and controlled vocabularies fall under information representation, so close collaboration between the digital curator and catalogers will be necessary. The decisions (and mistakes) made in the conceptualization stage can be corrected during implementation, but periodic assessments and evaluations will also be necessary to inform any changes to preservation strategies, metadata schema, records retention, and various outreach efforts toward faculty, administration, and community members.

Organizational dimensions of digital curation

6

Organizational dynamics are vital to digital curation, as the organization becomes a social, political, and intellectual context at various levels of collaboration. An educational institution's size, full-time enrollment, curricular interest in Digital Humanities (DH), organizational culture, and politics are significant factors in supporting the archives, special collections, and digital curation initiatives. The extent of humanities and DH scholarship at academic institutions translates into support for archives and special collections, as DH faculty and students will, at different stages of their projects, use digital collections at their institutions' archives and research libraries. Consequently, institutions with strong support for DH scholarship will lend more financial and political support for DH projects and digital curation in their archives and libraries.

Digital curation ranges from one department—usually a combined archives and special collections unit—to multiple units including the archives, technical services, information technologists, academic departments, and the community at large. The size of archives and range of digital curation efforts, thus, vary by the size and mission of parent institutions, and may consist of one team (of two or more members) to several interrelated multimember groups. In smaller American institutions, archives and special collections comprise one unit in academic libraries, but at larger institutions, they may be detachments to academic departments such as history, anthropology, popular culture, urban studies, architecture, or others with significant special collections. Small universities and colleges usually maintain small archives with one employee with the combined responsibilities of the archivist, records manager, curator, or manuscripts librarian and maybe one support staff where budget and politics permit. Limited technology skills, support, and very small collections generally result in small-scale digital curation programs. In contrast, very large universities may not only have separate archives for records management and university manuscripts collection but also designated archives and special collections serving individual departments as well.

Concerning workflow and collaboration politics, the single-unit scenario presents important advantages over multidepartmental ones: the former scenario enables smaller teams (e.g., archives) to focus their energies on cultural heritage materials, local history collections, and archival records. In contrast, the multidepartment scenario requires larger staff and closer working relationships, which often presents political and administrative challenges, but there is an advantage of having a broader knowledge base at collaborators' disposal. While organizational politics are beyond the scope of this chapter and the book, political dynamics significantly determine what and how expediently digital projects receive priority, funding, staffing, and other means of administrative support. The significantly larger staffing and technology support at these institutions would typically result in multiple extended and interrelated digital curation programs. At institutions investing minimally in organized curation

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initiatives, the staffing, financial, and technology support are also minimal, which will hinder if not stall curation.

Placing the digital curation lifecycle model within the framework of the parent organization can help curators conceptualize curation projects, identify the stakeholders and participants at various stages of the lifecycle, and implement a digital curation initiative. While many decisions come out of the archives, critical support must also come from various library departments, administration, and technologists. This chapter focuses on the organizational dimensions of digital curation, placing the digital curation lifecycle in a higher education institution's context. It will first address knowledge management aspects, since the digital content lifecycle model itself will benefit from active knowledge exchange among teams and stakeholders. The chapter does not explore knowledge management theory and practice extensively, but will discuss at length the role of knowledge architectures and knowledge transfer in digital curation in four distinct settings: archives, academic library, institution, and community. This approach will demonstrate the relationship of the digital lifecycle model to curation initiatives at various levels of the institution and surround communities.

Knowledge management in digital curation

Most recent publications place knowledge management within the context of corporations seeking to capture human capital and knowledge through technological means; however, knowledge management presents value to archives, libraries, and academic institutions. Due to the highly collaborative nature of digital curation in various organizational contexts, the exchange and transfer of knowledge among individuals, teams, departments, and organizations elevates the role of knowledge management in archives. Definitions for knowledge management are ubiquitous in the publications and conference programs. An early definition by Thomas Davenport is "the capturing, distributing, and effectively using knowledge" (as cited in Koenig, 2012, para. 2). In another definition, "knowledge management is the discipline of enabling individuals, teams and entire organisations to collectively and systematically create, share and apply knowledge, to better achieve their objectives" (Young, 2009, para. 2). The digital curation lifecycle (Digital Curation Centre, n.d.) integrates knowledge management in various regions of the model, emphasizing the need for creating content for purposes of use and reuse.

In the higher education context, the aim of universities has been to produce knowledge, but faculty are truly concerned about the implications of managing knowledge, intellectual property, and "academic capitalism" (Metcalfe, 2006, p. 5). While embedding practices from the corporate environment into some academic institutions may arouse curiosity as well as suspicion, there are benefits to incorporating knowledge management strategies in some teams of faculty, students, and information professionals engaging in sharing and transferring knowledge. For the present purpose, knowledge—including academic knowledge—will imply sharing descriptive information; personal knowledge of archivists, librarians, and faculty; and knowledge attained in the classroom. All intellectual property necessarily belongs to those who create knowledge in the various disciplines, and this is where academic institutions differ from corporations.

Knowledge sharing and transfer will, however, improve digital curation efforts as two or more individuals (archivists, librarians, faculty, students, and others) exchange operational knowledge in addition to transferring digital content. For instance, archivists develop knowledge of collections they process and would share with digital curators implementing long-term preservation plans using the digital curation lifecycle model. Some of that personal (or tacit) knowledge, including the documentations added in digital history projects, is present in the metadata record and will be useful to curators making decisions and taking appropriate preservation actions. Someone's personal knowledge of preservation issues, provenance, and missing descriptive data can help the department improve service at the institutional level.

Knowledge architectures for digital curation

From the knowledge management perspective, therefore, this chapter treats departments as knowledge architectures, which are vital to the organizational framework of digital curation. Knowledge architectures present the structural frameworks for the creation, management, exchange, and transfer of knowledge and is different from and yet somewhat related to information architecture, which is about the navigation of information spaces across multiple domains. For present purposes, the former is macroscopic, while the latter, microscopic. Within the curation space occupied by archivists, digital curators, and digital humanists, knowledge architectures entail the networked use of technologies and contents with their own respective information architectures. They also present a framework for hypertext content, linking meaningful data and information to support decision making at management levels on the one hand and analytic and interpretive work on the scholarly level on the other. In academic knowledge management context, there is a relationship between these two (operational and academic) aspects (Sabharwal, 2010).

Definitions for *knowledge architecture* vary by discipline engineering, information technology (IT), information science, etc., but the one offered by Rebentisch and Feretti (1995) is a starting point: "a characterization of the structure and the artifacts into which knowledge has been embodied in the organization, and describes the way an organization stores and processes information" (p. 10). The reference to artifacts is important, as archives frequently preserve artifacts in two- or three-dimensional and digital form as well. Applehans, Globe, and Laugero (1999) take a Web-based approach to knowledge architectures, which consist of three components—content, people, and technology—joined in collaborative relationships and are critical to maintaining workflows throughout organizations. "The knowledge architecture represents your company's formal recognition that it has important experience and expertise that it must preserve and use to its advantage" (p. 32).

Knowledge architectures have an important role in the digital curation lifecycle, as they interact at different times through the lifecycle model. From an archivist's perspective, "people" refers not only to those processing the collection but also to those who donate, access, preserve, teach, and publish. They are archivists, historians, catalogers, technologists, citizen scholars, and schoolteachers who manage knowledge

individually and collaboratively. The Web-based model for knowledge architecture presented by Applehans, Globe, and Laugero is applicable to digital curation, as it extensively relies on knowledge sharing on the Web. Along with the digital collections, archivists share data; information, or data in some context; and knowledge, a combination of data, information, experience, and heuristics needed to support decisions and actions; such a data-to-knowledge continuum (Ackoff, 1989; Zeleny, 1987) forms a part of the digital curation workflow. The combination of metadata, finding aids (which provide historical context), and internal databases containing operational knowledge outlines this continuum in the workflow involved in digital curation.

Archives present a distinct example of knowledge architecture. Archivists work with information related to provenance, processing, and preservation issues. In this context, knowledge architecture brings repositories, communities of practice, and the learning process into a relationship in order to facilitate knowledge access and exchange through which the community can contribute to new knowledge (Wallace, 2006). The role of archives in such knowledge architectures is vital to sustained learning because archives develop and maintain content in repositories necessary for learning and other areas of institutional operations. Manuscript collections contain primary sources needed for humanistic scholarship and universities' archives contain data, information, and knowledge to support administrative decisions. The relationships among the people in the various departments and teams, the technologies that they use, and the processes underlie knowledge management behind the scenes of digital curation, and the ability of these teams to create, access, transfer, and reuse digital contents. Whether addressed as such or not, academic institutions create archival records, which comprise institutional knowledge and require preservation. The digital curation of digitized records, datasets, and institutional scholarship in institutional repositories will ensure long-term access to such records through sustained support and collaboration.

Knowledge transfer in digital curation

Knowledge transfer is vital to collaboration and the digital curation workflow, but in an environment involving disparate teams (archivists, historians, technologists, administrators, broadcasters, and others) there are limits to what knowledge may be meaningful and useful across the organizational boundaries. For instance, how much of the descriptive or preservation metadata will be useful to administrators or technologists who often fail to consult archivists and librarians before making uninformed (or worse) decisions? The aim of knowledge transfer is to connect those who create knowledge to those who need it—academically and strategically. The digital curation lifecycle model specifies stages where preservation planning, community watch, and curation will benefit from knowledge transfer.

Milton (2005) categorizes various methods of knowledge transfer, which apply to digital curation across multiple organizational levels. The "connect and collect" approaches correspond to direct (face to face, correspondence, or dialogue over networks such as teleconferences) and indirect transfers (using databases, knowledge banks, and libraries), respectively. With regard to the sequence of projects, Milton

differentiates between serial and parallel transfers: in the context of digital curation, the former occurs in a series of projects undertaken in a single location (such as a department), whereas the latter, in simultaneously occurring projects across a broader organizational spectrum (e.g., across archives, academic departments, and community organizations). Hence, there are also near and far transfers, referring to the physical proximity of projects, but Milton emphasizes the asynchronous aspect of these two methods that are otherwise similar to serial and parallel transfers occurring in real time. Near and far transfers, therefore, necessarily rely on knowledge bases, databases, and digital repositories. These methods are vital to anywhere between small settings such as archives to larger ones involving multiple collaborating teams and departments. Digital repositories can facilitate knowledge transfer once institutions, legal counsel, and libraries resolve legal issues regarding intellectual property and copyright protection. Depending on the institution's size, technological capabilities (and support), and special collections, one or more of these transfer methods may be possible through the working relationships within and across departments. The effectiveness and scope of digital curation programs, therefore, strongly depend on supportive knowledge architectures. This chapter, therefore, underscores the importance of integrating digital curation of special collections into the archival workflow instead of circumventing the archives (as well as the legal and ethical aspects) like some eager but uninformed administrators may envision.

Organizational contexts for knowledge architectures

The following analysis considers four distinct contexts for knowledge architectures, which are: (1) the archives and special collections in a university's academic library, (2) the academic library, (3) the university's academic function, and (4) the broader local and global community. Each context presents a progressively larger organizational dynamic for digital curation, allowing a continuum for digital curation to develop from the starting point (the archives) to the destination (community audience), where curation begins to extend beyond the control of the initial curator. A context also represents an organizational level, with members of the teams interacting and collaborating on cross-departmental projects or tasks. Knowledge architectures are hence fluid because (1) the people may include members from multiple teams across departments and administrative units; (2) they may share various technology resources such as storage, software, and hardware; and (3) some contents may eventually be distributed throughout the institution and the community engages in continued and possibly joint curation of some digital contents. At each organizational level and beyond, the collaboration of various units can ensure the promotion of digital collections, but the collaboration involves more than the exchange of contents; it also involves the transfer of knowledge via metadata.

Archives and special collections

The digital curation of special collections, institutional records, and other archived material requires adequate trained staff and specific technologies to meet expectations

from university faculty, administrators, and other stakeholders. Given the specific mission, responsibilities, and capabilities of archives, the knowledge architecture is specific to carry out preservation, conservation, curation, and other archival functions. "People" includes such specialists as records managers, manuscript archivists, conservators, curators, digital media preservationists, catalogers and indexers (in larger facilities), and processing staff. In some minimally staffed archives, part-time employees and students do the processing—however strange that sounds, but it is true—which presents challenges to continuity and knowledge transfer. "Technology" includes productivity software, departmental shares, digital reformatting equipment, digital library software, some Web 2.0 applications for encoded archival description (EAD) finding aids, library guides, social networking sites, and other associated applications. "Content" includes finding aids, special collections, digital libraries, digital finding aids, virtual exhibitions, and various artifacts used in physical exhibitions. The workflows bringing the three components of knowledge architectures are vital to digital curation because of constant reliance and use of specific technologies and staff to provide access to digital collections. Digital curation requires such workflows in archives.

Figure 6.1 presents a model for knowledge architecture in the archives and special collections department. The source of knowledge in this scenario is the record-creating individual or corporation whose records serve as evidence to the organizational structure and activities over a specific period of time. The transfer of records can be a one-time or continuous process, and the method may be serial, parallel, near, or far transfer, depending on the donor–archives relationship and physical proximity (Milton, 2005). Once the archivists complete accessioning and processing the collection, they can initiate other knowledge transfers by contacting the cataloger to enhance the metadata record, outreach staff to promote the collection, and reference/instruction librarians to inform faculty and students about the collections. Knowledge transfer in larger organizations may also involve working with curators who plan for exhibitions and digital curators who develop digital collections and virtual exhibitions. Two-way communication is a vital aspect of knowledge transfer, but missing links can often cause even well-planned processes to halt.

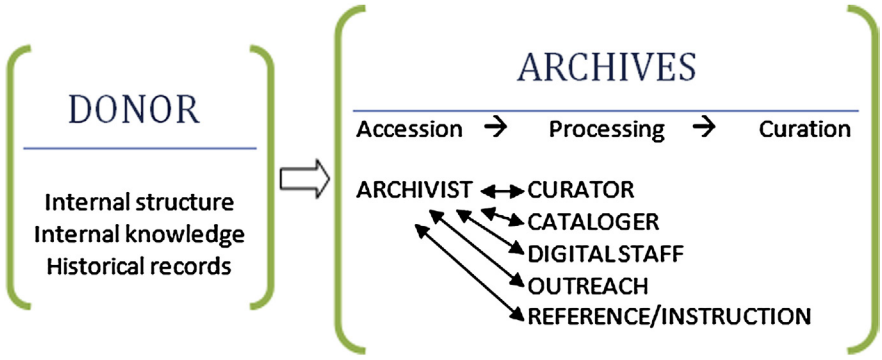


Figure 6.1 Archives as knowledge architecture.

The process begins with accessioning the collections with considerable knowledge about the background, provenance, and current condition of the collection. By routine, archivists conduct background research and personal interviews with the donor to develop a better understanding of the collection. This is an essential part of the appraisal in order for archivists to determine the relevance and suitability of the collection. As archival processing also involves arrangement and description, processing archivists develop an in-depth knowledge of the collection as well as the internal workings and structure of the donor organization or, if it is a personal donor, his or her environment being documented in those records. The finding aid is, therefore, more than an inventory of contents; it is an intellectual map to the collection with important contextual information presented in the biographical or historical sketch as well as in the "Scope and Content Note" sections.

The processing archivists' knowledge of the organizations or circumstances leading to the creation of historical records points to two values: evidential and informational. In the former case, records were vital to the operation of an organization. Once the organization ceases to function, those records begin to have informational value useful to researchers. This kind of knowledge significantly improves the quality of information researchers get from archivists during a reference transaction and helps curators present more accurate and extensive information at exhibitions. It also helps archives reach out to the appropriate audiences—faculty, students, and others—more effectively since they can identify those audiences. The knowledge gained at the initial stages of curation is vital to the successive stages where archivists share and transfer knowledge within the department. In many cases, the initial stages of curation—accession and processing—do not involve digital preservationists unless the collection in question is born digital supplied with metadata. The working assumption here is that smaller archives do not have a cataloger to help with metadata, but where that is the case, such a cataloger will bring value to the continued curation of archival collection because the catalog becomes an additional access point with truly relevant metadata. Depending on the extent of digital curation, an institution may be managing descriptive as well as administrative, technical, preservation, and copyright metadata.

For collections in print, film, magnetic, analog, or other tangible formats, knowledge transfer between the archival and digital staff may include any one or more of the aforementioned metadata classes, but knowledge about the background, provenance, processing, and physical condition of the collections will certainly help digital preservationists enhance their metadata records. For instance, items with missing or incomplete metadata records may translate into action items to clarify the identity of donors, people, or places on various media, or further processing issues. Collections with provenance information in the metadata record can assist researchers identify donors of other collections and resolve various enigma about the relationship of some collections in a specific area. Although many archivists have subscribed to the "More Content, Less Process" approach to make more content instantly available, one benefit of progressive/successive metadata enhancements is improved information retrieval and collocation through searches across the Web, union catalogs (like WorldCat or BASENet), finding aid databases such as ArchiveGrid, as well as digital repositories like DSpace, CONTENTdm, Fedora Core, Omeka, Digital Commons and others not listed here. Archivists'

knowledge in the metadata records of collections in these platforms is accessible in machine-readable forms using XML, OWL, HTML, and other schemas, which inform researchers about the condition, location, and accessibility of born-digital and digitized collections. Ongoing knowledge transfer is, therefore, vital to digital curation, and an archives knowledge architecture presents an ideal structural framework.

In the past decade, digital curation programs have gradually expanded toward using social networking and various Web 2.0 applications that allow archives to interact with their audiences. The use of blogs, Flickr, HistoryPin, Facebook, Twitter, Instagram, and other communities has allowed archives to curate their collections in ways that had not been considered before. Archivists with Web developing skills have created virtual exhibitions to prolong access indefinitely to the exhibitions and collections featured within. Since virtual exhibitions usually present the information at the physical exhibition, this relationship demonstrates the need to transfer knowledge between archivists and curators to a Web designer (who may be an archivist). The virtual exhibitions provide links to related digital collections, finding aids, library collections, and perhaps even social networks for visitors to connect with the collections and audiences in new ways. The discussions around the exhibits and topics demonstrate the relevance of archives to the public and enrich public discourse around these collections, providing visitors interesting insights and perspectives.

Academic library

Most archives in academic institutions operate within their parent institutions' research libraries, but some larger institutions have designated archives within their academic departments. Elsewhere, they may be entirely independent of the institutions' libraries. Independence from the main library may have advantages if those archives and special collections departments employ catalogers to integrate collections and knowledge into the main catalog record. In the majority of academic libraries in the United States, however, catalogers work in the technical services department, while digital project librarians work in either special collections or technical services. Any lack of a supportive knowledge architecture in such settings may present some challenges to the digital curation workflow therefore.

However, digital curation is not limited to the relationship of archives to technical services but public as well. Reference and bibliographical instruction may also be instrumental in promoting special collections since in most cases, the public encounters staff in these two departments first before contacting (or discovering the existence of) the archives. Most archives provide reference assistance at their facility, which allows archivists processing the collections to assist with very specific reference questions, but reference and instruction librarians may play a vital role in connecting researchers with related materials in the libraries holdings. Therefore, research libraries present a supportive multidepartmental knowledge architecture even if the locus of digital collection development and preservation remains in the archives.

An extensive digital curation program will not utilize all the reference and instruction resources, but will focus on what the librarians in these departments can contribute. Since many reference and instruction librarians use Web 2.0 tools such as Blackboard

and Libguides they have an opportunity to participate in digital curation to a reasonable extent. Such participation may range from adding simple links to the institution's digital collections within Blackboard, Libguides (or other similar platforms), and Web pages to activities such as blogging, tweeting, Facebook posting, and the link. Since subject librarians function as liaisons to academic departments, their contributions to digital curation may include adding appropriate links to certain resource guide, online courses, and sharing knowledge of these collections with faculty who may include them in their course plan.

Thus, the people of the knowledge architecture at this level includes archivists, faculty, reference, and instruction librarians whose knowledge of these resources focuses on the relevance of those contents to the curriculum. Descriptive metadata does not necessarily involve the use of MARC, Dublin Core, EAD, or other schema; in fact—as addressed in the chapter on the ontological aspect of digital curation—reference and instruction librarians interact with a virtual audience using social tags. The content remains the same: special collections in digital form, books, journals, Web resources, and reference materials. Technology includes electronic subject guides such as Libguides, Blackboard, blogs, and other social media channels.

Figure 6.2 presents instances of serial/parallel as well as near/far transfer (represented by the arrows) similar to those in the previous model with the difference that parallel transfers would be more common due to larger staffing, more multitasking, and more complex working relationships outside and among the archives/special collection units. Knowledge transfer is bidirectional in order to support a two-way communication and mutual feedback system that informs the digital curation process. For instance, input

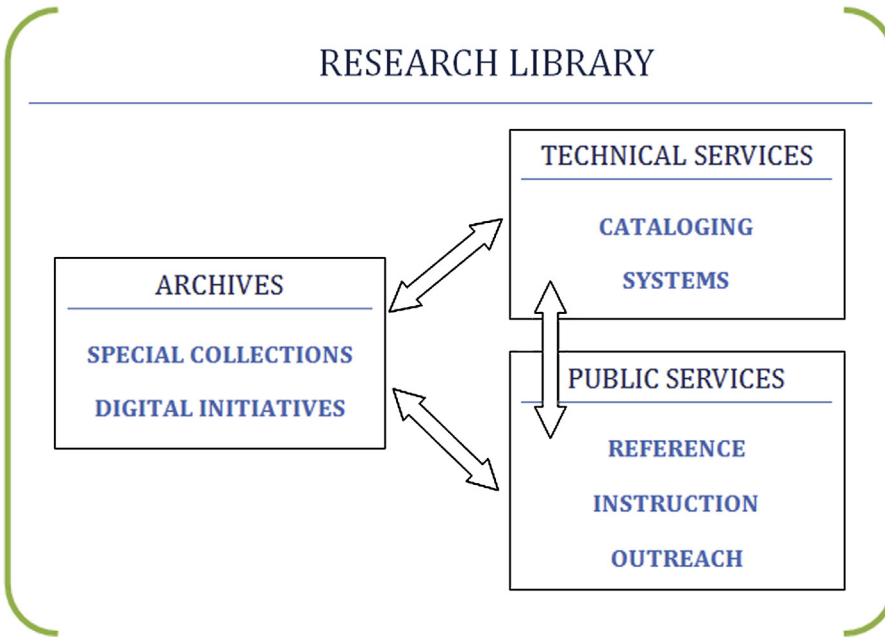


Figure 6.2 Academic library.

from catalogers, systems librarians, electronic resource librarians, and others in technical services can significantly enhance collaboration efforts with special collections and digital initiatives staff, resulting in new possibilities for digital curation. Continuous dialogue can bring these digital and electronic resources into two-way relationships. Some catalogs now allow users to tag and comment on the collections. The bidirectional relationship between archives and public services results from meaningful dialogue with instruction and reference librarians. Their role as mediators between content and public is potentially valuable; however, not all technical and public services units collaborate directly with archivists or digital curators; in such cases, the knowledge architecture is less supportive of digital curation.

Institution and curriculum

Beyond the archives and the library, institutions represent a dramatically different knowledge architecture partly due to curricular interests in Digital Humanities (DH). Faculty engaged in Digital Humanities research and teaching will seek collaboration with archives, digital experts, and technologists to develop thematic research collections such as those discussed in *The Companion to Digital Humanities* (Schreibman, Siemens, & Unsworth, 2004). The companion discusses several digital archives featuring the archival collections of the parent institution as well as textual materials authored by faculty. Pioneering efforts of the late 1980s in DH predate the World Wide Web and demonstrate the extent of institutional engagement. The university's multidimensional knowledge architecture consists of a number of smaller architectures—those of archives, special collections departments, and library units—in collaboration while sharing technologies and digital content. The curation of collections throughout the university continues with the faculty using, enriching, and contributing to the archives.

This dynamic is particularly noteworthy at institutions offering traditional, online, and hybrid online courses and where digital curation may well extend beyond the archivists' professional control. The curation of collections in these courses involves faculty, archivists, digital curators, and students using the digital collections for extended analysis and interpretive studies. The knowledge developed through curation in the curricular context differs considerably from the descriptive information available through the metadata records created by archivists. In the model discussed in Chapter 2 using Panofsky's framework for iconological study, archivists, metadata librarians, and catalogers can enhance the newly added information to the metadata record after students have completed research and discovered information related to previously cataloged and digitized collections.

An expanded knowledge architecture would involve the following people in digital curation: faculty, students, instructional technologists, IT staff, archivists, librarians, and various curriculum administrators. It is evident that this knowledge architecture results from several smaller architectures collaborating across departmental boundaries. The technology in use ranges from learning management systems such as Blackboard, Moodle, Sakai, iTunes University, and digital repositories and Web sites hosting thematic research collections. The content includes collections from the archives as well as the group projects and thematic research collections. This multidimensional

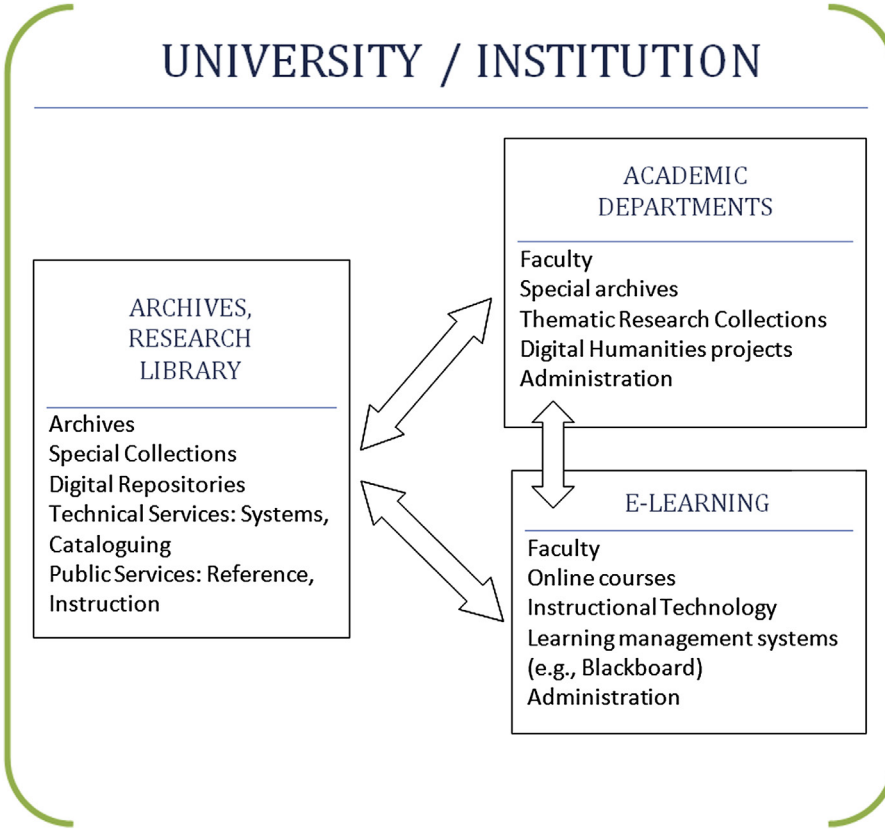


Figure 6.3 University or other institution.

architecture revolves around the structure of DH curriculum, and varies by the size of the institute and extent of support for DH education and scholarship.

Figure 6.3 illustrates an intricate combination of all types of knowledge transfers at a university or research institution too large for direct transfer. Such organizations are likely to cultivate indirect and far transfers. While archivists at the central repository and those in departmental archives may engage in direct, serial/parallel, and near/far transfers, faculty, administrators, and technologists may at best be using the digital library for information instead of interacting with the archivists. Such an extended reliance on indirect transfer, however, may have disadvantages because the archivist's knowledge and experience is missing at a level where data, information, and knowledge are expected to inform and support decisions. The faculty's awareness of special collections usually relies on direct transfers in the form of reference transaction and bibliographical instruction by librarians, but the use of electronic subject guides and learning management systems has gradually tipped the balance in favor of indirect transfer, even though the benefits of direct transfer have clear advantages. Finally, as universities and colleges invest more and more on online learning, knowledge transfers

are indirect and far with the exception of periodic face-to-face meetings that do not always afford the time to address details. The case between faculty and instructional technologists may be an exception where they may work directly, but in many cases, course administrators serve as mediators and coordinators. A solution to improve the quality of information from the archives is to bring archivists into direct working relationships with instructional technologists, and the process will benefit from the combination of direct, parallel, and far knowledge transfers. This approach would ensure that digital curation is possible with the information directly provided by someone from the archives and special collections departments.

Kapelos and Patrick (2012) describe collaboration among course instructors of architecture, subject librarians, and special collections at Ryerson University's Library and Archives Special Collections. The curation of visual resources showing architecture allows the transfer of knowledge to architecture faculty. A particularly important distinction between images of architectures and architecture itself is that the former relies extensively on the perspective and subjectivity of a photographer, whereas in the latter case, a large number of photographs present details allowing greater emphasis on architecture through a multitude of perspectives. Through the representation of reality through well-preserved, described, and curated photographs, the librarians and faculty were able to transfer some knowledge to students who then combined their esthetic experience with research to present new knowledge. With the new Web 2.0 technologies allowing new ways of visualizing architecture, faculty, archivists, and technologists can collaborate on linking the visualized space with information on the content. Good examples for this are the *HyperCities* and the Google *Art Project* projects that provide access to descriptive metadata of the linked image (painting, sculpture, architectural model, text) on the screen. In this case, knowledge transfer does not merely focus on the subject matter, but the technological application toward visualizing that information. Likewise, Viewshare allows visualizing historical information on maps and timelines, connecting the information to well-curated archival collections.

Community

The fourth and most expansive context for digital curation is the larger community—a broader hybrid knowledge architecture within and beyond the university. In the past, community outreach efforts of archives and libraries typically targeted the physical community surrounding the university, but in the past two decades, such outreach efforts have expanded to emerging virtual communities accessing the collections remotely. The community may include consortia of academic institutions with similar knowledge architectures. Multitype partnerships of dissimilar organizations (academic departments, special collection units, public libraries, historical societies, public broadcasting services, and museums) may comprise a larger knowledge architecture to work on such projects as Toledo's Attic. The direction of workflow varies among such organizations; for instance, the archives can digitally reproduce items from their collections for use in media productions, museum exhibitions, library outreach programs, and other events. The curation includes virtual exhibitions, which allows information to circulate on the Web and social media, thus directing visitors

back to these earlier venues or providing continued access to these collections long after the media programming, exhibitions, educational events, or library events end.

Local history and cultural heritage collections are getting more attention from old and young residents, which provides archives an opportunity to fill the missing gaps in knowledge about the digitized artifacts. It also provides archives to reach out to the community; therefore, digital curation opens doors to a mutually beneficial relationship between the archives and the community. Being involved with the regional events of the History Day also provides archives and special collections curators to work with public schools. Thus, the knowledge architectures present a structure and continuum for digital curation from the initial steps for developing archives to the collaborative curation of digital collections.

The people in this context include local historians, educators, broadcasters, news reporters, genealogists, museum curators, archivists, librarians, and technologists. Some may work as staff or volunteers at two or more institutions, which facilitates knowledge transfer across institutional boundaries. The content includes archived collections featured in exhibitions, public media productions, student projects, lecture series, and more. In the case of several academic institutions in consortial relationship, the contents may include several collections housed but collaboratively curated at these institutions. Technology used in these cases may include shared digital repositories, content management systems, and various media production systems.

Figure 6.4 presents the knowledge architecture (the community) in this context along with the relationships and workflows existing across those boundaries. This is not an exhaustive model, as several variations to this exist in practice. Conceptually, all the previous transfers apply to the community model, but in many cases, the units cease all or most forms of collaboration. A unique but often unstable combination of relationships supporting serial/parallel, indirect/direct, and near/far transfers among loosely affiliated libraries, historical societies, and public media may produce interesting results in digital curation. However, the lack of funding and other forms of support may render these relationships unsustainable and politically undesirable.

Knowledge transfer between the university and historical societies may be a combination of direct and near transfers because of faculty affiliation with the history or DH department and the historical society. In most cases, however, local history conferences present opportunities to exchange knowledge as a method of scholarly communication. Historians, humanities researchers, and technologists in the past have engaged in humanistic computing projects since the 1960s and the 1970s, and much later, involved librarians and archivists in various Text Encoding Initiative projects. Knowledge transfer in this case required combinations of indirect/direct, parallel, and near/far transfers.

University archives and special collections often collaborate on digital projects with other similar units in regional consortia formed around sharing technology and collections. In that case, knowledge transfer is indirect/direct, serial/parallel, and mostly far due to the distances. Periodic meetings present opportunities to exchange and transfer knowledge more directly, but there is no opportunity to complete projects at these meetings. The relationship of public libraries (with local history collections), public media establishments, and virtual museums mostly relies on indirect, parallel, and

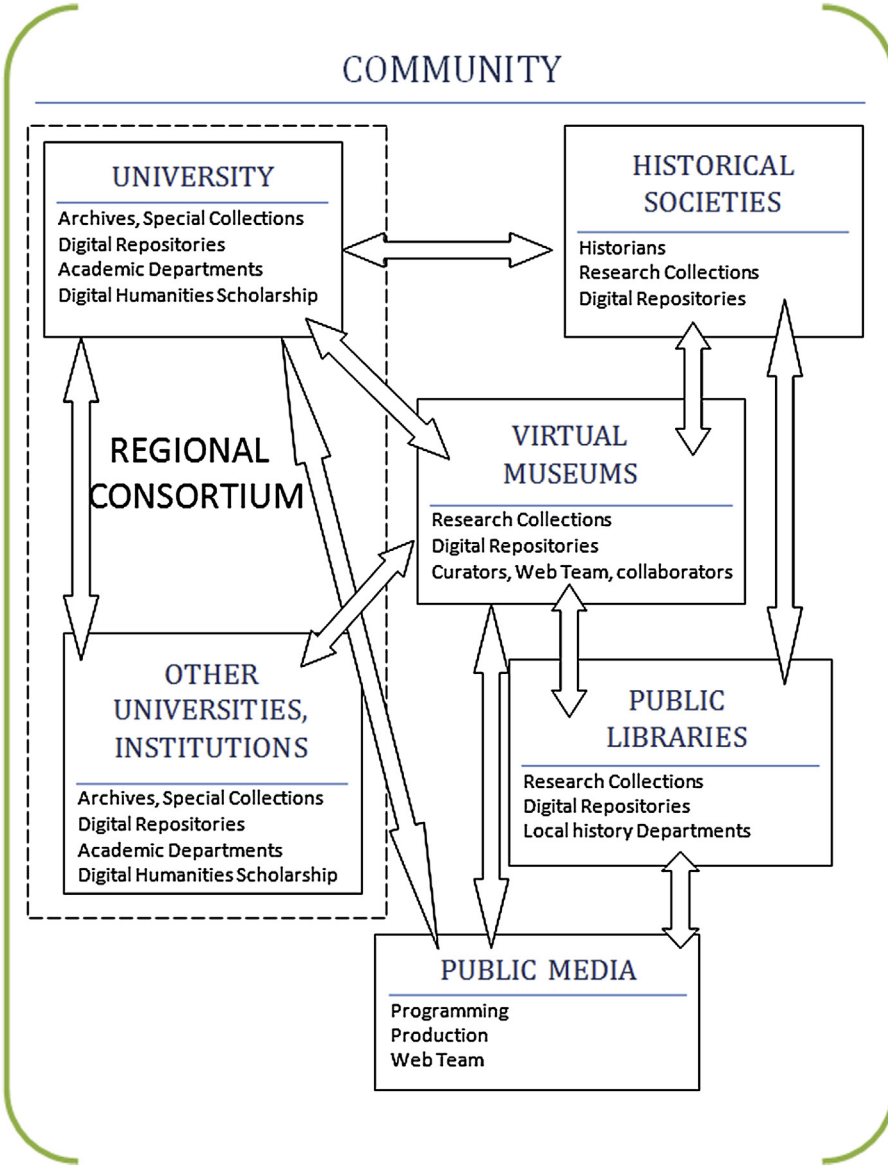


Figure 6.4 The community as knowledge architecture.

far transfers. A television station's ability to produce informational programs about local history often involved direct, serial, and near/far transfers in order to transfer collections and knowledge for producers. In the end, a link from the virtual museum provides viewers with access to the Web site containing the videos. Access to the public library's local history collections (physical and digital) uses similar knowledge transfer mechanisms in addition to the library's own curation programs for patrons.

The regional collaboration on the Toledo's Attic project since 1995 (see Sabharwal, 2013) illustrates such a multiinstitutional effort in sharing knowledge and collections.

History Day competitions present an example for collaboration among participating archives, history faculty, libraries, and schools in the community. Steman and Post (2013) describe the collaboration in detail in their article on Minnesota's History Day, which involves the Minnesota Historical Society, St. Cloud State University, Central Minnesota Libraries Exchange (a consortium of multitype libraries), University of Minnesota–Twin Cities, Hennepin County Library System, Minnesota Digital Library, and the public school system. Knowledge transfer (of history resources) was not the only positive outcome of the collaboration; it also helped libraries to improve services and resources and improved collaboration between schoolteachers and librarians. Most of all, since the area libraries have felt the impact of economic downturn, these organizations began to think creatively to deliver information and knowledge needed to succeed with the event. History Day is an outstanding opportunity for digital curation, as in many History Day events, student projects also include Web sites with historical material from archives and special collections. As students analyze and interpret the information for their projects, the result enriches the discourse on history and increases knowledge of and appreciation for history. The knowledge transferred to students is not only of historical events, places, and people but also of important digital collections containing primary sources.

Conclusion

While this chapter focused on the organizational dimensions of digital curation, it neither mandates the use of knowledge management techniques for digital curation nor does it present a prescriptive approach to transferring humanistic knowledge. The arrows, which indicate the direction of collaborative relationships, workflows, and knowledge transfer, can be logical, arbitrary, or practice based, but the organizational dynamics will ultimately determine how these collaborative relationships evolve. This chapter demonstrated that digital curation relies on organizational structures and dynamics that set such processes in motion but in different ways under different political, economic, and intellectual climates.

Social networks' impact on digital curation

7

Digital technology has significantly influenced various archival and library functions, such as the development of digital heritage collections and virtual museums. The digitization of selections from special collections, manuscripts, photographs, and other historical media has made it possible for worldwide audiences to access information about places, communities, people, and events. Social media has made it even easier for audiences to interact with the content and network with others to discuss their ancestry, inquire about places, and share information about upcoming events. Improvements in social media within the past decade have enabled users (including heritage institutions like archives and libraries) to network across these social media boundaries, resulting in cross-curation. These institutions can build significantly on these emerging social curation ecosystems to extend their outreach efforts and promote their collections.

Prior to the arrival of the World Wide Web, archives and museum had organized exhibitions, lectures, presentations, and other outreach activities. The Web has expanded the possibilities in outreach, and the virtual exhibitions allowed these institutions to do more while more remote audiences could also follow the activities at the archives and museums of their choice. The Web has also contributed to the proliferation of virtual museums leading to more creative sites and curation approaches as well as access to more knowledge on selected topics. The social networking sites, however, have succeeded in involving the public in a crowd-sourcing model for social curation. In "The Future of Museums," HASTAC Scholars Program (2012) reports that social networks have significantly transformed the relationship and dynamics between museums and audiences, allowing the former to engage the public in crowd-sourced curation. The same applies to Brooklyn Museum's *Click!* Exhibition (2009), which involved the public in the social curation process. In fact, the emergence of Web 2.0 and social networking sites has not only transformed heritage institutions, it has also led to the emergence of social curation ecosystems—that is, a broader framework for social, textual, and ontological relationships across multiple collections, networks, and knowledge domains.

This chapter focuses on implications of social networks for digital curation and addresses social curation ecosystems. Cross-curation results from the ability of social media to connect audiences across networks via automatic feeds and human-initiated actions, increasing the virality of topics, which in turn outlines a social curation ecosystem where actions in one region of the system can have varying effects elsewhere. This approach helps enriching the discourse on any topics circulating in social networks but greatly depends on the size of the engaged audience.

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Cross-curation, social curation ecosystems, and cultural heritage

In scientific context, cross-curation (as a data quality control mechanism) aims to "ensure that [data] curation standards are comparable across all participating databases" (CORDIS, 20113, para. 2). Thus, cross-curation brings quality and consistency to data curation, which is akin to curating metadata and maintaining metadata standards for interoperability in the information sciences. Elsewhere—in art and education contexts,—"cross-curation" is more analogous to the archival context of digital curation as discussed throughout this book. The work of the La Criée Center for Contemporary Art (n.d.), for instance, is involved in cross-curation of contemporary art across municipal, regional, national, and international boundaries while forging cultural connections between the City of Rennes in France and other cities across the globe. Likewise, a museum or archive may not only offer exhibition for the local visitors but may also place a limited amount of digital copies on its Web site for remote visitors to view and discuss in various social networks such as blogs, Facebook, Twitter, or other venues. Cross-curation is a strategy that allows heritage institutions to curate their collections diverse audiences with three effects on their curation activity:

1. through the additional interrelated channels it enriches the public discourse on the collections (albeit reaching well beyond the immediate control of archives);
2. it underscores the social, textual, and ontological relationships across these channels; and
3. it extends archival outreach to audiences far beyond the traditional range of outreach programs in archives and museums.

In social media, cross-curation refers to the simultaneous curation of heritage collections in multiple channels of communication, resulting in *social curation ecosystems*. Social curation ecosystem describes an ecosystem of social networks focused on digital curation using social media. Chapter 6 of this book focuses on knowledge architectures of such an ecosystem, which evolves around the relationship of content, technology, and people. Ecosystems are frameworks for the social, textual, and ontological relationships within such knowledge architectures spanning anywhere from one to multiple social networks where content and knowledge circulate. Karl Gregory Jones (2011) describes curation ecosystems as "web content which is cataloged, cross-referenced, rated, filtered, and otherwise managed by human beings… How useful any one tool might be depends heavily upon the intended use… however I think this form of curation is actually closer to aggregation" albeit this is not resulting from algorithms in artificial intelligence but a combination of that as well as human creativity, selectivity, and intelligence. In her blog, Bray (2011) writes,

> *A curator pulls together [or aggregates] and oversees collections of materials. The Internet, Web 2.0 tools and social media has expanded the traditional role of publisher to almost anyone. The role of curator is changing too. Anyone can 'curate' online material, pulling together their own collections.*
>
> *para. 1*

She points to the social curator's role in such an ecosystem. The importance of social, textual, and ontological relationships is vital to social curation ecosystems as any one of the three can serve as an access point to the other two in the hypertextual environment.

For archives and special collections, works by Theimer (2010) as well as by Whittaker and Thomas (2009) discuss the use of Web 2.0 tools and social media for archival outreach purposes. Lynne and Thomas point out that social networks are the virtual forms of old personal, professional, and collegial networks that fostered verbal and written feedback at conferences, meetings, correspondence, and informal gatherings—they may differ in venue from traditional channels of communication but may nonetheless be equally effectual. While social networks thrive through asynchronous interaction, they also bring members with mutual acquaintances into new relationships that proliferate in time. The meaning of "friending," following, re-tweeting may signal personal and/or intellectual affinity between members, but as Lynne and Thomas write, "friendship on these sites is really just a jargon for people who have chosen to connect to one another for personal or professional reasons, and does not necessarily denote close emotional relationship" (p. 5). Lynne and Thomas as well as Theimer discuss the use of social media by institutions that can create their own accounts and pages with followers and friends. In both cases, the institution may reciprocate friending or following, but their main purpose is to communicate general information, post upcoming events, share content, and maintain relationship with their audiences and supporters.

For archives, libraries, and museums, social networks present new channels for curating special collections. Theimer (2010) describes the use of discussion boards in Facebook as well as the "Reviews" application used by the Ohio Historical Society, which allows the public to provide feedback, rating, and free-form reviews. The rising popularity of social media for curation purposes is an important factor for heritage institutions, which are likely to combine digital and traditional curation. Furthermore, the coexistence of multiple curation channels leads to the cross-fertilization of hitherto isolated domains, leading to what this chapter calls *cross-curation*. This is where ideas, identities, perspectives, and knowledge cross over from one channel of communication (lecture series, presentations) to another (social media) shared among members of single and multiple communities and social networking sites. As topics go "viral" through sharing, re-tweeting, and following, the responses can be anywhere from minimal to exponential, depending on activities in channels across multiple social networks.

Hypertextuality and ontologies in social media

Cross-curation evolves in multichannel communication through social contacts in the networks. The hypertext environment allows curators to link textual materials from different Web sites, digital repositories, blogs, and research databases (especially the open-access ones without login requirement). This allows curators and readers follow hyperlinks from one textual material to related texts and construct a mental image of the text. George P. Landow (2006) has addressed networks of textual units called the *lexias* (after Roland Barthes) with various uni- and bidirectional relationships, allowing readers to construct the supernarrative. The social connections within networks may be helpful in pointing

researchers to certain materials—therefore, researchers' critical skills in evaluating those sources will be necessary for greater credibility—the social and textual relationships among members and texts evolve and emerge along parallel paths, but will not overlap entirely. This is because humans would not be able to socialize with a database or digital repository as they could with others in their own or other social networks. However, textual relationships can form between materials presented in blogs, Facebook walls, digital libraries, Web sites, and Flickr galleries through social connections. The relationships will not form autonomously; human intelligence will play a role in this process, but this is also where ontological relationships enter the framework especially where the organization of information uses controlled vocabularies (as in databases, digital libraries, and catalogs) and social tagging (as in social network). Recent professional literature has explored the semantic gap created by the disagreement surrounding the use and preference of one representation approach over the other (see Avery, 2010; Gruber, 2007; Peterson, 2006).

Emerging semantic technologies such as linked data, discovery services, and various artificial intelligence solutions may close this gap, which has philosophical and cultural (generational) roots, to a significant extent. The construction of an ontology that reinforces existing links and allows researchers to discover new ones will form yet another set of relationships on top of the social and textual ones. Thus, digital collections with rich metadata containing multiple subject terms (standard and local) may connect researchers to related works in a library's print and/or electronic collections as well as material in social networks especially if those subject headings also appear as social tags. The main point about such a multitier system is that each aspect (social, textual, or ontological) can become an entry point to the other two. In this framework, cross-curation is not merely a social activity (exchanging opinions and other information), but one that allows researchers to identify concepts, texts, and the relationships that hold them together. This understanding echoes Tom Gruber (2009) who defines ontology as "'explicit specification of a conceptualization,' which is, in turn, 'the objects, concepts, and other entities that are presumed to exist in some area of interest and the relationships that hold among them.'" ("Ontology," para. 3). Some texts may help researchers, archivists, and librarians enrich the metadata records and findings aids in ways that were not always possible before the availability of these technologies and platforms. In turn, readers can make new connections among scattered texts, and present to their own social networks—physical or digital. This chapter presents the framework consisting of these three aspects, and demonstrates that cross-curation plays an important part in a heritage institution's strategy to use social network for digital curation.

Social network theory, hypertextuality, and cross-curation

The sociological analysis of social networks by Charles Kadushin (2012) presents an important framework for understanding the relationships among the various intertextually related domains. The social relationships and intertextuality, in turn, shape the approaches taken in digital curation and may inform the strategy taken by archives

to promote their collections in particular social networks. Kadushin views social network theory as one that applies to any communities from small groups to networks of global proportions. Archives planning outreach on such scales may benefit from various fundamental components of social network theory, as it may speak volumes about internal organizational cultures, politics, and hierarchies. The relationships forming around "cross-curation" are not only hypertextual in nature but cultural, social, professional, and technological also. Another chapter in this book focuses on the hypertextual aspects as the cultural and intellectual foundation for the digital humanities. This chapter focuses on the social–technological foundations that evolve along mutual interests in social interaction and intellectual exchange across communities and their membership. The following analysis aims to demonstrate that digital curation significantly depends on social dynamics forming along the boundaries of social networks, their intellectual profiles, and technological capacities. A discussion of various cultural heritage sites, digital repositories, and thematic research collections described in the *Companion to Digital Humanities* will emphasize the variety in the range of digital curation programs surrounding each resource. The Toledo's Attic experience with cross-curation exemplified efforts in this direction.

Kadushin's theory of social networks outlines a fundamental ontological structure where nodes represent concepts with various types of relationships among them. A *network* is a "set of relationships" (2012, p. 14) that can also be expressed in formal mathematical, logical, and ontological terms. He adds, "more formally, a network contains a set of objects [or concepts] (in mathematical terms, nodes) and mapping or descriptions [or other specifications] of relations between the objects, or nodes" (p. 14). Kadushin's framework considers the following dyadic and triadic relationships. Dyads (pairs) build on simple (nondirectional), directed (unidirectional), and symmetric (bidirectional) relationships. As in social networking sites, simple relationships are unavoidable as soon as there are two or more members in a group within which there is no relationship among the members. Directed relationships are unidirectional, nonmutualized connections where one member directs interest to connect with another or multiple others in a network who do not reciprocate that interest. This is common in social network sites such as Facebook, LinkedIn, ResearchGate, and others. In contrast to the previous two models, symmetric relationships are bidirectional, mutualized, and potentially productive from a curation perspective because members in such relationships are more likely to share ideas and collaborate. However, two participants will not be successful where a substantial critical mass is necessary to produce certifiable curation results.

The triads require three members one of whom may be either neutral (as in simple relationships) or intermediary who alters the relationship dynamics. LinkedIn relationships constantly identify a third person who may introduce the unacquainted acquaintances to one another, thus increasing the relational complexity in triads. The triadic relationships could be simple, uni-, and bidirectional, and Kadushin (2012) identifies 16 triadic relationship types based on the configuration of these relationships between each two members, and the relationships could be symmetrical or asymmetrical (this type occurs in most cases) with varying implication for digital curation. Symmetrical triads with reciprocal bidirectional relationships in social media sites mean that

members follow, like, and share materials with others in the triad, and this can positively affect the virality statistics that Facebook presents to members. Although groups of three may also be insufficient where a critical mass is required, the juxtaposition of several symmetric triads can significantly factor into the virality statistics. Although this chapter does not investigate the sociological aspects of these relationships, it is important to remember that nodes do not simply represent individuals but smaller and larger groups (comprised of multiple dyads and triads), institutions, communities, cities, states, or unions thereof.

Kadushin discusses various qualitative and quantitative characteristics of social networks, which have varying implications for digital curation. For instance, homophily—two or more members "sharing characteristics that match in a proportion greater than expected in the population from which they are drawn or the network of which they are a part" (p. 18). Thus, while two or more members may casually follow or friend one another, particular chemistries may evolve simply because of mutual interests and similar experiences. Many groups form in Facebook, Flickr, and other virtual communities allowing these groups to focus on topics and experiences. The *Experience Project* site is an eminent example of such a social network, while other social networks simply focus on a narrower topics related to age, gender, ethnic, linguistic, and other demographic groups.

Density, centrality, size, distance, multiplexity, and segmentation are important quantitative aspects that also factor into the success and impact of digital curation. The characteristics may be very useful for heritage institutions to consider certain strategies for curation. Density results from the division of the number of direction actual connections by the number of possible connections and is vital to visibility and the transmission of ideas; in brief, it is an important factor in digital curation because there is a direct relationship between curation, visibility, and density. Digital curation can have a greater impact on audiences in social networks with greater density, which will boost virality. Centrality has many benefits as the centrally located pivotal node is equally accessible by all other nodes and can coordinate all—or most—relationships. If heritage institutions position themselves centrally, their ability to interact with most audiences, stakeholders, and the public, the effectiveness of their curation program will also improve. Likewise, a centrally located and highly visible digital collection can also receive more attention; thus, strategic planning is vital to the success of digital curation. Centrality and distance are related concepts, therefore, since the shortest paths are more efficient than longer ones not merely because of distance but also because there are fewer variations to shorter paths with fewer intermediators. Distance, therefore, is constructed in terms of zones of friends, friends of friends, friends of friends of friends, which constitute first-, second-, and third-order zones. The transmission and sharing of ideas and content will always be more efficient over shorter distances—that is, fewer zones—but multiplex relationships may compensate for long-distance relationships, as such relationships evolve over several strands or channels. For instance, two individuals may have relationships in multiple social networks, professional associations, or even community organizations, which is vital to cross-curation more particularly. Thus, a heritage institution may not only curate its collections in Facebook or Pinterest but through multiplex syndication it can also

curate collections across the boundaries into Twitter and Instagram, which enables it reach as many different audiences. Multiplex syndication is possible in Pinterest, Facebook, Instagram, and other social networking sites.

The notion of boundaries, however, indicates that networks are not monolithic entities but ones that evolve through segmentation. Since notes represent diversity of individuals, organizations (such as archives, libraries, museums, and historical societies), and governments, they are all different in their hierarchy, organizational culture, politics, and contents. Core and peripheral groups in larger organizations exhibit unique characteristics and interests, and among social scientists (anthropologists and sociologists), such groups constitute subcultures. For instance, a group of military veterans at a university may be interested in curating photographic and oral history collections related to wars of the past, focusing on local heroes. Another one may be focusing on disability history. They may be different from the core (or primary) group of archivists and historians focusing on collections related to regional industrial and commercial history. Such dynamics shape the curation (and cross-curation) of collections because each group may have its distinct social networking site with members intensely focused on certain topics.

The theoretical framework for digital curation—with a significant emphasis on social curation and cross-curation—has three interrelated and integrated dimensions: social, textual, and ontological. The social dimension builds on human connections, resulting in greater virality of discussion topics in participating groups. These connections span multiple social networks like Facebook, Twitter, Tumblr, or Scoop.it, and extend discussions to several communities, eventually driving traffic to the core content located on Web sites, virtual exhibitions, digital repositories, and special collections. Textual relationships may have roots in connections across social networks, as such, relationships may affect a person's contributions to several networks (such as WordPress, Tumblr, BlogSpot, Scoop.it, Facebook, etc.), but can also result from multiple postings by unacquainted authors with a mutual interest on the topic.

Finally, the ontological relationship speaks to the organization of content over the social networks. Social networks like Tumblr, Instagram, and Scoop.it offer generous space for social tags, which enable curators of special collections to extend the use of standard (Dewey, Library of Congress, or other) and local subject terms describing local collections. This approach allows users to locate collections using those subject terms and learn to include them to describe newly discovered resources in those social networks. Therefore, the interrelationships of the social, textual, and ontological links among networks, contents, and participants in narrowly defined but more integrated contexts is important, as one dimension can effectively inform and enrich the other two. Social connections within and across social network may lead to sharing texts, subject terms (metadata), new contents, and new social connections. Likewise, the discovery of texts can lead to new social connections and enriched metadata through the addition of new social tags, which may inform metadata enhancement in certain cases with inadequate knowledge on the content. The following discussion will demonstrate how these relationships evolve and influence one another in some narrower curatorial ecosystems.

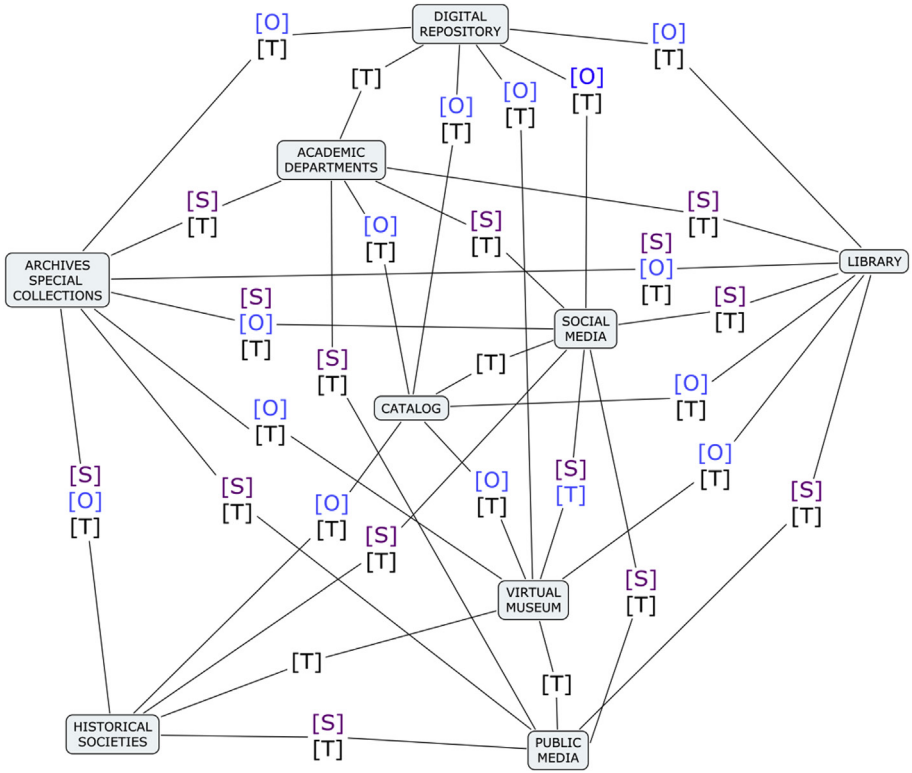


Figure 7.1 Curation and cross-curation of heritage collections.

Figure 7.1 presents a conceptual model for digital curation—including cross-curation showing the combination of social [S], textual [T], and ontological [O] relationships in one matrix. The uni- and bidirectional relationships enable the flow of sharing and curating, but the absence of arrows between any two nodes illustrate that some level of awareness may exist but there is no mutual interest in collaboration. Research and human intelligence can make those connections however. In practice, however, most of these pieces do not always and necessarily fall into place because of the internal politics, organizational culture, and technology competency (or interest) of those involved. The nodes represent combined—social, textual, ontological—nodes with simple, uni-, and bidirectional relationships to others.

Kadushin's framework is particularly helpful to demonstrate those relationships but this chapter aims to expand on those ideas in order to present a comprehensive picture of a system of curation going beyond the social relationships alone. For instance, parallel to the social interaction, there are also textual relationships between the contents curated by archives and virtual museums, or between public media content and the local history collections of the public library collections. Likewise, the virtual museum may feature content created by the public library, historical society, archives, and the public media. Finally, with the help of standard classification and social

tagging, researchers will also be able to make connections across these ontological systems, which outline a fundamental ontological framework for expanding visitor's knowledge and understanding of the domain of local history. While standard classification may help librarians make connections between library catalogs and digital repository metadata records, making similar connections across the social ontologies of Facebook, Instagram, Pinterest, and other social networks will be much more challenging if not impossible because these systems cannot interlink.

The nodes and relationships in these three levels will not match necessarily, as social connections may be richer in one relationship whereas textual and ontological relationship may be more visible elsewhere. Model in Figure 7.1 also shows whether some relationships represent a combination of all three types ([S][O][T]) or fewer ([O][T]), which illustrate the principle that while social relationships do not exist between databases, catalogs, or repositories strong textual and ontological ones may. Likewise, there are insignificant ontological relationships between historical societies and public media, but there is a rich social and textual relationship. Although the model does not indicate the directions of the relationships, any of these can be simple, uni-, and bidirectional. Nodes on each level may nonetheless be the access point to the other two; thus, cross-curation is an important concept from the viewpoint of information discovery.

Social networking and Web 2.0 tools in archives

In the past decade, several social networking sites and tools to curate digital collections and meta/data have emerged. These sites and tools allow users to aggregate, arrange, and display them in thematic contexts with the help of timelines, maps, clusters, and even in simple narrative forms. The networking sites and visualization tools discussed in the remaining part of the chapter enable Digital Humanities scholars to collaborate, visualize, and develop narratives for stronger intertextuality throughout the system of relationships. With the help of social and ontological relationships, scholars can enhance existing texts or create new ones in the course of their research, using the archives and libraries.

Facebook

Founded by Mark Zuckerberg, Dustin Moskovitz, Eduardo Saverin, and Chris Hughes in Cambridge, Massachusetts, in 2004. The curation of photographs originated in its predecessor, Facemash, founded by the same group in 2003, with the aim of compiling and ranking the photographs of attractive people ("History of Facebook," 2014). As a natural fit for curating photographs (of all sorts), Facebook has been used by heritage institutions for purposes of displaying photographs, information about their organization, and event information. The wall, now combined with the timeline, allows institutions to interact with a growing audience. The wide range of applications and features can support single-channel as well as cross-curation. It is, for instance, very easy to push feeds into Twitter, which automates notifications. Likewise, Facebook can also receive feeds from such sites as Twitter, Instagram, Pinterest, HistoryPin, and others.

Thus, each new item in any of these social networks can appear in a short announcement in Facebook. In fact, audiences can establish relationships with institutions on two levels: as a Facebook member and/or a Page created by the institution as a member or by someone else. In the former case, a bilateral agreement is necessary for relationships to exist; otherwise, any one party can initiate, refuse, or terminate a relationship. In the latter case, a member may like a page and post on its wall in order to interact with the institution or other members, but ultimately the institution may decide to block an obnoxious member, preventing him or her from posting unwanted messages or media. Facebook allows customization with apps, which have different purposes for different users. The photo-tagging feature, for instance, may allow visitors to add names to photographs. Most users tag friends' names but it is just as easy to add names of people who never had a Facebook account (like Attila or Gilgamesh) or places, buildings, events, and so on. These tags go into a database upon entry and are matched against future tagging, which establishes an ontological relationship with another site, a block of text, or a photograph. Facebook, therefore, serves as a node where all three types of relationships enmesh—social, textual, and ontological—and it becomes a venue for cross-curation.

A highly visible example of curating archival collection via Facebook is the U.S. National Archive's Facebook site. The main page provides access to historical materials including photographs, YouTube videos, events, and live streaming. The timeline organizes discussions, announcements, and comments in a reverse chronological order, providing connections to other agencies and organizations related to the national archives. Photo tagging is not an option, which limits this Facebook site to two relationship: social and textual. An August 6, 2012 posting showing President Lyndon B. Johnson as he signed the Voting Rights Act in the Capitol Rotunda in front of Martin Luther King, Jr. on August 6, 1965 ("U.S. National Archive," Facebook) has historical roots in (the source of textual relationship to) historical documents in the Lyndon Baines Johnson Library and Museum. The social relationship between the Archives and the Facebook community enables the agency to curate its collections in this channel, which helps it preserve and promote the documents at the same time. On the one hand, less traffic and physical handling (except by conservators and researchers) will result in slower deterioration of the historic documents; on the other, the information is available in summarized as well as digital complete form. The public has access to these posting and researchers may inspect the documents upon request, which lends continuity to the curation process.

Instagram

Launched in 2010 by Kevin Systrom and Mike Krieger, Instagram is a social network that focuses on photo and video sharing ("Instagram," 2014). Users can either upload media directly from a mobile device such as an Android or iPhone, or via the Instagram site on a computer. Instagram offers space for more extensive narrative than Twitter or Facebook with tagging feature that support a more efficient way to collocate related material, given that the creators of the images are diligent in describing their photographs. Images tagged as #nationalarchives will allow anyone to view images

tagged as such, but the lack of consistency in this matter will inevitably mean the omission of many otherwise interesting images or may mean the addition of otherwise irrelevant material. Instagram also allows adding geospatial tagging, which associates images with a location with photographs by others. This will allow users to discover other photographs associated with a specific geographical location, given the degree of diligence with previous efforts by others. These two methods refer to ontological and geospatial collocation with keyword and geographical data, respectively, which data aggregators and visualization tools like Storify and Dipity could gather for their maps. When typed into the body of descriptive text, the keywords are instantly hyperlinked for mobile device users. Like Facebook, Instagram supports social, ontological, and textual relationships, as demonstrated in the discussion on Toledo's Attic earlier in this chapter.

Museums such as the Brooklyn Museum and the San Francisco Museum of Modern Art have utilized Instagram to capture events and building space used for exhibitions. It provides visitors an opportunity to share visual experiences with friends and casual followers, and means additional free promotion for the institutions. Where institutions ban the use of cameras around exhibits, taking photos of exhibits will be a challenge, but taking pictures of less sensitive areas, lobby, exterior, and other public spaces will still contribute to cross-curation, relaying on social, textual, and ontological relationships. Archival collections present different opportunities and challenges for Instagrammers; while exhibitions allow visual access to artifacts and photographs, many primary source documents contain multiple (up to hundreds or thousands of) pages of just text, which may provide important historical, political, or cultural context for images and objects displayed but may only in part be accessible via this particular social media.

While Instagram has a large community of users with strictly personal content, the instantaneous and direct aspect of Instagram content can make Instagram very useful for the curation of cultural heritage, local history, and historical preservation. Instagrammers can visit sites and events and instantly upload their images to Instagram with the appropriate tags. As mentioned earlier, a Toledo-based user has been spending significant time to develop a formidable series on Toledo Rephotography, which imposes historic photographs from the Toledo Public Library over a recently taken photograph, supplied with some historical notes on the location and the buildings' history, tags, and geographical location. Anyone clicking or touching these links will see a map with pins to photographs taken in the same location or nearby. Historical preservationists may take advantage of this technology, which turns valuable individual efforts into crowd-sourced local knowledge. From time to time, Instagrams may surprise viewers with photos that may reveal the original designs of the beautiful historic buildings covered up to promote modern designs.

Incidentally, Instagrammers sharing the popular fascination with decaying structures, urban decay, urban experience (urbex), and similar themes are photographing interiors of buildings off limits to public access. At the risk of being found while trespassing, these Instagrammers often publish very valuable and timely data on interiors of building that city governments or the well-connected owners are hiding from the public. The imagery can be very useful to historical preservationists who can use

these images to urge the preservationist and government agencies to restore buildings. Despite the potential value of such photographic evidence, Instagrammers are not consistent with providing geospatial and other descriptive information along with the images, but the aesthetic appeal of decay in structurally compromised urban structures evidently earns these contributors the admiration and recognition from other urbex and urban decay enthusiasts. For instance, Industrial Decay is an exhibition of decaying industrial structures shown in Instagram and Flickr. In a Twitter thread tagged #ifihadglass, Waller (2013) labels it "Documentary done through the eyes of an (urbex) explorer," and provides link to the Flickr exhibit. Aside the intended purpose of "urbex" photography in Instagram, Twitter, and Flickr, archivists, historians, and preservationists may recognize the cultural an intellectual—as well as the artistic—value in these popular activities. From the perspective of cross-curation, Instagram supports social, textual, and ontological relationships (in this order of priority) among communities and contents.

Pinterest

Launched in 2010 and founded by Ben Silbermann, Paul Sciarra, and Evan Sharp ("Pinterest," 2014), Pinterest uses a pinboard approach to sharing photographs and limited descriptive details. It allows members to follow activities and repin images from other members' pinboards, add or enhance the descriptions, and post on Twitter and Facebook. Additionally, both Facebook and Twitter allow access to Pinterest, which may open up other possibilities for cross-curation. The relationships include textual and social ones, not ontological, as there is no social tagging feature. The pinned photographs are accessible through a content and metadata harvesting method Pinterest uses. It does not load or copy the original; it merely produces a mirror instance that fully credits the source. The authors of all original works retain their intellectual ownership. Pinterest allows users to organize selected images into thematic folders, which may allow others to follow those boards. A significant advantage of Pinterest is that it facilitates the curation of Web sites, virtual exhibitions, digital archives, and other image repositories (such as Flickr or Tumblr) without downloading the images. The debates surrounding intellectual property of content creators have raised valid questions as to who owns the content on Pinterest (see Tsukayama, 2015), and the public access policies of access by institutions. Cultural heritage institutions may include what is already on their Web sites (those resolutions will not produce high-quality prints), most of which is in the public domain, while artists, including graphic designers and photographers, will need to establish creative commons licenses for their work before or immediately after displaying watermarked images.

Binghamton University Archives (n.d.) in New York State present one cause for using Pinterest to curate historical photographs, special collections, and covers of rare books. The pinboards are organized alphabetically and the captions serve as brief descriptions of photographs with links to the original site. This method allows the institution to curate special collections almost effortlessly while connecting visitors to other collections of the library and the university. This application demonstrates a typical use of Pinterest, and may present no copyright issues where the images are either

minimal (such as book covers) or have been released into the public domain. Similarly, the University of Maryland also displays some of its Special Collections on Pinterest, but it also re-pins images of related content from other heritage institutions such as the British Museum. Pinterest presents an easy way to cross-curate content because Pinterest users can simply repin images form Flickr, Facebook, Web site, Tumblr, and other image-oriented social networks. From a preservationist's point of view, the strategy may save deteriorating books from unnecessary handling while providing access to invaluable information, most of which is in the public domain.

WordPress/Blogger

WordPress as social media and/or platform with a large array of plug-ins, can play an important role in curation. A curator can create extensive narratives—not an option in Facebook, Twitter, and other platforms for microblogging—related to collections and exhibitions. WordPress is available as a content management system as well as a blog service, the latter of which is of interest for purposes of this chapter. As a blog, WordPress offers space for extensive writing, image sharing, and cross-curation to other social networks such as Facebook, Twitter, Pinterest, Tumblr, Google Plus, StumbleUpon, Digg, and LinkedIn. The difference in the culture of these communities have implications for cross-curation because the members of those communities interpret the information differently, thus adding a new thematic dimension to the information in the collections. Archives and museums can use blogs to provide background on the collections exhibited physically and digitally. Curators can play a new role in shaping public memory. The blog site National Archives in the United Kingdom (n.d.) presents a scenario for other archives to consider blogging. It is essentially a Web site with tabs for essential information on the resources and services, but the purpose of the information is informational and educational. The National Archives offer workshops, videoconferences, podcasts, professional development, and virtual learning opportunities for archivists. The Records tab provides instruction and guidance on locating records in the system. Under *Archives sector*, visitors can search various collections. The blog provides extensive support for the curation of archival records and special collections for informational, educational, and professional uses. The Bebe Miller Archive blog (2011) presents a different scenario for curation, as it appears to speak about the professional archival perspectives on curating collections while it curates collections. The scope of curation, however, is limited to the Bebe Miller Archive preserved at the Ohio State University's Jerome Lawrence and Robert E. Lee Theater Research Institute. The Smithsonian Institute's Archives use blogs to educate visitors about the collections and the scientific contexts to inspire continued research and learning. The "Bigger Picture" blog also presents visitors the process of developing and maintaining the archives and metadata. For instance, "What are You Watching?" presents in some detail the details of curating digital video for social media while addressing the preservation and technical issues as well. These three blogs (many more exist) present scenarios for macroscopic and microscopic curation approaches, respectively, as the former presents information on all the collections of an archival institution while the latter focuses on one single collection.

While many institutions simply post information on the development and historical background of their collections, blogs are suitable to invite discussion from blog followers with potentially unique knowledge of the collections. Such a crowd-source curation brings local knowledge, individualistic perspectives, and missing data into the public discourse, which informs the public and engages the institution in a process of enriching any metadata that requires updating or completion.

Scoop.it!

Launched in November 2011 by cofounders Guillaume Decugis and Marc Rougier, Scoop.it! (2011) has taken online curation into a new direction. As a social media curation tool, it allows users to aggregate textual and media content from Web sites and other social networking sites into a thematically arranged collection. It allows two-way cross-curation by accepting feeds from the various social media channels already used for interaction with diverse audiences on the one hand and then share the newly developed composite narratives (or "scoops") in those social networking channels, which can support automatic feeds into others. Scoop.it sends feeds into social networks such as Facebook, Google Plus, Tumblr, LinkedIn, WordPress, Pinterest, StumbleUpon, and Twitter, all of which present additional feeds into other channels such as Digg for instance. Thus, a feed in Facebook may also produce a tweet in Twitter given that curators choose the correct settings within Facebook or Twitter or both. Additionally, Scoop.it allows curators to add their own insights, which presents readers with perspectives, historical background, and other contextual information, which may ultimately inspire new narratives in Scoop.it or elsewhere such as blogs. Thus, for some researchers, it may present some primary source material (pending the quality and scope of the selected stories) for new writing.

Archives' and museums' experience with Scoop.it may be sketchy due to its recent arrival, but a blog posting (Cullingford, 2012) on the use of Scoop.it for archives and special collections demonstrates similar potentials for long-term use compared to other established social networks. To this effect, she describes another Scoop.it site titled *Girl in the Moon* as a reasonable model for heritage institutions considering Scoop.it. The collection of articles from various news sources, library Web sites, and related social media informs readers of special collections and archives available for academic researchers, citizen scholars, and aspiring students interested in local history. Scoop.it embraces all three types of relationships: textual, social, and ontological. The social connections to other sites and their users allows sharing various narratives very easily, but the ontological relationships must be explicit and visible in the narrative as to allow researchers to use those subject terms for continued research in databases, catalogs, and digital repositories.

Google

Unlike the stand-alone social networks, Google presents an integrated arsenal of social media in combination with galleries such as Picasa and Panoramio, maps, analytics, blog, video (YouTube), social network (Google Plus), and interaction curation spaces

such as the *Art Project* (n.d.). In fact, if everyone used Google service only, this would be the perfect solution. In reality, however, heritage institutions are reaching out to audiences that are more diverse in those choice of social networks and collaboration tools. Panoramio allows commenting, sharing on Twitter, and adding a geolocation for the image, which allows map users to discover those images along with others associated with the same location. The Google Art Project provides an extensive platform for the combined use of multiple Google services, including maps and street view technology to navigate the virtual museum space (emulating Second Life), blogs, media galleries, Google Documents, and other features.

The *Google Art Project* (Google Cultural Institute, n.d.) provides a comprehensive model for museums that this platform for exhibiting images. *Stories of the Holocaust* (which covers Anne Frank's life along with other holocaust stories), *History is a Mystery*, and the destruction of Hiroshima and Nagasaki present archives possible models for displaying their historical records and special collections. Museums with art exhibits have utilized the interactive technology in Google Street View but in a three-dimensional representation of the exhibit space, which visitors can experience through the screen of a computer or mobile device. A map informs viewers about the geographical location of the institution, but visitors can also navigate the interior of the institution. Clicking or tapping on hotspotted locations of the space can allow visitors to move between rooms of the museum and touching the image displays details about the art exhibit. Archives with rich artifact collections can emulate a museum environment by using interactive panoramas or by using the *Google Art Projects* Web site for their virtual exhibitions. Using the maps, visitors can effortlessly explore the geographical space of a city within which the museum is located, and slide into the museum space itself—provided that it uses the associated technologies.

Storify

Storify (2014) is a service of Livefyre and is offered as both free and commercial service. Unlike social networking sited, Storify as a Web 2.0 tool that allows curators to aggregate texts, links, and hypermedia from various social networks in an order that suits unique interests and needs. It allows users to drag and drop from a pool of text and media retrieved from a search of multiple social networks such as Facebook, Twitter, Instagram, Google+, YouTube, Flickr, and others. Users can also draw on material from their own social network accounts. Facebook and Google Plus also allow developing stories in their respective structure, so these platforms may allow users to view the supernarrative in a more tangible but brief form as the length limitations will prevent curators to copy large texts into a posting or series of postings. Alternatively, links to large bodies of texts and media collections can take visitors to those sites. The various applications and widgets allow users to integrate functions such as RSS, Twitter, and other feeds into the timeline. Storify presents a unique way of visualizing the supernarrative on the screen. Thus, a curator preparing a virtual exhibition on a celebrity or politician will be able to aggregate narratives and hypermedia from a large array of social networks to support the theme of the exhibition under development. The current Storify screen displays featured stories from prominent contributors (news media, independent journalists, others).

Scripto

Scripto (2014) is a freely available open-source community transcription tool released in 2012 by the Roy Rosenzweig Center for History and New Media with National Endowment for the Humanities and National Archives and Records Administration's National Historical Publications and Records Commission funding. Scripto engages the public in a crowd-sourced transcription of historical (handwritten) documents and multimedia files such as oral history interviews. It uses wiki technology to facilitate community editing and feedback. Institutions with limited funds for supporting digital humanities projects may use this technology to engage communities of volunteer transcribers, editors, researchers, scientists, oral historians, and others to process and publish documents. Since this tool integrates well into WordPress, Drupal, and Omeka, curators can feed updates to several social networks about the progress and findings of the projects. Once completed, the transcription may have textual relationships to other historical documents, and if the editors supply subject keywords for the transcriptions, they can establish ontological relationships as well. Finally, curators may release the transcription to fill in the holes in the interviews that could not be cross-referenced with any texts or photographs but may exist in someone else's memory. Archivists and researchers may further collaborate on producing a text encoding initiative document to add special tagging to the transcription, which created a new set of ontological and textual relationships with other similar documents. Thus, Scripto is a social text production tool that facilitates collaboration in the digital humanities environment. As a cross-curation tool, its effectiveness is maximal when curators publish the text on Web sites, blogs, and certain pieces of the text and media in social networking sites, and engage the public to inspect, comment, interpret, analyze, and further curate it.

Dipity

Launched in 2007 (Dipity (2011) is a visualization tool that allows users to construct timelines, maps, lists, and flipbooks containing material in Facebook, Google, Twitter, Pandora, YouTube, and other accounts. The timelines can them be shared on Facebook, Google+, Twitter, Digg, and other social networks, and are useful tools to teach history and social sciences. While it seems to draw mainly from social media, users can present links to special collections, finding aids, Web archives, virtual exhibitions, and other primary source materials that historians and humanists can use for analysis and interpretation. The types of connections researchers can build in Dipity are primarily social and textual, as there is no place to add tags—for standard subject terms or social tags. Once curators view the material in their home domain, however, adding tags becomes a possibility, which allows cross-curation using ontological relationships. The social tags created in the home community, however, do not appear in the Dipity timelines. Curators can use Dipity to build textual and/or social relationships between the various events and the historical documents and social networks. Although Figure 7.1 did not include Dipity or Storify, these tools can effectively bridge some gaps in the vital connections needed for effective cross-curation. They will not only allow researchers and students to fill in the gaps in knowledge of some

historical events, places, people, or institutions, but will also learn to create simple timelines for presentation.

StumbleUpon

Launched in November 2002 by Garrett Camp, Geoff Smith, Justin LaFrance, and Eric Boyd ("StumbleUpon," 2014), StumbleUpon is a discovery service with an integrated peer-sourcing recommendation system that brings content to users based on personal interests. StumbleUpon recommends Web sites and other Web contents (such as images and videos), which curators can organize into lists based on topics of interest. In his blog, Visser sees a combination of Facebook and Google+ in StumbleUpon and an ideal tool for museums to create online scrapbooks. It is "a cabinet of curiosities of the Web" (Visser, 2012) with randomly discovered sites through the stumble process. Archival collections contain many scrapbooks, so the format is familiar to curators interested in extending this approach to the Web, except that in StumbleUpon, the recommendations for new content can come from the service as well as other Stumblers. A stumble button increases the visibility of Web pages in this service, which allows museums to promote their own institution's collection while adding other materials to their StumbleUpon site. A StumbleUpon button allows visitors and curators alike to add images, articles, and other Web content to their sites. Comments to the blog posting from museums like the Smithsonian Institute and others present various curatorial perspectives related to discovery, duration of time spent on a site—possibly determined from analytic data—and visitor loyalty. For curators using StumbleUpon as part of a broader digital curation strategy, the service supports social, textual, and ontological relationships.

Tumblr

Started by David Karp in 2007 ("Tumblr," 2014), Tumblr combines microblogging and other forms of social networking. It allows members to post images, links, quotations, video, and text. It uses a method similar to Pinterest, Scoop.it, and Stumble, which allows users to click on the button in a menu bar in order to harvest the content of the site into those services. Like StumbleUpon, Tumblr also supports social, textual, and ontological relationships, and serves as suitable platform for the curation of archival and special collections. The Center for Legislative Archives (a part of the U.S. National Archives) maintains a Tumblr site, *Congress in the Archives* (n.d.), with video clips on such timely topics as the 1963 march on Washington, DC, commemorating Martin Luther King's "I have a dream" speech and presenting the pamphlet distributed at the event. The facsimiles of historic letters can serve as primary sources while skeptical researchers will want to examine documents at the archives. Each blog posting presents readers subject term and opportunity to comment and share the postings on other social networking sites. Like most other sites, Tumblr readily supports cross-curation.

The Special Collections & Archives, University of Iowa uses Tumblr (2014) to curate special collections, announce events, and allow visitors to post comments. The site also supports social tagging. On the right-side navigation bar, the site receives feed from Twitter with more real-time conversations among a wider audience. Mount

Holyoke College Archives and Special Collections' Tumblr site (2014) presents more links to related Tumblr, Pinterest, and other social networking sites and digital collections. The support for social tags allows archivists and librarians to reinforce the ontological relationships between site(s) and the collections they curate, which allows researchers to locate research material with subject terms added to blog postings. The expansive space allows curators to post extensive narratives on featured collections as visible on the Miami University Special Collection's Tumblr site, which has links to other social networking sites such as Twitter, Flickr, and Pinterest.

YouTube

Launched in 2005 by Chad Hurley, Steve Chen, Jawed Karim, and acquired by Google in 2006 ("YouTube," 2014), YouTube is a video sharing social network, which allows uploading video content as well as discussing the content. The videos are universally accessible from most devices, and can be used to educate and inform, and lead viewers to full versions of the program. Some contents in the public domain do remain freely accessible to the public by the producers' decision. For instance, special collections and archives of state institutions may release some videos into the public domains and use YouTube for curation purposes where the public can access and comment on them. The Special Collections Department at Iowa State University (n.d.) has published digital copies of historical films on YouTube with brief descriptions. This strategy helps curators in many respects: while it provides instant access to the information on the films to anyone researching the topic, it also helps the institution preserve the original media, which may be in an unstable or unplayable condition—a realistic situation for many film collections. Therefore, digitization is an important part of preservation granted that there is a digital master stored in a suitable environment for long-term access.

Many archives also use YouTube to educate the public on the use and access of archival and special collections. Instructions on searching, social tagging, and assessments of archival collections are available on YouTube. Harvard University, University of Chicago, and other institutions have produced instructional videos to help students, faculty, staff, and the public navigate the complex structures of archival collections. YouTube is also where public television stations have published documentaries using photographs and other sources in the public domains. Like other social networking channels, YouTube supports cross-curation through social, textual, and ontological relationships. The U.S. National Archive's YouTube site presents many historical documentaries along with the finding aid and social tags for clearer organization. Moreoever, transcripts are also available for some videos, and this lends the site remarkable accessibility and makes it a valuable research tool. Links to external content (archival collections, other social networking sites, etc.) can easily take visitors to the desires collection, albeit as with archives, most researchers may ultimately consider traveling to the archives to access specific records.

Toledo's Attic: an experiment in cross-curation

With the continuous growth in social networking sites across the globe and the widening gamut of technologies, it is reasonable to ask which ones and how many site to

use as part of a diverse social curation strategy. Several factors motivate (or discourage) heritage institutions to establish, maintain, and expand their presence in social media—most importantly, staffing and the support for outreach. In most cases, the choice of social media is limited to a few choices, such as Facebook, Twitter, Flickr, and Pinterest while some also peruse Tumblr, WordPress, Picasa, and others. Surveys of the community may help in such decisions. Perhaps, the archives with one of the most extensive use of social media is the U.S. National Archives. The social media directory lists the following social media: Blogs, Citizen Archivist (Flickr), Facebook, Foursquare, Github, Google+, Historypin, IdeaScale (Have Your Say), Our Archives Wiki, Pinterest, RSS Feeds, Storify, Twitter, Tumblr, Ustream, and YouTube. Moreover, the directory presents the distribution of these sites as used by their individual regional repositories almost all of which use Facebook with Twitter following a second most widely used social network. Only the main site (and the one with most employees) uses 14 social networks, and attending to all requires planning and coordination. Such an arsenal of social media will support extensive cross-curation of digital collections and related outreach programs.

The planning of the Toledo's Attic project in Northwestern Ohio began in 1995, and since 1997, the project has become one of the many pioneering digital humanities endeavors across the country. The social curation program began using the AppleTalk feature—essentially a comment box—allowing community interaction with the site. Between 2009 and 2013, it has also become an experiment for cross-curation using not only one social network—Facebook—but several others like Twitter, Instagram, Pinterest, Google Plus, HistoryPin, and more. The social curation ecosystem evolved as new social, textual, and ontological relationships formed among the networks and the Web site. During the initial stages, Delicious (social bookmarking site) and Google Earth (the community for sharing and curating geospatial data) were strong candidates, but the former has diverted from its earlier mission while HistoryPin and Whatwasthere.com were much more suitable for curation purposes. This was a gradual expansion however, but the underlying concept is that the automatic and instantaneous feeds between any two allow messages to reach larger and more diverse audiences (Figure 7.2).

The social curation ecosystem in Toledo's Attic

The hyperlink relationships are the basis for the conceptual model presented in Figure 7.1 where any one of the several channels can be an access point to the others. For example, someone who views *Toledo Stories* produced by WGTE Public Media (or WGTE Public Broadcasting as it was called earlier) will discover the textual relationships between the material in the program and the sources of the Ward M. Canaday Center, Maumee Valley Historical Society, or the Toledo Public Library. Likewise, if someone joins or likes Toledo's Attic on Facebook will discover the social and textual relationships to these institutions. However, when someone discovers the area's heritage collections through the Public Library's *Images in Time* collection, the Canaday Center's digital repository, or the University of Toledo Library catalog, the relationship among these resources is ontological because of shared subject terms associated

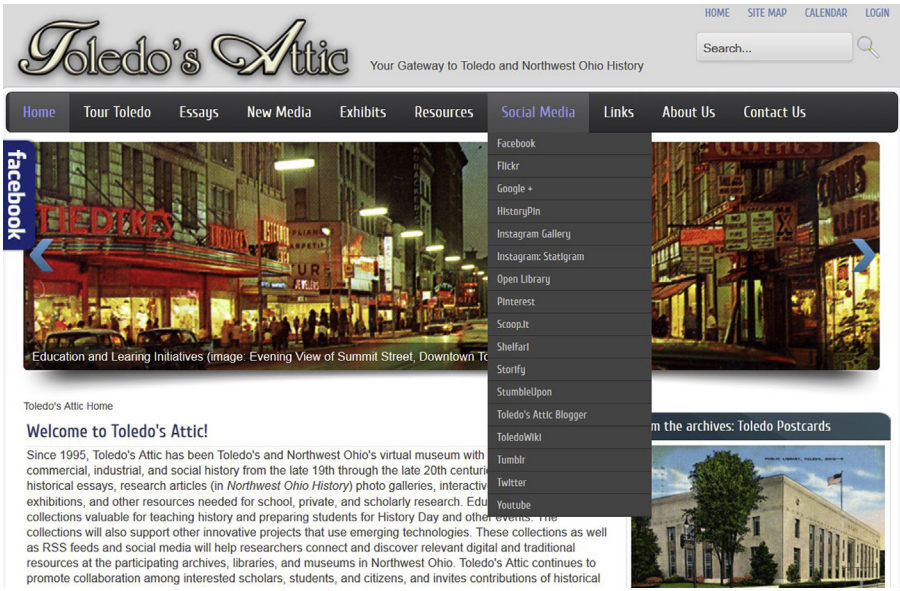


Figure 7.2 Social media in Toledo's Attic.
Source: toledosattic.org.

with the material inside. Some textual relationships may emerge between full-text collections in the repository, virtual exhibitions, and blog stories using those resources for investigative purposes. With blogs linked to other social networks such as Google Plus, Twitter, Facebook, Flickr, Pinterest, and others, the connections between the textual, social, and ontological dimensions favored a full-scale cross-curation approach. Another eminent example for cross-curation is Toledo Rephotography (Morales, n.d.) with the creator taking and posting photographs of local historical landmarks in Instagram with detailed historical descriptions of the photographed sites. Toledo Rephotography illustrates all three types of relationships in cross-curation, as it presents texts with relationships to other (more extensive) texts, keyword tags, and it connects to other Instagram and Twitter communities.

Figure 7.3 focuses on a part of the larger digital curation map presented in Figure 7.1, and shows the node–link relationships between the various curation areas for the Canaday Center and Toledo's Attic. The emerging information space no longer focuses on a single domain or Web site, but multiple interlinked ones. The illustration demonstrates two methods of cross-curation: the "link" method allows curators to place links to key locations to facilitate navigation between specified sites. In this setup, all social networking sites for Toledo's Attic provide a link to the Web site, which reinforces the centrality of Toledo's Attic as a Web site where all the collections reside. "Link and Feed" relationships exists between two or more sites allowing automatic cross-curation; for instance, a post on the Toledo Facebook wall will produce a tweet, or a Pinterest activity generated feeds in Facebook and/or Twitter. Someone

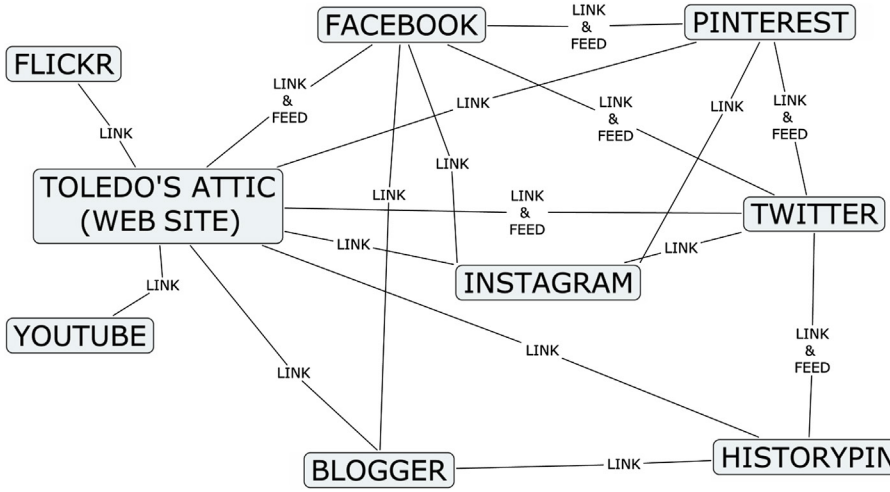


Figure 7.3 Node–link relationships in the social networks used for digital curation.

Table 7.1 Linking forms in the Canaday Center' information space

Canaday Center Web site→Finding aid indices (unidirectional, many-to-many)
Canaday Center Web site→Past exhibitions, virtual exhibitions (unidirectional, one-to-many)
Canaday Center Web site→Digital Resource Commons (unidirectional, one-to-many)
Canaday Center Web site→Toledo's Attic (unidirectional, one-to-one)
Canaday Center Web site→Canaday Center Facebook page (unidirectional, one-to-one)
Finding aid indices→HTML, PDF/A, EAD finding aids (bidirectional, many-to-many)
Finding aid indices→Digital Resource Commons (unidirectional, many-to-many)
Finding aids→Toledo's Attic (unidirectional, one-to-one)
Finding aids→Related finding aids (bidirectional, many-to-many)
Digital Resource Commons→Finding aids (unidirectional, one-to-one)
Digital Resource Commons→Virtual exhibition (unidirectional, many-to-one, one-to-one)
Digital Resource Commons→MARC records in library catalog (unidirectional, many-to-many)
Toledo's Attic Web site→Resources (unidirectional, one-to-many)
Toledo's Attic Web site→Virtual exhibitions (unidirectional, one-to-many)
Toledo's Attic Web site→Social media (unidirectional, one-to-many, and via news feeds)
Social media sites→Toledo's Attic Web sites (unidirectional, one-to-one)
Social media sites→Other social media (bidirectional, many-to-many, and via news feeds)

posting in Instagram can feed those activities through Twitter and Facebook, which shows in the Web site's Facebook and/or Twitter feed boxes as well.

Table 7.1 presents the relationships hold the social curation ecosystem together for Toledo's Attic. Although the UTOPIA digital repository has replaced the Digital Resource Commons experimentations with social curation predate the migration to UTOPIA. The same concept governs the inclusion of UTOPIA in the present.

Conclusion

Social media plays an increasingly vital role in the digital curation of heritage material. While the tools used in social curation have not been a part of standard archival practice, these networks have significant implications for outreach practices whereby global audiences can interact with the institution, information, content, and other in those networks. Cultural heritage institutions can turn their social curation ecosystems to their advantage, as they can use various automatic feed mechanisms to "cross-curate." The decentralized hypertextual nature of social networks and the human dynamics in social networking will enable curators to emphasize the textual, social, and ontological relationships, which turn just about any channel, collection, or virtual exhibition into an entry point to the collection.

Preserving the data from social media will present varying challenges to the institutions' curators. Will they be able to apply a single standard for capturing content and metadata from very diverse and heterogeneous data environments? Perhaps, a service such as ArchiveSocial or developing an in-house method, using BitCurator or other software may work better at institutions supporting open-source platforms. Will historians and humanists be able to use such data to answer questions in their fields? Will archives create finding aids for these resources the way they have with manuscript collections? A blog may after all be the future manuscript in the works toward the next great monograph, film, or stage production. The value of knowledge is not limited to specific media—it is all in the mind of the creator who in the future will likely to use digital technology to store, archive, and curate his or her personal material.

Afterword

The conceptualization and implementation phases of digital curation initiatives present a continuum throughout the digital curation lifecycle. Conceptualization considers a broad range of institutional factors, which will shape the digital curation initiative from the start to as long as the institution's support lasts. Such factors take into consideration the disciplinary and curricular interests in digital curation (sciences, humanities, and arts), the politics (stakeholders, financing, and reporting lines), and community relations. It is important to remember that both conceptualization and implementation need to be scale realistically to the environment from which it draws support. Including the chapters on digital humanities and history reflects the long-standing relationship of archives to these disciplines, but a similar book with emphasis on the sciences might have been equally conceivable. Likewise, the discussion on knowledge architectures and collaboration may have taken different departments and communities into account. The use of social media has also been on the rise and an increasing number of archives—including the national archives of the United States and United Kingdom—are using social media to support social curation. The social curation ecosystem for each institution varies and is unique to the environment it engages. There are no monolithic approaches and solutions here.

This book presented a model for digital curation that considers a broader theoretical and practical framework in order to embrace a diversifying cultural, intellectual, and technological environment. Archives continue to remain extremely important for their role in preservation, conservation, and curation of digital collections—which include cultural heritage collections, digital humanities data, electronic publications, and even social media. At the same time, new technologies and disciplines emerge and make extensive use of popular technologies and little dialogue with repositories interested in preserving digital ephemera. The decision to archive tweets was received with skepticism about intruding private spaces for instance. A similar move in the future may also aim to archive Facebook, Instagram, Flickr, and other social media committed to sharing memory and cultural identity among their membership. The ability of smaller archives to absorb the onslaught of social media and the storage requirements for such data are limited. Only national and regional initiatives may be able to accommodate such projects. Smaller institutions must focus on their internal collections however while following standards and best practices to ensure optimal quality, interoperability, and mutual discovery.

Many curators may find themselves on the epistemic gateway where they will expect to provide access to both while enabling researchers to migrate across disparate worlds of records in print and digital form. This is, of course, until (and *if*) we do reach a stage of "digital utopia" where presumably everything will be accessible via digital technology of all sorts—not just desktops but mobile technologies alike.

In that spirit, it is quite okay to visualize an environment with digitally networked watches, eyewear, or even vehicles on four wheels sending notifications about nearby places, institutions, or monuments of historical and cultural significance once all archival collections become digitally accessible via some recommender system, pervasive technology, or other pieces of ubiquitous computing technology in this "digital utopia." There are the so-called techno-realist archivists and humanities scholars who rightfully raise valid questions about such visions or the implications thereof to Humanities scholarship. Is this true scholarship, or are we allowing technology to dictate the course of Humanities scholarship? Much of the debates in the digital humanities evolve around this single question. There are reasons for concerns about preservation of and access to digitized and born-digital content mainly because digital content is ephemeral as are the technologies used to produce them. Even if the digital collections and data survive into future decades, other questions remain: what hardware and software will be around to access the information if the market forces render these collections inaccessible and, therefore, useless? Archivists have mastered preservation and conservation of print and film collections, but the maintenance and repair of analog and digital media has, thus far, been beyond the scope of archival training at most universities. Perhaps, there should be more hands-on training to maintain film projectors, video players of all types, and computers. Better relationships with IT departments may have resulted in archives receiving such equipment in reasonably good condition, but at most institutions, this may be a challenge.

Yakel, Conway, Hedstrom, & Wallace (2011) address the difference in which digital natives and older generations use information. There are also generational differences in the expectations of what is available on the Internet. These generations now include "digital natives" and "digital immigrants." Yakel et al. (2011) write, "A new generation of professionals is needed who are comfortable working in hybrid (digital and analog) environments and are capable of managing media-neutral information throughout its life cycle" (p. 23). Given the place of archives as providers of information in both digital and pre-digital formats, some "digital immigrants" may be in a better position to fill this space. A growing number of technology users believe that everything worth to know is on the Internet, which is the ideal place to connect with narratives, media, and communities of people committed to sharing knowledge and memory in social networks. There is no problem in this: Flickr, Instagram, Facebook, Pinterest, Twitter, and emerging social networks play an important role in developing community and personal digital archives, but are they committed to long-term storage and preservation of that data? Facebook, for instance, enables account owners to download personal data, which include textual and nontextual (media) data. The Library of Congress has recently archived billions of tweets. ArchiveSocial is a new service committed to archiving legally sound social media records, which is a serious step to close some gaps between social media and ancient history. Also, just about anyone can be meticulous about saving personal data, creating screenshots, and initiating a lifelong sequence of storing and migrating to new formats. Someone in their 40s, 50s, and 60s today (the so-called "digital immigrants") have the benefit of knowing and having worked with digital and predigital technologies and media formats. Their knowledge and insights will be crucial in framing a realistic environment for accessing knowledge across the epistemic threshold between

digital and nondigital worlds. A new digital curation curriculum (Yakel et al., 2011), therefore, may strongly benefit from the skills and knowledge of digital immigrants.

The misinformed understanding that everything is on the internet has implications for the Digital Humanities community. Some are concerned about abundance and information overload (of information on the Internet) while others have such concerns about trends in the digital humanities that there is too much emphasis on computation, visualization, and building all things digital, and not enough critical discourse on scholarship or technologies that truly support qualitative (even narrative) history or humanities. Yet, archivists are rightfully more aware of content that is disappearing (or have already disappeared) from public records because of the media discontinuity. The concerns about quantitative history and humanities are that it is far too dictated by the positivistic nature (or limitations) of computing. The fact that blogging and social media can go far beyond by mitigating the effects of what older database-driven approaches did not do or accommodate: to represent knowledge in narrative, hypertextual, and intertextual forms.

The ephemeral nature of social media is still of concern to curators. A common standard for exporting and archiving heterogeneous disparate data in various social media may support progress in digital curation practice, but then the next question will relate to logistics of capturing and storing these in digital repositories. Unless there is a common standard for disparate systems to follow in spite of what the Open Archives Initiative Protocol for Metadata Harvesting environment has been offering, some (presumably open source) systems will support data sharing through greater interoperability while others will not. The digital curation standards were a response to the early discussion on digital preservation in the early 2000s; some institutions will be able to commit to those standards with a dedicated IT department while others seek commercial solutions that may not even meet the necessary interoperability standards. There is hype about everything digital; yet culturally significant collections (Colorado Digital Project, OhioLINK Digital Resource Commons, even the *My History is America's History* Web site) disappear due to changing political landscapes, declining budgets, and shifting demographics. Even if society were better-off with print copies, the question of born-digital and now deteriorating and inaccessible content would still linger. Committed archivists and digital curators will remain as mediators between history and researcher, between memory and the public, as long as their resources and working environments support. In the end, many will continue to feel that history still belongs to those who pay more for it.

Alas, this book has only addressed what is the tip of the iceberg. It praises what archivists have one in the past, and looks forward to how the field can leverage technology to preserve future memories. The growing visibility of archivists in their communities, institutions, and even in regional and national media can make a significant impact on using technology to preserve history, memory, and identity. And there is great hope for seeing more opensource, open access, and open standard solution to abridging the gap between two worlds. So, do not throw away your photos, videos, cassettes, and LPs: you have history in your hands and those memories are still yours!

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